

Mathematical Structures

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Preface

These notes are for the module MTH4110, *Mathematical Structures*, taken by students on mathematics or joint honours courses at Queen Mary, University of London, in the first semester.

The course description is:

This module is intended to introduce students to the concerns of mathematics, namely clear and accurate exposition and convincing proofs. It will attempt to instil the habit of being "precise but not pedantic". The module covers an informal account of sets, functions and relations, and a sketch of the number systems (natural numbers, integers, rational, real and complex numbers), outlining their construction and main properties.

The syllabus is:

- Definitions, theorems, proofs, counterexamples. Examples of proofs. How to construct proofs, and how to recognise false proofs.
- Sets, subsets, operations on subsets. Functions; injective, surjective and bijective functions. Relations, including equivalence relations. Counting finite sets (including the number of r -subsets of an n -set). Infinite sets; countable and uncountable.
- Natural numbers and induction. Integers and rational numbers with a sketch of the construction. Real numbers (treated as infinite decimal and binary expansions) including some completeness properties. Countability of the rationals and uncountability of the reals.
- Complex numbers. The complex plane with cartesian and polar coordinates; addition and multiplication. Statement of the Fundamental Theorem of Algebra.

Further details can be found on the [School's module page](#) or on the [module organiser's page](#).

The set books for the course are

- Tim Gowers, *Mathematics: A Very Short Introduction*, Oxford University Press, Oxford, 2002
- Kevin Houston, *How to Think like a Mathematician*, Cambridge University Press, Cambridge, 2009

Other useful resources for the course include

- [10 ways to think like a mathematician](#) by Kevin Houston
- [Book of Proof](#) by Richard Hammack
- [Some Theorems of the Day](#) by Robin Whitty
- [Sudoku, logic and proof](#) by Catherine Greenhill

In the text, items in [red](#) are cross-references to other parts of the notes, while items in [magenta](#) are references to external websites (which may open in your browser).

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Chapter 1

Introduction

This document is a compendium of the notes for a new module MTH4110, *Mathematical Structures*, at Queen Mary, University of London. This is a first-semester module for all mathematics and joint honours students whose home department is mathematics. The aim of the module (described in more detail below) is to introduce mathematics and explore some of its major concerns, as well as the basic objects of mathematics (sets, functions, and numbers).

Each chapter of the notes will be accompanied by study notes, and by supplementary material, which you are recommended to read. It will tell you a bit more about some things covered in the main chapter. In this composite version, the study notes and supplementary material follow on from the main text of the chapter.

This chapter will tell you about

- (a) a mathematics degree;
- (b) what is mathematics?;
- (c) what are proofs, and why are they important?;
- (d) some mathematical terms;
- (e) mathematical notation, including the Greek alphabet.

In the supplementary material you will find

- (a) descriptions of some of the mathematicians met in the chapter;
- (b) the reason why 1 is not a prime.

We start with a few words about what mathematics is, how to do it, and why a good degree in mathematics is worth the effort.

1.1 A mathematics degree

A degree in mathematics is recognised as being a good thing. Maths graduates have varied careers: as codebreakers, financial traders, operations research specialists, teachers, engineers, policy wonks, actuaries, ... – even mathematicians! Many recent surveys show that mathematician is among the top few professions for overall satisfaction with job and life.

You might like to look at [Jobs Rated](#), which gives Mathematician as the most satisfying job of 2009, followed by Actuary, Statistician, Biologist, Software Engineer Another site which describes some of the benefits of a mathematics degree is the [Oxford College wiki at Emory University](#).

Employers like maths graduates not just because they can solve a few equations. In a mathematics degree you will learn skills that will be useful to you no matter what you do with your life: you will be able to construct a convincing argument, and to see flaws in other people's arguments; you will be aware of the cultural importance of mathematics, the way it underpins so much of modern life, and the resources it has to offer for improving life in future; and, with the practice you get in writing solutions and projects and in giving presentations, you will be able to persuade others that what you say is worth listening to.

By enrolling for a degree here, you have entered into a contract with us. We will provide you with excellent teaching, resources, and advice. Your part of the contract is simply to put in the effort to take advantage of this. Studying is a full-time job: we expect you to put in ten hours per week on each of your modules, including lectures, tutorial classes, and your own work on your lecture notes and the homework problems. If you do this, you will get the added bonus that exams will hold no terrors for you since you will be well prepared for them.

We provide lecture notes for our modules. But these are not a substitute for attending the lectures. You are expected to attend and to take your own notes. Later, you may check your notes against the versions on the Web; they won't be the same, but you will get some new insights about the material.

The bottom line is: you are not at school any more, but you are an adult and are expected to take responsibility for your actions. We will help you study in the most effective way, but we will not stand over you while you do it.

1.2 This module

This module has three aims.

The first is to introduce you to some of the basic objects of mathematics, namely *numbers*, *sets* and *functions*. We start with sets in the next chapter because you will meet them right at the beginning of Introduction to Probability, and be-

cause you already have some idea what numbers and functions are; but later on we will look at these as well, and try to make your ideas a bit more precise.

The second is to introduce the idea that mathematics is more than the properties of numbers, sets, functions, and all the other things you will meet in your degree course: vectors, matrices, integrals, curves, infinite series, and so on. The central thing in mathematics is the notion of *proof*, which means giving a convincing argument for our claims; we will look at proofs, and see techniques for constructing proofs, and how to spot something which looks like a proof but isn't.

The third aim is to get you involved. Mathematics is not a spectator sport, and you will not do well unless you take part! There will be plenty of exercises to try. We expect you to give a short presentation to your tutor group, and there will be plenty of suggested topics for these presentations.

I will not always tell you what you should have written in your homework solutions. One of my colleagues said,

Watching your friend do a work-out at the gym will not make you any fitter.

Listening to Ashkenazy playing *The Moonlight Sonata* will not make you a better pianist.

So why do you think that looking at my model solutions will improve your mathematical ability?

1.3 Mathematics

Paul Erdős (1913–1996) was a Hungarian mathematician who was one of the most famous mathematicians of the twentieth century. (His name is pronounced, roughly, “air-dish”.) He said,

The purpose of life is to prove and to conjecture.

I think this is a good description of mathematics, but we need to understand the words “prove” and “conjecture”.

First, let's see another definition of mathematics. Chambers' Dictionary says,

mathematics *n sing or n pl* the science of magnitude and number, the relations of figures and forms, and of quantities expressed as symbols.

That tells us something of what mathematics is about, but doesn't give a clue about how to *do* mathematics.

What makes mathematics different is that we can *prove* our assertions. A proof is an argument or a piece of reasoning which leaves you absolutely convinced of the truth of the statement being proved.

Here is an example of a proof. This is one of the most famous of all mathematical proofs. First, so that we know what we are talking about, here is a definition:

Definition 1.1 A *prime number* is a whole number greater than 1 which is divisible by itself and 1 only.

Another way of saying the same thing is: A *prime number* is a whole number $n > 1$ such that, if $n = ab$, then either a or b must be equal to 1; that is, we cannot write $n = ab$ where both a and b are bigger than 1 (and smaller than n).

Definitions are like dictionary entries for mathematics; they say precisely what the vocabulary we use means. The definition of a prime number tells us several things:

- 1 is not a prime;
- 2 and 3 are primes;
- $4 = 2 \times 2$, so 4 is not a prime;

and so on. We will come back later to the reason why 1 is not a prime; for now, this is our definition.

Theorem 1.2 *There are infinitely many prime numbers.*

Notice we use the word *Theorem* for the statement which we are going to prove. To begin with, you are not sure whether the prime numbers go on for ever or not. They begin

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, \dots$$

but however many you write down, can you be sure that there is another one still to be found? Yes, and here is the proof.

In the lectures, we spent a long time on this proof, and at the end, everybody said they understood it. If you are reading this, and were not in the lectures, it is your responsibility to follow it carefully and check that you are convinced!

Proof The proof is a *proof by contradiction*. That means, we assume the opposite of what we are trying to prove, and show that this leads to an impossible conclusion.

So we assume that there are only finitely many primes. Let us call them $p_1, p_2, \dots, p_r$, where r is the (assumed finite) number of primes.

Now let N be the number obtained by multiplying all the primes together and adding 1:

$$N = p_1 p_2 \cdots p_r + 1.$$

Now either N is a prime or it is not; we will see that both alternatives lead to an impossibility.

If N is prime, then it must be one of $p_1, p_2, \dots, p_r$; but this is impossible since it is bigger than all of these primes.

If N is not prime, then it must have a prime factor, which is necessarily one of $p_1, p_2, \dots, p_r$. But by its construction, we see that N divided by p_i leaves a remainder of 1, since p_i is a factor of $p_1 p_2 \cdots p_r$. Another way of saying the same thing is that the two numbers $p_1 p_2 \cdots p_r$ and $p_1 p_2 \cdots p_r + 1 = N$ are both divisible by q (the first because $q = p_i$ for some i , the second because we chose q to divide N), and so their difference (which is 1) is also divisible by q , which is not possible.

So we hit a contradiction in either case. The only way out is that our original assumption (that there are only finitely many primes) is wrong, and so we have proved that there are infinitely many primes. $\square$

Notice the symbol $\square$ we put at the end of the proof, to indicate that the argument has concluded. If you have been following it, you should not need this, since it should be obvious when the argument is finished and we have achieved our aim.

The notion of proof by contradiction makes some difficulties. The basic idea is this. There are only two alternatives (the primes are finite, or the primes are infinite), one of which must hold; we showed that the first alternative is not possible since it leads to nonsense, so the second alternative must be the case.

Another way to view the proof goes like this. Suppose that someone claims to have found all the primes, and shows you a list (probably a very thick book) which contains every single prime. Then, using Euclid's argument, you can demonstrate that his list must be incomplete. You may not be able to actually *find* a prime which is not in the book, since the calculations may be too hard; but the argument shows that such a prime must surely exist.

You can read another account of Euclid's theorem [here](#) on the Theorem of the Day website.

If mathematics were simply theorems and proofs, it would be a dead subject. The reason it is not is that there are things we think might be true but cannot yet prove. Such a statement is known as a *conjecture*. Here is a famous conjecture known as *Goldbach's Conjecture*.

Conjecture 1.3 *Every even number greater than 2 can be written as the sum of two prime numbers.*

For example, $4 = 2 + 2$, $6 = 3 + 3$, $8 = 3 + 5$, $10 = 3 + 7 = 5 + 5$, and so on. (The last of these shows that there may be more than one way of writing a number as the sum of two primes.) Goldbach made this conjecture in 1742, in a letter to

the Swiss mathematician Leonhard Euler (pronounced “oiler”); in his reply. Euler said that he was convinced that it was true although he couldn’t give a proof.

The conjecture has been checked for all numbers up to around a million million by computer; but no amount of checking makes a *proof*, a convincing argument that the statement is true. However, in the centuries since Goldbach made this conjecture, mathematicians have made some progress on it, and it is quite likely that it will be proved within your lifetime.

In 2001, the Greek author Apostolos Doxiadis published a novel called *Uncle Petros and Goldbach’s Conjecture*. The publisher offered a prize of a million pounds if the conjecture was proved within a year. The prize was not claimed.

Mathematics is a living subject because mathematicians do both of the things that Erdős mentioned: we construct *proofs*, that is, we find convincing arguments; and we make *conjectures*, or inspired guesses.

1.4 Reading a proof

Think about Euclid’s proof in the last section. How was it for you? Was the argument entirely clear, or did you have to stop and think carefully about some steps? Or was it all a bit of a mystery to you?

I tried to take the trouble to write it as clearly as possible. Mathematics writers don’t always do this, so you should expect to have to work hard at proofs. But you should get into the habit of doing this. We will say more about reading proofs, constructing proofs, and recognizing false proofs later in the course.

The main message now is:

Don’t take anyone else’s word for it: demand proof!

Read the first of Kevin Houston’s [Ten ways to think like a mathematician](#) (entitled “Question everything”) now!

1.5 Some problems with the proof

This section is a bit more difficult; you may want to skip it at first reading and come back to it, but you should at least try to read it before passing on.

You have to work a bit to follow the argument in Euclid’s proof. But there are some things where maybe you just don’t get it, where I seem to have jumped to a conclusion without stopping to convince you.

Perhaps the most serious problem was when I claimed

If N is not prime, then it must have a prime factor.

Why is this true?

According to our convention, the number 1 is not prime, and does not have any prime factors either. But N is not equal to 1. Remember that N is obtained by multiplying together all the primes in the list (which we assume to be complete) and adding 1.

So we have to justify the fact that every number bigger than 1 has a prime factor (if it is not itself prime).

If N is not prime, then it is the product of two factors each smaller than N ; say $N = a \times b$, where a and b are smaller than N (and greater than 1). If either a or b is prime, then we have found a prime factor of N . If not, then each is a product of smaller factors, say $a = c \times d$ and $b = e \times f$; so $N = c \times d \times e \times f$. Each of c, d, e, f is a factor of N , so if any of them is prime, then we have found our prime factor. Otherwise, ...

You can probably see what happens. Either we find a prime factor at some stage (and we can stop), or the argument goes on for ever, finding smaller and smaller numbers: $N > a > c > \dots$.

This is impossible:

Lemma 1.4 *There cannot exist an infinite sequence of natural numbers, each one smaller than the one before it.*

Proof Natural numbers have the property that the gap between any two is at least 1: that is, if $b < a$, then in fact $b \leq a - 1$.

Suppose that we have an infinite decreasing sequence

$$a_0 > a_1 > a_2 > \dots$$

Then $a_1 < a_0$, so $a_1 \leq a_0 - 1$. And then $a_2 < a_1$, so $a_2 \leq a_1 - 1$; combining this with the previous inequality, we see that $a_2 \leq a_0 - 2$. Continuing the process, we find that $a_n \leq a_0 - n$ for all n . But then, putting $n = a_0 + 1$, we see that $a_n \leq -1$, which is impossible, since we supposed we were only using natural numbers (which are 0 or larger). $\square$

1.6 Some mathematical terms

Mathematicians use some strange words such as *theorem* and *surjective*, and also use familiar words such as *ring* and *tree* in unfamiliar ways. As we saw earlier, we need to have definitions so that we know what we are talking about. So here are my attempts at giving definitions of some words you will meet very frequently.

Theorem, proposition, lemma, corollary These words all have the same meaning: a statement for which we have a proof. Normally the statement is immediately followed by its proof.

But the four words have slightly different shades of meaning. A *theorem* is usually a very important statement; some are given names like *The Fundamental Theorem of Arithmetic* or *Euclid's Theorem*.

A *proposition* is like a theorem but not quite so important. If you have proved a new fact which wasn't known before, and you are not quite sure how important it is, then perhaps you should be modest and call it a proposition.

A *lemma* is a theorem which is not very important in its own right, but acts as a stepping stone in the proof of a much more important theorem. The theorem in the last section, asserting that there are no infinite decreasing sequences of natural numbers, appears to be a lemma or step in the proof of Euclid's Theorem. In fact, it is a very important property of natural numbers, and I would be justified in calling it a theorem, as we will see later. This does indeed sometimes happen, and many results which started off as lemmas were later recognised as being important theorems in their own right.

A *corollary* is a theorem which is an easy consequence of another big theorem. Here is a corollary of Euclid's theorem:

Corollary 1.5 *There is a prime number bigger than $10^{10^{10}}$.*

Proof There are infinitely many primes, so they must go on beyond $10^{10^{10}}$. $\square$

Here is another example. Suppose that you have just proved a theorem which says that, if $N = 2^{2^k} + 1$, then any prime factor of N has the form $2^{k+1}s + 1$. Then as a corollary you could conclude that 257 is prime. For $257 = 2^{2^3} + 1$, and so any prime divisor would have the form $16s + 1$ for some integer s , and so would be at least 17. But the product of two such factors would be at least $17 \times 17 = 289$.

Proof As we have seen, a proof is an argument which establishes the truth of a theorem beyond any doubt. A proof may refer to theorems which have already been proved. So the structure of mathematics grows, with bricks resting securely on other bricks. We use the symbol $\square$ to mark the end of a proof.

Definition A definition of a mathematical term explains what that term means. We gave a definition of a prime number earlier on. The definition must establish clearly what a prime number is, so that we can tell whether any number we are looking at is prime or not.

Conjecture We met this term also. A conjecture is a guess by a mathematician about what might be true. It is a challenge for other mathematicians to prove; once proved, it becomes a theorem. Or perhaps the original guesser was wrong, and someone comes up with a proof that the conjecture is false.

Can you discover what the Cameron–Erdős conjecture states, whether it has been proved or disproved, and if so by whom?

Sometimes these terms are misused. A famous example of this is *Fermat’s Last Theorem*. The French mathematician Pierre de Fermat asserted that the following theorem is true:

Theorem 1.6 (Fermat’s Last Theorem) *There do not exist positive integers x, y, z, n , with $n > 2$ such that*

$$x^n + y^n = z^n.$$

Fermat did more than conjecture this theorem. He wrote, in the margin of a maths book that he happened to be reading at the time, that it was true, and he had a truly wonderful proof of it, but the margin of the book was too small to contain it! For hundreds of years, nobody could reconstruct Fermat’s proof, and it is generally believed now that he was mistaken in thinking that he had a proof; so the statement was really a conjecture, and might have been called *Fermat’s Last Conjecture*. Then in 1995, Andrew Wiles (with help from Richard Taylor) came up with a proof, so now we should really call it *Wiles’ Theorem*. But the name that was used ironically for hundreds of years has stuck, so Fermat’s Last Theorem it remains.

Fermat was responsible for another famous conjecture. When talking about corollaries, I mentioned numbers of the form $2^{2^k} + 1$. Fermat conjectured that these are all prime. He was right about the first few:

$$\begin{aligned} 2^{2^0} + 1 &= 2^1 + 1 = 3, \\ 2^{2^1} + 1 &= 2^2 + 1 = 5, \\ 2^{2^2} + 1 &= 2^4 + 1 = 17, \\ 2^{2^3} + 1 &= 2^8 + 1 = 257, \\ 2^{2^4} + 1 &= 2^{16} + 1 = 65537 \end{aligned}$$

are all prime. But the Swiss mathematician Leonhard Euler disproved Fermat’s conjecture by showing that $2^{2^5} + 1$ is divisible by the prime 641. Since then, several other Fermat numbers have been shown not to be prime, and not a single further prime of this form has ever been found.

I will give Euler’s ingenious disproof of Fermat’s conjecture in the supplementary material.

1.7 Mathematical notation

Here is some notation which you will meet frequently in your mathematics degree. Much of this will be explained later in the module: keep these pages for reference.

Numbers

Notation	Meaning	Example
$\mathbb{N}$	natural numbers	$1, 2, 3, \dots$ (some people include 0)
$\mathbb{Z}$	integers	$\dots, -2, -1, 0, 1, 2, \dots$
$\mathbb{Q}$	rational numbers	$3, -\frac{1}{2}, \frac{22}{7}, \dots$
$\mathbb{R}$	real numbers	$\frac{1}{2}, \sqrt{2}, \pi, \dots$
$\mathbb{C}$	complex numbers	$a = bi, i^2 = -1$
$ x $	modulus	$ 2 = 2, -3 = 3$
$\lfloor x \rfloor$	floor	$\lfloor \frac{8}{3} \rfloor = 2$
$\lceil x \rceil$	ceiling	$\lceil \frac{8}{3} \rceil = 3$
a/b or $\frac{a}{b}$	a over b	$12/3 = 4, 2/4 = 0.5$
$a \mid b$	a divides b	$4 \mid 12$
$\binom{m}{n}$	m choose n	$\binom{5}{2} = 10$
$n!$	n factorial	$5! = 120$
$\sum_{i=a}^b x_i$	$x_a + x_{a+1} + \dots + x_b$	$\sum_{i=1}^3 i^2 = 1^2 + 2^2 + 3^2 = 14$

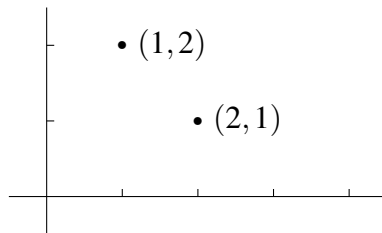
Be careful not to confuse a/b (which is a number, a divided by b) with $a \mid b$ (which is a *relation*, so is either true or false. In the next table, you will meet $A \setminus B$.

In the notation $\sum_{i=1}^3 i^2$, the symbol i is called a “dummy variable” because the name we give it doesn’t matter; replacing i by m or y would not change the value.

Sets

Notation	Meaning	Example
$\{\dots\}$	a set	$\{1, 2, 3\}$ NOTE: $\{1, 2\} = \{2, 1\}$
$x \in A$	x is an element of the set A	$2 \in \{1, 2, 3\}$
$\{x : \dots\}$ or $\{x \mid \dots\}$	the set of all x such that ...	$\{x : x^2 = 4\} = \{-2, 2\}$
$ A $	cardinality of A (number of elements in A)	$ \{1, 2, 3\} = 3$
$A \cup B$	A union B (set of elements in either A or B)	$\{1, 2, 3\} \cup \{2, 4\} = \{1, 2, 3, 4\}$
$A \cap B$	A intersection B (set of elements in both A and B)	$\{1, 2, 3\} \cap \{2, 4\} = \{2\}$
$A \setminus B$	set difference (set of elements in A but not B)	$\{1, 2, 3\} \setminus \{2, 4\} = \{1, 3\}$
$A \triangle B$	symmetric difference (“exclusive or” of A and B)	$\{1, 2, 3\} \triangle \{2, 4\} = \{1, 3, 4\}$
A'	complement of A	inside a universal set Ω $A' = \Omega \setminus A$
$A \subseteq B$	A is a subset of B (or equal)	$\{1, 3\} \subseteq \{1, 2, 3\}$
$\emptyset$	empty set (no elements)	$\{1, 2\} \cap \{3, 4\} = \emptyset$
$\mathcal{P}(A)$	power set of A	$\mathcal{P}(\{1\}) = \{\emptyset, \{1\}\}$
$f : A \rightarrow B$	f is a function from A to B	$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$
$f : a \mapsto b$	f maps a to b	$f : 3 \mapsto 9$

The order of the elements in a set is of no consequence; the sets $\{1, 2\}$ and $\{2, 1\}$ are equal because they have the same members, namely 1 and 2. We use the symbol (x, y) to mean the *ordered pair* with first element x and second element y ; so $(1, 2) \neq (2, 1)$. Think of $(1, 2)$ as representing the point with X-coordinate 1 and Y-coordinate 2 in the plane; then $(2, 1)$ represents a different point.



The Greek alphabet

In the course of an argument, we frequently run out of letters, and reach for a different alphabet to provide us with a few more. The Greek alphabet is most commonly used for this purpose.

You don't need to learn this; keep it for reference. You may want to practice writing these letters by hand. Be careful to distinguish between ζ (zeta) and ξ (xi), and between ν (nu) and υ (upsilon).

Apologies to Greek students: you may not recognise this, but it is the Greek alphabet that mathematicians use!

The symbols Σ for “sum” and $\prod$ for product are based on Σ (capital sigma) and Π (capital pi), but are not quite the same as these.

Name	Capital	Lowercase
alpha	A	α
beta	B	β
gamma	Γ	γ
delta	Δ	δ
epsilon	E	ϵ
zeta	Z	ζ
eta	H	η
theta	Θ	θ
iota	I	ι
kappa	K	κ
lambda	Λ	λ
mu	M	μ
nu	N	ν
xi	Ξ	ξ
omicron	O	$\omicron$
pi	Π	π
rho	P	ρ
sigma	Σ	σ
tau	T	τ
upsilon	Υ	υ
phi	Φ	ϕ
chi	X	χ
psi	Ψ	ψ
omega	Ω	ω

1.8 Study skills: Take nobody else's word for it

I'll just reiterate here what I said earlier. What makes mathematics different is that you don't have to believe the theories I (or anyone else) tell you; you can work it out for yourself and see if it is right. You should get into the habit of doing that with everything you are told.

This study skill means that you have two responsibilities:

- (a) When you are reading some mathematics, you should read it critically and make sure that you follow the arguments. If there is something that you don't follow, even after working at it, you should ask your friends, your tutor, or your lecturer to explain it. In particular, you should be very careful with words like "obviously" and "clearly". Mathematicians sometimes use these words when they are too lazy to explain what they are saying!
- (b) When you are writing mathematics, whether it is an answer to a coursework or exam question or a project essay, you must explain everything, so that your readers can understand it. When you have written something, read it over critically as if it were some else's work; do you find it convincing? If not, have you left out something important?

1.9 Supplement: Paul Erdős

The life of Paul Erdős is remarkable; it is told in the book *The Man who Loved Only Numbers*, by Paul Hoffman. (Incidentally, it is not true that Erdős loved only numbers, as the book makes clear.)

He worked in number theory, on many questions including some on the distribution of prime numbers; on probability; on combinatorics; on analysis; and on set theory and logic.

Most remarkably, he spent the second half of his life with no fixed abode, travelling the world to attend mathematics conferences or stay with friends and colleagues, to share mathematical ideas and do mathematical research with the people he met.

He wrote mathematical papers with over 500 co-authors, a far greater number than anyone else ever achieved. In fact, he was a pioneer in collaborative mathematical research, which increased enormously during his lifetime. It is still true that there are solitary geniuses who sit in their attics producing revolutionary insights; but it is much more common that a group of mathematicians sit at a table drinking coffee, or go for a walk together, and toss mathematical ideas around, and at the end, a new piece of mathematics emerges.

Erdős' propensity for collaboration led to a game among mathematicians, the *Erdős number*, which is rather like the Bacon number for film actors. Erdős himself has Erdős number zero. Anyone who has written a paper with Erdős has Erdős number 1. Someone who has a paper with a co-author with Erdős number 1 has Erdős number 2, and so on. It is estimated that there are, in round figures, 400000 mathematicians in the world who have published at least one mathematics paper; of these, about two-thirds have an Erdős number, ranging from 1 to 13. Someone who has not written any papers with co-authors, or who works in a small group of mathematicians who never publish papers with outsiders, will not have an Erdős number.

Now that Erdős has left, no more mathematicians can achieve an Erdős number of 1, of course; but anyone else has the possibility of getting a smaller Erdős number as the result of a new collaboration. You can read further information about this on the [Erdős number project website](#).

If you want to read my account of my own collaboration with Paul Erdős, you can find it [here](#).

1.10 Supplement: Euclid

Euclid was a Greek mathematician who lived in Alexandria in Egypt about 2300 years ago. Probably he taught mathematics there. His book, the *Elements*, is probably the most successful textbook ever written, having been in constant use for more than two thousand years.

Euclid used the “theorem–proof” format for his book. He began with axioms (self-evident truths such as “the whole is greater than the part”) and postulates (asserting that certain properties were satisfied by points, lines, and circles drawn in the plane), and on the basis of these he deduced all the geometric knowledge of his time in a rigorous fashion, setting the standard for mathematics books ever since.

Euclid's method had a great effect on the seventeenth-century English philosopher Thomas Hobbes. He was in a library one day where Euclid's book lay open at the forty-seventh proposition in the first book. Hobbes read it and said “That's impossible”, so he read the proof. This referred him back to an earlier result, which referred him to a still earlier result, and so on back to the axioms and postulates. At the end, Hobbes was completely convinced of the truth of the result he had thought impossible, and had fallen in love with geometry.

1.11 Supplement: Pierre de Fermat

Fermat was a lawyer in the south of France in the seventeenth century, but is best remembered as an amateur mathematician.

He is most famous for *Fermat's Last Theorem*, discussed in the main text. But he had many other achievements. He corresponded widely with other French mathematicians, delighting in setting them challenges. It was an exchange of letters with Blaise Pascal which is now regarded as establishing probability theory as a mathematical discipline. This is described in Keith Devlin's book *The Unfinished Game: Pascal, Fermat, and the Seventeenth-Century Letter that Made the World Modern*. Pascal had been given a problem by a friend who was a gambler. While Pascal found a "recurrence relation" which would give the solution to the problem, Fermat introduced the notion of the *sample space* for an experiment, the set of all possible outcomes, and the technique of assigning numerical probabilities to the outcomes, which is at the foundation of probability theory today. You will meet this in the module Introduction to Probability.

As promised in the notes, here is Euler's argument disproving Fermat's conjecture that, for any natural number k , the number $2^{2^k} + 1$ is prime. Euler showed:

Proposition 1.7 *The number $2^{2^5} + 1 = 2^{32} + 1$ is divisible by 641.*

Proof The proof depends on the fact that the number 641 can be written in two different ways.

First, $641 = 625 + 16 = 5^4 + 2^4$. Multiplying this number by 2^{28} , we see that 641 divides $2^{28} \cdot 5^4 + 2^{32}$.

Second, $641 = 640 + 1 = 5 \cdot 2^7 + 1$. Now we use the principle that $x + 1$ divides $x^4 - 1$ for any integer x : this is because of the algebraic factorisation

$$x^4 - 1 = (x + 1)(x^3 - x^2 + x - 1).$$

Putting $x = 5 \cdot 2^7$, we see that $641 = 5 \cdot 2^7 + 1$ divides $5^4 \cdot 2^{28} - 1$.

So 641 divides the difference between these two numbers: that is,

$$641 \text{ divides } (5^4 \cdot 2^{28} + 2^{32}) - (5^4 \cdot 2^{28} - 1) = 2^{32} + 1,$$

so Fermat's guess that all such numbers are prime is wrong. □

Notice that if Fermat had been right, then a general proof would have been required; but a single example is all it takes to disprove the conjecture. Such an example is called a *counterexample* to the conjecture. We will say more about counterexamples later in the course.

1.12 Supplement: Why is 1 not a prime?

The essential thing about a prime is that it has no factors except itself and 1. That is, p is prime if it cannot be written as a product $p = ab$ where the two factors a and b are each greater than 1 and smaller than p .

So far, 1 satisfies this requirement. Why did we decide that 1 should not be prime? After all, it would make no difference to Euclid's theorem: if there are infinitely many primes, this conclusion would not be affected if we allowed in one more.

The answer lies in another important theorem, which we will talk about later in the course. This theorem is so important that it has been given the name *The Fundamental Theorem of Arithmetic*. You can see another account of it [here](#) on Theorem of the Day.

Theorem 1.8 *Any positive integer can be factorised into prime numbers in a unique way, up to the order of the factors.*

The last phrase is put in because obviously the prime factors of a number can be written in any order: thus,

$$1001 = 7 \times 11 \times 13 = 7 \times 13 \times 11 = 11 \times 7 \times 13 = \dots$$

But the theorem says that we cannot also have $1001 = 29 \times 37$, for example.

Now for this theorem to be true, it is crucial that 1 is not a prime. If we allowed 1 to be prime, then any number could be factorised in many ways:

$$1001 = 7 \times 11 \times 13 = 1 \times 7 \times 11 \times 13 = 1 \times 1 \times 7 \times 11 \times 13 = \dots$$

In the final analysis, it is a matter of convention. When we make a definition, we are free to make it anything we like. As Humpty Dumpty said to Alice in *Through the Looking-Glass*,

When I use a word, it means exactly what I want it to mean, neither more nor less.

But our definitions should be convenient: if we allowed 1 to be a prime, then we would have to hedge in the statement of the Fundamental Theorem of Arithmetic to say that the factorisations are unique as long as we don't include the prime 1. Mathematicians have found it convenient to say that 1 is not a prime, and opt for the simpler statement of the theorem.

So definitions are conventions, just as the rules of football are conventions. There is nothing to stop you running onto the football field and playing with a different set of rules; but you might collect a red card rather soon!

Exercises

The exercises include “puzzles” and “projects”. The projects are expected to involve a greater investment of your time, and may involve some computing.

1.1 There are 26 sheep and 10 goats on the boat. How old is the captain?

1.2 When the mathematician Gauss was in primary school, his teacher asked the class to add up all the numbers from 1 to 100. Gauss saw that, if he took the sum

$$S = 1 + 2 + 3 + \cdots + 99 + 100,$$

and wrote it down reversed,

$$S = 100 + 99 + 98 + \cdots + 2 + 1,$$

then each pair of numbers in the two sums adds up to 101. So

$$2S = 100 \times 101 = 10100,$$

and $S = 5050$. Your job is to turn this into a general proof that the sum of the natural numbers from 1 to n is $n(n+1)/2$.

1.3 (Puzzle) Is it possible to have a party at which no two people have the same number of friends as each other?

1.4 (Project) Try to give either a definition or a description of mathematics. You should argue for your definition with examples of mathematics, in theory or in everyday life.

1.5 (Project) We can adapt Euclid’s method for showing that primes go on for ever to find an explicit infinite sequence of primes, as follows:

- Let p_1 be an arbitrary prime.
- If $p_1, p_2, \dots, p_n$ have been found, let $N = p_1 p_2 \cdots p_n$. If N is prime, set $p_{n+1} = N$; if not, let p_{n+1} be the smallest prime factor of N . So, for example, if $p_1 = 3$, then $3 + 1 = 2 \times 2$, so $p_2 = 2$; then $3 \times 2 + 1 = 7$ is prime, so $p_3 = 7$; and so on.

(a) Show that all the primes $p_1, p_2, \dots$ constructed in this way are different.

(b) Take $p_1 = 2$. Use MAPLE (or another computer algebra system) or a calculator to find $p_2, p_3, \dots, p_8$.

- (c) Show that the prime 2 always occurs in the sequence, no matter what the starting prime is.
- (d) Does the prime 3 always occur? [I don't know the answer to this; I offer a cash prize for the best attempt. If you can't prove it, you could try using MAPLE to investigate the question for a range of primes.]

Chapter 2

Sets

This chapter is about sets. It tells you

- (a) what is a set?;
- (b) relations between sets;
- (c) operations on sets;
- (d) Venn diagrams;
- (e) counting in finite sets, including the Binomial Theorem.

Under Study Skills, I discuss how to use lecture notes. The supplementary material contains

- (a) why sets give trouble if carelessly used;
- (b) is 0 a natural number?;
- (c) the origin of “blackboard bold”;
- (d) John Venn, inventor of Venn diagrams;
- (e) more counting.

One of the main features of abstraction in mathematics is that we can solve, not just a single problem, but a class of problems. One of the main tools for doing this is being able to treat a whole class of things as if it were a single object. If this sounds paradoxical and difficult, it is: no other idea has given mathematicians more trouble. But the basic ideas of *set theory* are simple, and we will start our journey with a gentle introduction. We also do some counting related to finite sets, culminating in the Binomial Theorem.

2.1 What is a set?

Few words in English have more meanings than the little word “set”. My Chambers dictionary gives 33 meanings as a noun, together with many more as a transitive verb, intransitive verb, or adjective. Somewhere in the list is the mathematical meaning; but I prefer to quote the first two meanings given:

a group of persons or things, *esp.* of a type that associate, occur, or are used together or have something in common; a clique, coterie, exclusive group.

In these two meanings we have the idea that any collection of objects of any kind can be gathered together to form a set, but that usually we have either of two conditions:

- the objects gathered together have something in common;
- there is a rule for distinguishing the objects in the set from outsiders.

Much of the difficulty that mathematicians have with the notion of “set” arises from the precise interpretation of these conditions. Do we have complete freedom in how we make up a set, or should there be some rule for forming it?

So it is actually not possible to define a set; it is a basic concept on which the rest of mathematics rests. Try to define a set; you will probably come up with something like “a set is a group of objects” or “a set is a collection of things”, and then you are faced with explaining what a group or collection is. Let us agree for now that we understand what a set is.

The most basic feature of a set is its elements. Here is the notation we use for this.

Definition 2.1 Let A be a set. We write $x \in A$ (read “ x is in A ”, or “ x is a member of A ”, to denote that the element x belongs to the set A . We write $x \notin A$ for the negation of this, that is, x does not belong to A .

The mathematical notation for building a set is to write a list of its elements and enclose them in “curly brackets” or “braces” $\{ \}$. For example,

- $\{2, 3, 5, 7\}$ is the set consisting of the first four prime numbers;
- $\{\text{red, orange, yellow, green, blue, indigo, violet}\}$ is the set of (English) names for the colours in the rainbow;
- $\{1, \{2\}\}$ is an odd sort of set, since one of its elements is a number, and the other is a set. (This may seem odd, but we will allow such strange things to be sets.)

Thus, if $A = \{2, 3, 5, 7\}$, then $3 \in A$ but $4 \notin A$.

Small sets like this are easy to specify, but for large sets there is a problem. The set consisting of the first four prime numbers is $\{2, 3, 5, 7\}$, but what about the set consisting of the first four million primes? I can calculate the first four million primes very quickly with a computer, but writing them down as a set will be hard. A mathematician wanting to talk about this set would say,

Let P be the set consisting of the first four million prime numbers.

This specifies quite clearly what we are talking about, and allows us to reason about it. Thus, for example, 101 is a member of P but 1001 is not. [You should stop to check this.] Being able to specify a large collection by a single letter is what gives set theory its power.

There are a couple of other ways in which we specify sets.

A few very important sets (mostly sets of numbers) have standard names:

- $\mathbb{N}$ is the set of natural numbers;
- $\mathbb{Z}$ is the set of integers (positive, negative, or zero);
- $\mathbb{Q}$ is the set of rational numbers or fractions $\frac{m}{n}$ (where m and n are integers and $n \neq 0$);
- $\mathbb{R}$ is the set of real numbers (numbers which can be expressed as infinite decimals);
- $\mathbb{C}$ is the set of complex numbers.

We will describe these sets of numbers more precisely later in the module.

There is a problem right away:

Is 0 a natural number?

This is one of a few important questions on which mathematicians cannot agree. In these notes I will almost always take the view that the natural numbers are the positive integers, starting at 1: thus, I could write informally

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

But I may not always follow this convention! More on this in the supplementary material.

You may also be wondering why we choose the letters we do for these numbers. Surely the integers should be $\mathbb{I}$? Well, $\mathbb{Z}$ stands for *Zahlen*, the German word for numbers, and set theory in its modern form was invented by the German mathematician Georg Cantor. $\mathbb{R}$ for real numbers and $\mathbb{C}$ for complex numbers

are obvious, but if $\mathbb{R}$ is the real numbers, what shall we call the rational numbers? Well, every rational number is the quotient of two integers, so we use $\mathbb{Q}$ for quotients.

Now there is another way of describing sets. Let A be a set, and let P be a property which objects may or may not have. Suppose that the notation $P(x)$ means “object x has property P ”. Then we write

$$\{x \in A : P(x)\}$$

for the set of all those members of A which have property P .

For example, we can say that a natural number n is even either by saying that 2 divides n , or by saying $n = 2k$ for some natural number k . Then the set of even natural numbers can be written as

$$\{x \in \mathbb{N} : x \text{ is divisible by } 2\}$$

or as

$$\{x \in \mathbb{N} : x = 2k \text{ for some } k \in \mathbb{N}\}.$$

Note that we do not write $\{x : P(x)\}$ for all the objects satisfying property P . There is a good reason for this, discussed in the supplementary material.

2.2 Relations between sets

Definition 2.2 Two sets are *equal* if they have the same members. Note that the members of a set come in no particular order. Thus, in our first example above, $\{3, 5, 7, 2\}$ is the same set; we write

$$\{2, 3, 5, 7\} = \{3, 5, 7, 2\}.$$

There is a special set called the *empty set*, which is the set having no members at all. This set should be treated with care, since it often defies our intuition; but reasoning carefully about it is possible, as we will see. We call it *the empty set* because there is only one empty set:

Proposition 2.3 *Let E_1 and E_2 be empty sets. Then $E_1 = E_2$.*

Proof E_1 and E_2 have the same members (namely, none at all)! □

Computer scientists sometimes distinguish between, say, the empty set of numbers and the empty set of strings, or of subroutines, or whatever. The above proposition shows that in mathematics there is no such distinction. Brian Butterworth, an expert on how children learn mathematical concepts, said,

Although the idea that we have no bananas is unlikely to be a new one, or one that is hard to grasp, the idea that no bananas, no sheep, no children, no prospects are really all the same ... is a very abstract one.

The most important relation between two sets is the subset relation:

Definition 2.4 Let A and B be sets. Then A is a *subset* of B , written $A \subseteq B$, if every member of A is also a member of B .

We sometimes write $A \subset B$ to mean that $A \subseteq B$ but $A \neq B$. Read this as “ A is a *proper subset* of B ”. This means: every member of A is a member of B , but there is at least one member of B which is not a member of A .

Note that $\subseteq$ is like $\leq$ for numbers: it permits the two sets to be equal. Thus, it is true that $3 \leq 4$, and it is also true that $3 \leq 3$; on the other hand, $3 < 4$ is true but $3 < 3$ is false.

Similarly $\{2, 5\} \subseteq \{2, 3, 5, 7\}$, and indeed $\{2, 5\} \subset \{2, 3, 5, 7\}$; and $\{2, 3, 5, 7\} \subseteq \{2, 3, 5, 7\}$ is true but $\{2, 3, 5, 7\} \subset \{2, 3, 5, 7\}$ is not true. Also,

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

In fact, in each of these cases, it would also be correct to write $\subset$ in place of $\subseteq$. For -1 is in $\mathbb{Z}$ but not in $\mathbb{N}$; $\frac{1}{2}$ is in $\mathbb{Q}$ but not in $\mathbb{Z}$; $\sqrt{2}$ is in $\mathbb{R}$ but not in $\mathbb{Q}$ (we will see a proof of this later); and $\sqrt{-1}$ is in $\mathbb{C}$ but not in $\mathbb{R}$.

Here are a couple of simple properties of this relation.

Proposition 2.5 (a) *The empty set is a subset of every set.*

(b) *Let A and B be sets. Then $A = B$ if and only if both $A \subseteq B$ and $B \subseteq A$ hold.*

Proof (a) Is it true that every member of the empty set belongs to the set A ? Yes, because there are no members of the empty set, and so there is nothing to check! The statement could only fail if some member of the empty set does not belong to A ; and this is clearly impossible. (This is a first example of the slipperiness of the empty set.)

(b) Remember that $A = B$ means that A and B have the same elements; this means that every member of A belongs to B and every member of B belongs to A . In other words, $A \subseteq B$ and $B \subseteq A$. The argument reverses; if $A \subseteq B$ and $B \subseteq A$, then every member of A is in B , and *vice versa*; so A and B have the same members, and $A = B$. $\square$

Important remark The second part of this proposition reminds us that, in order to show that two sets A and B are equal, we have two jobs to do:

- (a) show that every element of A belongs to B ;
- (b) show that every element of B belongs to A .

We can't get away with just one of these jobs.

The notion of subset gives us a new way of constructing sets:

Definition 2.6 Let A be a set. The *power set* of A , written $\mathcal{P}(A)$, is the set whose members are all the subsets of A .

For example, if $A = \{1, 2, 3\}$, then

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

It is a set whose elements are sets; but as we saw, there is nothing wrong with that!

Corollary 2.7 (to Proposition 2.5) For any set A , $\emptyset \in \mathcal{P}(A)$.

Proof What we are proving is that the empty set is a subset of A ; but this is what we showed in Proposition 2.5. $\square$

2.3 Operations on sets

There are several operations for building new sets out of old ones.

Definition 2.8 Let A and B be sets.

- (a) The *union* $A \cup B$ (read “ A union B ”) consists of all elements which lie in either A or B (or possibly both).
- (b) The *intersection* $A \cap B$ (read “ A intersection B ”) consists of all elements which lie in both A and B .
- (c) The *set difference* $A \setminus B$ consists of all elements which lie in A but not in B .
- (d) The *symmetric difference* $A \triangle B$ consists of all elements which lie in A but not B , or in B but not A ; in other words, which lie in either A or B but not both.

For example, if $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$, then

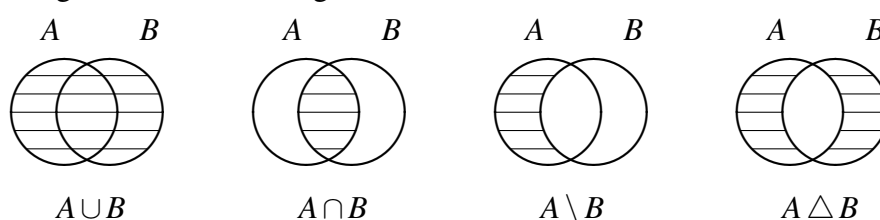
(a) $A \cup B = \{1, 2, 3, 4\};$

(b) $A \cap B = \{2, 3\};$

(c) $A \setminus B = \{1\};$

(d) $A \triangle B = \{1, 4\}.$

We can picture sets in what is known as a *Venn diagram*: each set is enclosed by a curve. The figure below shows the four combinations defined above as the shaded region in the Venn diagram for two sets.



Proposition 2.9 For any three sets A, B, C , we have

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C);$

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$

Proof Before we begin, remember the important remark in the previous section. We are trying to prove that two sets are equal, so we have to show that every element of the first belongs also to the second, and also the other way round.

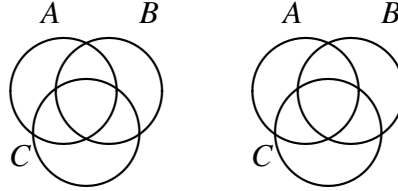
(a) An element belongs to $A \cup (B \cap C)$ if it belongs either to A or to both B and C . If it is in A , then it is in both $A \cup B$ and $A \cup C$, and so it is in their intersection. On the other hand, if it is in both B and C , then it is in both $A \cup B$ and $A \cup C$, and so in their intersection. In either case, it is in $(A \cup B) \cap (A \cup C)$.

The other way round, suppose that X belongs to $(A \cup B) \cap (A \cup C)$. Then x is in both $A \cup B$ and $A \cup C$. If $x \in A$, then certainly $x \in A \cup (B \cap C)$. On the other hand, if $x \notin A$, then we must have $x \in B$ and $x \in C$; so $x \in B \cap C$, and then $x \in A \cup (B \cap C)$. So the assertion holds in either case.

(b) Over to you for this one! □

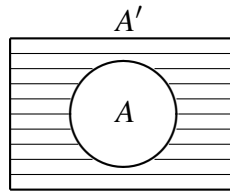
There is more in the supplementary material about proving things like this for sets. One technique which is very useful for finding out what is going on involves Venn diagrams. Here are two Venn diagrams for three sets A, B, C . Take one of the two equations in the proposition and work out carefully on the first diagram which set is represented by the expression on the left, and on the second diagram which set is represented by the expression on the right. They should be the same.

A question for you. Do you think that an argument with diagrams like this makes a proof? (This question is really asking: are you convinced by the argument?)



There is another notion, which only applies if all our sets belong to a “universal” set Ω (Greek capital Omega). You will meet this situation in Introduction to Probability, where Ω is the *sample space*, the set of all possible outcomes of an experiment, and an *event* is a subset of the sample space.

To draw a Venn diagram for complement, we imagine that the universal set is a box which includes everything we are interested in. Then if A is represented by the inside of a curve, A' will consist of everything outside.



Definition 2.10 Let Ω be a universal set. Then the *complement* of a subset A of Ω , written A' , is the set difference $\Omega \setminus A$, that is, the set of all elements of the universal set which are not in A .

Proposition 2.11 (De Morgan’s Laws) (a) $(A \cup B)' = A' \cap B'$;

(b) $(A \cap B)' = A' \cup B'$.

Proof (a) Suppose that $x \in (A \cup B)'$. This means that x is not in $A \cup B$, so it is not in either A or B . So it is in A' and in B' , and hence in their intersection $A' \cap B'$.

The other way round, suppose that $x \in (A' \cap B')$. Then x is not in A and also not in B ; so it is not in either of these sets, i.e. not in $A \cup B$; so it is in $(A \cup B)'$.

(b) Your turn. □

2.4 Finite sets

A lot of mathematics is about counting. What is this?

When you count a flock of sheep, or a class of children, you point at each one in turn and say, “One, two, three, ...”. In other words, you match up the sheep in the flock (or children in the class) with the natural numbers $1, 2, 3, \dots, n$ for some n , and say that the number of sheep (or children) is n .

Now you won’t be surprised to learn that there is a theorem which says that if you match up the elements of a set with the numbers $\{1, 2, \dots, n\}$, and also with the set $\{1, 2, \dots, m\}$, then $m = n$; that is, however we count the set, we always get the same result. Of course, in practice, if the children are running around, you might mis-count and get the wrong number. There is also a legend that, if you count the Rollright Stones (a circle of standing stones in Oxfordshire) twice, you will get different answers. But it is very believable that apart from things like this, the general result is true.

So we can make a definition:

Definition 2.12 (a) A set is *finite* if the elements of the set can be matched up with the natural numbers $1, 2, \dots, n$ for some n .

(b) If A can be matched with the numbers $1, 2, \dots, n$, then the *cardinality* of A is n , and we write $|A| = n$.

Note that the symbol $|A|$ for the cardinality of a set A (the number of elements of A) looks the same as the symbol $|x|$ for the *modulus* or *absolute value* of the real number x , defined by

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Don’t get the two notations confused.

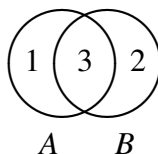
For example, $|\{2, 3, 5, 7\}| = 4$, while $|\{1, 2, 3\} \cap \{2, 3, 5, 7\}| = 2$.

Our next result relates the cardinalities of two sets A and B to those of their union and intersection.

Proposition 2.13 *Let A and B be finite sets. Then*

$$|A \cup B| + |A \cap B| = |A| + |B|.$$

Proof The Venn diagram for A and B will probably help you to understand this proof. I have numbered three regions of the diagram.



Now consider the equation in the Proposition. On the left, we have $|A \cup B|$ (which counts the elements in all three regions) added to $|A \cap B|$ (which counts the elements in region 3 only). On the right we have $|A|$ (which counts regions 1 and 3) added to $|B|$ (which counts regions 2 and 3). So on each side we have counted the elements in region 3 twice and those in regions 1 and 2 once; so the answers are equal. $\square$

This is the first case of a result known as the *Principle of Inclusion and Exclusion*. It can be written in the form

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

I will state the second case with just a hint of the proof; I hope you can see the pattern, which does indeed extend to the general case.

Proposition 2.14 *Let A, B, C be finite sets. Then*

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|.$$

Draw the Venn diagram for the three sets and mark the seven regions. Now, for each region, see how many times the points in that region are added up in the calculation on the right.

Another thing we can count is the power set of a finite set.

Proposition 2.15 *Let A be a finite set, with $|A| = n$. Then the power set of A has cardinality $|\mathcal{P}(A)| = 2^n$.*

Proof Let us label the elements of A : say $A = \{a_1, a_2, \dots, a_n\}$. Now we can describe any subset S of A by answering n questions about it:

- Is $a_1 \in S$? (Yes/No) (2 possibilities)
- Is $a_2 \in S$? (Yes/No) (2 possibilities)
- ...
- Is $a_n \in S$? (Yes/No) (2 possibilities)

Since we can answer Yes or No to each question, there are 2^n possible answers, each of which specifies a subset, and every subset arises just once. So there are 2^n subsets. $\square$

For example, if $n = 3$, the answers Yes, No, Yes to the three questions specify the subset $\{1, 3\}$; the answers No, No, No specify the empty set.

We can refine this count a bit. Given a set A with $|A| = n$, how many subsets of A of cardinality k are there?

Definition 2.16 (a) The *binomial coefficient* $\binom{n}{k}$ is the number of k -element subsets of a set of size n .

(b) The *factorial* of a natural number n , written $n!$, is the product of all the integers from 1 to n . By convention, we define $0! = 1$.

Thus, $\binom{4}{2} = 6$: the 2-element subsets of $\{1, 2, 3, 4\}$ are $\{1, 2\}$, $\{1, 3\}$, $\{1, 4\}$, $\{2, 3\}$, $\{2, 4\}$, $\{3, 4\}$. And $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$.

Proposition 2.17 For $0 \leq k \leq n$, we have

$$\binom{n}{k} = \frac{n \cdot (n-1) \cdots (n-k+1)}{k \cdot (k-1) \cdots 1} = \frac{n!}{k!(n-k)!}.$$

Proof We choose a k -element subset of the n -element set A by choosing, one at a time, k distinct elements $a_1, a_2, \dots, a_k$ from A . Now

- there are n choices for a_1 , since it can be any element of A ;
- there are $n - 1$ choices for a_2 , since it can be anything except a_1 (which has already been chosen);
- ...
- there are $n - (k - 1) = n - k + 1$ choices for a_k , since it can be anything except $a_1, a_2, \dots, a_{k-1}$.

We multiply these numbers together to get the total number of choices of $a_1, a_2, \dots, a_k$.

But we have over-counted the number of subsets. Suppose that $k = 3$, $n = 10$. The choices $a_1 = 1, a_2 = 3, a_3 = 7$, $a_1 = 3, a_2 = 7, a_3 = 1$, and several more will all give rise to the subset $\{1, 3, 7\}$, since the order in which its elements are chosen doesn't matter. So we have to divide by the number of different orders in which these k elements could be chosen from the same set. Now, just as before,

- there are k choices for the first element a_1 ;
- there are $k - 1$ choices for the second element a_2 ;
- ...
- there is just one choice for the last element a_k .

So we have to divide the number we obtained before by $k \cdot (k - 1) \cdots 1$ to get the number of subsets. This gives the fraction shown in the theorem.

Finally, consider the fraction $n!/k!(n-k)!$. The numerator $n!$ is the product of all the numbers from 1 to n . The term $(n-k)!$ in the denominator is the product of all the numbers from 1 to $n-k$; these cancel many of the factors in $n!$, leaving just $n \cdot (n-1) \cdots (n-k+1)$. So the two fractions in the theorem are equal. $\square$

The last expression is quick to write down and is easy to remember, but is usually not the most efficient way to calculate. For example, what is $\binom{100}{2}$? The last formula gives $100!/(2!98!)$. But

100! = 933262154439441526816992388562667004907159682643816214685929638952175999932299156089414639761565182862536979208272237582511852109168640000000000
 2! · 98! = 1885378089776649549125237148611448494761938752815790332698847755458941414004644759776595231841545823964721170117721692085882529513472000000000000

and you probably don't want to do the division sum! The first formula gives

$$\binom{100}{2} = \frac{100 \cdot 99}{2 \cdot 1} = 50 \cdot 99 = 4950.$$

Later in the module, we will see another way of calculating the binomial coefficients, *Pascal's Triangle*.

We made a convention that $0! = 1$. With this convention, the proposition tells us that $\binom{n}{0} = n!/(0! \cdot n!) = 1$. This is sensible because the only subset of size 0 is the empty set, and we saw that there is only one empty set. In particular, $\binom{0}{0} = 1$: the empty set contains the empty set as its unique subset.

You may be wondering why the numbers $\binom{n}{k}$ are called binomial coefficients. A *binomial* is an expression with just two terms, and the binomial coefficients arise in a famous formula for working out the n th power of a binomial:

Theorem 2.18 (Binomial Theorem) *For any non-negative integer n , we have*

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

Before we prove the theorem, look at the formula. It contains the notation $\sum_{k=0}^n$. The symbol $\sum$ means “sum” or “summation” and tells us we have to add up a number of terms. There is a variable k which goes from the lower value $k = 0$ to the upper value $k = n$; the formula tells us to calculate the terms for each value of k and add them up.

For example, suppose that $n = 3$.

- Putting $n = 3$ and $k = 0$ we obtain $\binom{3}{0}x^3y^0 = x^3$.
- Putting $n = 3$ and $k = 1$ we obtain $\binom{3}{1}x^2y^1 = 3x^2y$.
- Putting $n = 3$ and $k = 2$ we obtain $\binom{3}{2}x^1y^2 = 3xy^2$.
- Putting $n = 3$ and $k = 3$ we obtain $\binom{3}{3}x^0y^3 = y^3$.

So the theorem says

$$(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3.$$

Proof We have

$$(x+y)^n = (x+y)(x+y) \cdots (x+y)$$

where there are n factors $x+y$ in the product. To work it out, we would have to expand all the brackets. When we do so, we would obtain many terms; but each would be a product of x s and y s, where the total number of x s and y s would be n . A term containing k y s and the rest x s would be $x^{n-k}y^k$; so we have to count how many such terms there are.

We obtain a term $x^{n-k}y^k$ by choosing k of the n factors $(x+y)$ and choosing y from these factors and x from the others. The number of ways of choosing k of the n factors is $\binom{n}{k}$. So the corresponding term in the expansion is indeed $\binom{n}{k}x^{n-k}y^k$. Adding up all the terms gives the result of the theorem. $\square$

We will return to this theorem and give another proof later in the module.

If we put $x = 1$ and $y = 1$ in the Binomial Theorem, we obtain

Corollary 2.19

$$2^n = \sum_{k=0}^n \binom{n}{k}.$$

This ties together things that we have already seen. Let A be a set of cardinality n . On the right of the equation, the k th term is the number of subsets of A of cardinality k (by Proposition 2.17, and the summation tells us to add up all these terms; this gives the total number of subsets of A , which is 2^n , by Proposition 2.15.

2.5 Study skills: Lecture notes

The printed lecture notes are not the same as the textbook, and they are not the same as the lectures. It is very important that you should make your own notes, in your own words, and understand what is in them.

Research on learning shows that most people take in information better if they see it unrolled in front of them, and if they actually use the information they are learning. This suggests that the best way to benefit from lectures is to take your own notes during the lectures, make them as good as you can, and go over them later to make sure that you understand everything. This was the basis of my study technique when I was a student, and it worked for me. You will find that the effort is definitely worthwhile!

On the subject of lectures, there are two basic rules:

- Don't be afraid to ask a question. If you are puzzled by something, chances are that other people will be puzzled as well. I am happy for you to ask a question at any point.
- Please make sure that your phone is switched off, and that you do not cause a disturbance; noise can be very disruptive of other people's concentration.

The printed notes come out after the lecture, so if you really don't understand what is in your own notes, you can look there to get another view of the same point. If it is still unclear, don't just leave it; ask your tutor, or one of your friends, or me, to explain it to you.

Using a textbook is a different skill. The order of material in the textbook will be different from that in the lectures; some material may not be covered at all, or may be covered in a different way; the notation may be different. You should get to know the textbook well enough that different order of material and different notation is not a problem for you. Above all, learn to use the index of the book to find the material you need. Bear in mind that if the term "commutative", say,

occurs in the index, it will probably just point you to a definition; there may be further discussion of the concept elsewhere in the book.

There are also many resources on the Internet which you may find useful. You may certainly use these. But please remember two things:

- Printing out a page or two from Wikipedia and clipping it into your folder is *not* the same as understanding it. You have to work just as hard on this material as you do on your lecture notes. Remember Study skills 1; don't believe something just because Wikipedia says so!
- The University has rules against plagiarism. You are not permitted to take someone else's work and pass it off as your own. If you quote even a definition taken from the Internet in work submitted for credit, you should say clearly where you took it from.

2.6 Supplement: Trouble in the foundations

After set theory was developed, and some of the basic results established, by Georg Cantor in the late nineteenth century, some problems began to emerge. Some of them appeared rather technical at first, but Bertrand Russell came up with a paradox which seemed to put the whole enterprise in doubt. He asked the following simple question:

Let S be the set of all sets which are not members of themselves. Is S a member of itself?

Let us take this slowly and carefully.

We saw already that a set can have members which are themselves sets. We had the simple example of the set $\{1, \{2\}\}$, and also the notion of the *power set* of a set A , which is the set of all subsets of A . Neither of these sets is a member of itself. (For example, $\{1, \{2\}\}$ is a set with two elements; one of them is a number, and the other is a set with one element.) So presumably most sets (all but a few abnormal ones) will not be members of themselves. Indeed, you probably can't think of a set that is a member of itself; and there is a good reason for that!

However, in a slightly different context, an adjective may describe itself or it may not. Here are some examples.

- "short" is a short word, but "long" is not a long word.
- "English" is an English word, but "French" is not a French word.

Now we can write down a paradox similar to Russell's in this context. Suppose we invent two new words:

- *autological* refers to a word which describes itself;
- *heterological* refers to a word which does not describe itself.

Now ask: is the word “heterological” autological or heterological? If it is autological, then it describes itself, and so it is heterological. On the other hand, if it is heterological, then it does not describe itself, so it is not heterological. Either answer leads us to an impossibility!

The situation is just the same in Russell’s paradox. Let S be, as Russell said, the set of all those sets which are not members of themselves. We agree that most normal sets belong to S , though there may be a few contrived ones (like the words we invented above) which give trouble. But now ask Russell’s question. Is S a member of itself?

- If S is a member of itself, then (by the definition of S) it is a set which is not a member of itself.
- If S is not a member of itself, then it does not qualify for membership in S because it is a member of itself.

Again, we have an absurdity in either case.

We are forced to the conclusion that S is an impossibility; there should be no such set. That is, although we can write down the equation

$$S = \{x : x \notin x\},$$

it does not define a set.

I am not going to describe here how mathematicians rebuilt the foundations of set theory to avoid Russell’s paradox. This has been done, to most people’s satisfaction.

But there is general agreement that not everything that looks like a set is one, and we have to be careful. There is no problem with writing, as I did in the notes,

$$S = \{x \in A : P(x)\},$$

for some property P ; but we cannot just write $S = \{x : P(x)\}$.

Another rule is that a set can be specified if we have a definite rule for which elements belong to it and which don’t; but if not, there may be some trouble. Bertrand Russell (again) gave the following example.

Suppose that you have a drawer containing infinitely many pairs of shoes. Can you form a set of shoes containing one from each pair? Yes, very easily: just choose the right shoe from each pair. But could you do the same with infinitely many pairs of socks? There is no obvious rule to help you make your choice, so this is more problematic.

If you want to learn more about Bertrand Russell, I strongly recommend the book *Logicomix*, by Apostolos Doxiadis and Christos Papadimitriou.

2.7 Supplement: Is 0 a natural number?

People have been counting for tens of thousands of years. Archaeologists have found bones with scratches on them, almost certainly used as tallies, dating back around 40000 years. With the invention of agriculture about 10000 years ago, counting became even more important, since the farmer (and also the king's tax-man) needed to know how many sheep he had in his flock.

But for this purpose, the counting numbers begin at 1; a prehistoric farmer or taxman would look at you strangely if you tried to say that a flock with no sheep needed to be recorded in some way. They would almost certainly take the view that there was no flock, and so nothing needed to be recorded!

The number zero was probably invented in India about 1500 years ago, and reached the West by way of the Islamic civilisation.

So, from a historical point of view, 0 is not a natural number.

In much more recent times, logicians have tackled the problem of constructing the natural numbers from scratch. It would be very convenient if we had “standard” sets containing each possible number of elements; we could use these in much the same way as a standard ruler is used for measuring length. If we can match our flock of sheep with the standard 37-element set, then we have 37 sheep.

The logicians start with what they have to hand, which is the empty set. This is the standard (indeed the only) set containing no elements. Then we can use this set to build the set $\{\emptyset\}$ containing one element, and then the set $\{\emptyset, \{\emptyset\}\}$ containing two elements and so on.

The process seems very tedious. But if $0 = \emptyset$, then we see that $1 = \{0\}$, $2 = \{0, 1\}$, and so on: *each natural number is the set of all smaller natural numbers*. This procedure works only if the natural numbers start at 0.

However, it lends itself to some confusion. A famous story concerns the logician Alfred Tarski who, going on holiday once, worried that he had forgotten one of his suitcases. So he looked at his luggage and said,

“Zero – one – two – three – four; good, I have all five of my suitcases.”

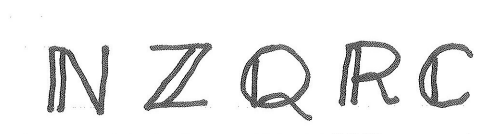
2.8 Supplement: Blackboard bold

We used the symbols $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ for the systems of natural numbers, integers, rational numbers, real numbers, and complex numbers.

Originally, these number systems were shown in mathematical writing by bold-face letters **N**, **Z**, and so on. But mathematicians writing on blackboards with chalk cannot easily draw bold-face letters, and you probably can't easily draw them in your lecture notes either. So they developed the habit of writing them in a rather stylised way on the board or on paper.

Eventually, printers caught up, and developed a font of letters to enable what was written on the blackboard to be printed out (or, now, shown on a computer screen). These letters are called *blackboard bold*.

However, now there is a problem in the other direction. It is not clear how you should best draw these letters if you are writing them down yourself. I have written out by hand the letters used for the number systems and scanned them in below. They are a bit less fussy than the printed versions.



These letters are mostly used for number systems, but you might meet them in other modules with different uses. For example, in Probability, the letter $\mathbb{P}$ is used to denote probability, so that $\mathbb{P}(A)$ is the probability that the event A occurs.

Many mathematicians (myself included) prefer a blackboard to any of the alternatives such as whiteboard, overhead projector, or data projector, for talking about mathematics. In the **Isaac Newton Institute** in Cambridge (where Andrew Wiles announced his proof of Fermat's Last Theorem in 1993), there are blackboards everywhere, including the lift and the toilets!

2.9 Supplement: John Venn

The inventor of the Venn diagram was John Venn, a Fellow of Gonville and Caius College in Cambridge (an institution of which I am a member). He was an expert on logic and probability, and introduced the diagrams in a book on *Symbolic Logic* in 1881.

Venn also had an inventive streak; among other things he made a machine for bowling cricket balls.

The College commemorates many of its famous Fellows by stained-glass windows in the dining hall. Among the people commemorated are Francis Crick (who, with James Watson, discovered the structure of DNA), the statistician R. A. Fisher, James Chadwick (who discovered the neutron), and mathematician George Green.

Here is a photo of the Venn window, taken by Dr Basia Bogacka of our department.



2.10 Supplement: More about binomial coefficients

A binomial coefficient counts something (the number of subsets of cardinality k of a set of cardinality n), and we also have a couple of formulae for it. So, not surprisingly, when we come to prove things about binomial coefficients, there are several ways we can go about it. Here are a couple of examples.

Proposition 2.20 *For any natural numbers n and k with $0 \leq k \leq n$, we have*

$$\binom{n}{k} = \binom{n}{n-k}.$$

First proof By definition, $\binom{n}{k}$ counts the k -element subsets of an n -element set, say A , while $\binom{n}{n-k}$ counts the sets of cardinality $n-k$. If we can pair up the k -element sets with the $(n-k)$ -element sets, we will know that there are the same numbers of each.

The pairing is easily done. If B is any subset of A of cardinality k , then its complement $A \setminus B$ is a subset of cardinality $n-k$; and *vice versa*.

Second proof

$$\binom{n}{n-k} = \frac{n!}{(n-k)!(n-(n-k))!} = \frac{n!}{(n-k)!k!} = \binom{n}{k}.$$

Proposition 2.21 *If $0 < k < n$, then*

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

First proof You are a teacher, and have a class of n children. You have to pick a team of k children for the school Sports Day. There are obviously $\binom{n}{k}$ possible teams that you might pick.

Now consider one particular child, A. How many teams contain A? If we include A in the team, we have to choose the remaining $k - 1$ team members from the remaining $n - 1$ members of the class, which can be done in $\binom{n-1}{k-1}$ ways.

How many teams don't contain A? This time we have to pick the entire team of k , and must pick them from the $n - 1$ pupils different from A. So there are $\binom{n-1}{k}$ ways to make the choice.

We have accounted for everything, so the equation in the proposition is proved.

Second proof

$$\begin{aligned}
 \binom{n-1}{k-1} + \binom{n-1}{k} &= \frac{(n-1)!}{(k-1)!(n-k)!} + \frac{(n-1)!}{k!(n-k-1)!} \\
 &= \frac{k \times (n-1)!}{k!(n-k)!} + \frac{(n-k) \times (n-1)!}{k!(n-k)!} \\
 &= \frac{n \times (n-1)!}{k!(n-k)!} \\
 &= \binom{n}{k}.
 \end{aligned}$$

The chain of equalities needs some explanation!

In the first line we have simply used the formula for the binomial coefficient in terms of factorials.

In the second line, we used the fact that $(k-1)! = k!/k$ and $(n-k-1)! = (n-k)!/(n-k)$. The aim in this line is to make the two fractions have a common denominator. Think of the factorial as the product of many integers; we see that most of the factors in the denominator are already common to both, but we have an extra factor $(n-k)$ in the first denominator and k in the second.

In the third line we have brought the two fractions over the common denominator $k!(n-k)!$, and noticed that the numerators have a common factor of $(n-1)!$, while what is left is $k + (n-k) = n$.

In the last line, we used the fact that $n \times (n-1)! = n!$, and then recognised the formula for $\binom{n}{k}$.

Exercises

2.1 Let A be the set of odd positive integers, and B the set of integers greater than 10. For each of the following sets, say whether it is finite or infinite; if finite, list all its elements, and if infinite, list the first ten elements.

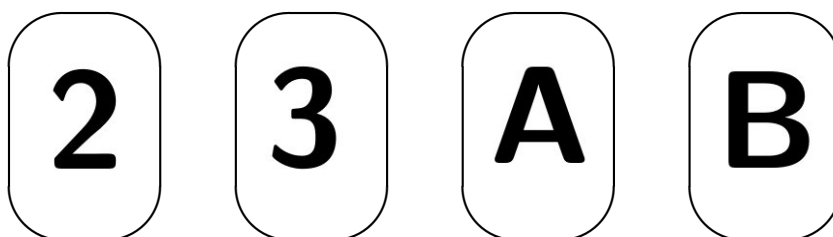
$$(a) A \cup B; \quad (b) A \cap B; \quad (c) A \setminus B; \quad (d) B \setminus A.$$

2.2 Let

$$\begin{aligned} A &= \{n \in \mathbb{N} : \text{the last digit of } 2^n \text{ is } 8\}, \\ B &= \{n \in \mathbb{N} : n + 1 \text{ is divisible by } 4\}. \end{aligned}$$

Show that $A = B$.

2.3 Each card in a pack has a number on one side and a letter on the other. Four cards are placed on the table:



You have to test the following hypothesis:

A card which has an even number on one side has a vowel on the other.

You are allowed to turn over two cards. Which cards should you turn? Why?

2.4 (Puzzle) A block of chocolate has 20 squares, arranged in five rows of four. How many breaks do you have to make to separate all 20 squares?

Can you generalise this to a block with m rows of n squares?

2.5 (Project) Let $\mathcal{P}(A)$ be the power set of A . We are going to define two operations on $\mathcal{P}(A)$, which we will call *addition* and *multiplication*; though they look very different from the usual operations on numbers, we will see that they share many of the properties of these. For this question, arguments with Venn diagrams are acceptable if they are clearly explained.

The definitions are simple:

- $A + B = A \triangle B$, the symmetric difference of A and B ;
- $A \cdot B = A \cap B$, the intersection of A and B .

Prove that

$$(a) \quad A + (B + C) = (A + B) + C \quad \text{and} \quad A \cdot (B \cdot C) = (A \cdot B) \cdot C;$$

- (b) $A + B = B + A$ and $A \cdot B = B \cdot A$;
- (c) $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$;
- (d) there is an element O such that $A + O = A$ for any A [what is O ?];
- (e) if O is as in the previous point, then for any A there exists B such that $A + B = O$ [what is B ?];
- (f) there is an element I such that $A \cdot I = A$ for any A .

Chapter 3

Infinity

In this section we look at infinite sets, and see:

- (a) a subset of an infinite set may have the same number of elements as the whole set;
- (b) nevertheless, there are different sizes of infinite set (in particular, a set may be countable or uncountable).

Under study skills, we see that writing in sentences is more understandable than writing chains of formulae connected by equality or implication signs (two signs particularly liable to misuse).

The supplementary material discusses Galileo, who first realised point (a) above.

3.1 Infinite sets

Definition 3.1 A set is *infinite* if it is not finite.

This means that, if A is an infinite set, and we choose elements $a_1, a_2, \dots, a_n \in A$, then we have not used up all of A (or else we would have matched A up with the set $\{1, 2, \dots, n\}$, and it would be finite); so $A \setminus \{a_1, a_2, \dots, a_n\} \neq \emptyset$ for any elements $a_1, a_2, \dots, a_n \in A$.

The prototype of an infinite set is the set $\mathbb{N}$ of all natural numbers. However far you count, there is always a next number. We saw in Chapter 1 that the set of prime numbers is also infinite; however many primes you discover, there is always another.

3.2 Countable sets

Counting is how we discover how large a set is. We saw in the last chapter that counting a finite set A is equivalent to matching A with the set $\{1, 2, \dots, n\}$ for some natural number n ; if this can be done, then the cardinality of A is n (that is, A contains n elements).

Definition 3.2 A set A is *countably infinite* (or just *countable*) if its elements can be matched up with the natural numbers.

In other words, A is countable if we can label its elements $a_1, a_2, \dots$, without ever coming to an end, so that every element of A is labelled a_n for some natural number n , and every natural number labels an element of A . This amounts to the same thing as writing the elements of A in an infinite sequence:

$$A = \{a_1, a_2, \dots\}.$$

For example, the set P of prime numbers is countably infinite. For we can label the primes in order of size: $p_1 = 2$ (the smallest prime), $p_2 = 3$ (the first prime after p_1), $p_3 = 5$, and so on.

This simple idea allows us to show:

Proposition 3.3 *Every subset of $\mathbb{N}$ is either finite or countably infinite.*

Proof Let A be a subset of $\mathbb{N}$. Proceeding as with the primes, we let a_1 be the smallest element of A , a_2 the next element of A after a_1 , and so on. Either we run out of elements of A at some point (in which case $A = \{a_1, a_2, \dots, a_n\}$, and A is finite, with cardinality n), or we don't (in which case the process goes on for ever, and we match A with all the natural numbers, and it is countably infinite). $\square$

Actually we have shown something quite remarkable here. Take an infinite subset of A ; for argument's sake let us suppose we take the set of all even natural numbers. This set is obviously smaller than $\mathbb{N}$, since it contains only half of all the natural numbers; but on the other hand it is just as big as $\mathbb{N}$, since it can be counted by matching it with all the natural numbers:

$$\begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & \dots \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \\ 2 & 4 & 6 & 8 & 10 & \dots \end{array}$$

So, when we get to infinite sets, the rule that “the whole is greater than the part” no longer applies: part of an infinite set can be as large as the whole set. This paradox was discovered by Galileo, two centuries before Cantor (see the Supplementary Material).

By the same token, sets which are apparently much larger than $\mathbb{N}$ are also countable:

Theorem 3.4 (a) *The set $\mathbb{Z}$ of integers is countable.*

(b) *The set $\mathbb{Q}$ of rational numbers is countable.*

Proof (a) We can write the integers as an infinite sequence as follows:

$$\mathbb{Z} = \{0, 1, -1, 2, -2, \dots\}.$$

So the matching between $\mathbb{N}$ and $\mathbb{Z}$ looks as follows:

$$\begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & \dots \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \\ 0 & 1 & -1 & 2 & -2 & \dots \end{array}$$

We can write a formula for the n th integer z_n according to this correspondence: check that

$$z_n = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even,} \\ -\frac{n-1}{2} & \text{if } n \text{ is odd.} \end{cases}$$

(b) This is a bit trickier. First we write the rational numbers out in a square table (with some repetitions). Write the integers along the top of the table (according to the scheme we just saw): $0, 1, -1, 2, -2, \dots$. Then write the natural numbers in their usual order down the side: $1, 2, 3, 4, 5, \dots$. Now in the row indexed by natural number n and column indexed by integer m , we write m/n . The table starts like this:

$$\begin{array}{cccccc} 0/1 & 1/1 & -1/1 & 2/1 & -2/1 & \dots \\ 0/2 & 1/2 & -1/2 & 2/2 & -2/2 & \dots \\ 0/3 & 1/3 & -1/3 & 2/3 & -2/3 & \dots \\ 0/4 & 1/4 & -1/4 & 2/4 & -2/4 & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \end{array}$$

Every fraction m/n occurs in the table. Of course, the same number occurs many times: $2/1 = 4/2 = 6/3 = \dots$, for example.

Now we turn the square table into a sequence by following successively the north-east to south-west diagonals:

$$\begin{array}{cccccc} 0/1 & & 1/1 & & -1/1 & & 2/1 & & \dots \\ & \swarrow & & \swarrow & & \swarrow & & \swarrow & \\ 0/2 & & 1/2 & & -1/2 & & 2/2 & & \dots \\ & \swarrow & & \swarrow & & \swarrow & & \swarrow & \\ 0/3 & & 1/3 & & -1/3 & & 2/3 & & \dots \\ & \swarrow & & \swarrow & & \swarrow & & \swarrow & \\ \dots & & \dots & & \dots & & \dots & & \dots \end{array}$$

We end up with the sequence

$$0/1, 1/1, 0/2, -1/1, 1/2, 0/3, 2/1, -1/2, 1/3, 0/4, -2/1, 2/2, -1/3, 1/4, 0/5, \dots$$

Now go through this sequence omitting any rational number which has occurred before. We end up with a sequence containing every rational number precisely once:

$$0, 1, -1, 1/2, 2, -1/2, 1/3, -2, -1/3, 1/4, \dots$$

So the set of rational numbers is countable. $\square$

3.3 Uncountable sets

The natural numbers form the prototypical infinite set. You may be wondering whether perhaps every infinite set is countable. In fact, as Cantor discovered, there are sets which are larger than countable, for example, the real numbers.

I will anticipate a construction we are going to see in more detail later on: any real number can be represented by an infinite decimal. The representation is unique, except for a small amount of ambiguity for numbers which end with all 9s or 0s, for example,

$$1.247999\dots = 1.248000\dots$$

The ambiguity won't be a problem here; we simply choose to use the form which ends with all 0s rather than all 9s.

Theorem 3.5 *The set $\mathbb{R}$ of real numbers is uncountable.*

Proof First, what does the statement mean, and what do we have to prove? It means that there is no way of arranging the real numbers in a sequence so they can be matched up with the natural numbers. So we have to show that, if someone claims to have a sequence containing all the real numbers, we can point to at least one real number that he forgot. This is a very famous argument known as *Cantor's Diagonal Argument*. There is an account of it, in a slightly different setting, [here](#) on Theorem of the Day.

Let us suppose we have a sequence of real numbers, say

$$\begin{aligned} x_1 &= 0.8800427\dots \\ x_2 &= 7.0883949\dots \\ x_3 &= 3.5603216\dots \\ x_4 &= 2.6554699\dots \\ x_5 &= 1.0689319\dots \\ x_6 &= 9.4892587\dots \\ &\dots \end{aligned}$$

We have to show that this sequence doesn't contain every real number. So we construct an infinite decimal, and show that the real number y which it represents is not in the list.

We do this as follows. Look at the first digit (after the decimal point) of x_1 , the second digit of x_2 , the third digit of x_3 , and so on:

$$\begin{aligned} x_1 &= 0.\mathbf{8}800427\dots \\ x_2 &= 7.0\mathbf{8}883949\dots \\ x_3 &= 3.56\mathbf{0}3216\dots \\ x_4 &= 2.655\mathbf{4}699\dots \\ x_5 &= 1.0689\mathbf{3}19\dots \\ x_6 &= 9.48925\mathbf{8}7\dots \\ &\dots \end{aligned}$$

Write down the number having these as its decimal digits:

$$0.880438\dots$$

Now we change the digits according to some rule. For example, if the digit is equal to 8, then change it to 5; if it is not 8, then change it to 8. The precise rule doesn't matter, as long as every digit changes and we don't change a digit to 9 (this might end up with a number ending with all 9s, which we didn't allow). Call this number y :

$$y = 0.558885\dots$$

We claim that y is not in the list, and prove our claim by contradiction. Suppose that y is in the list; say $y = x_n$. But this is impossible, since by construction the n th digit of y is different from the n th digit of x_n . So our claim is proved, and with it, the theorem. $\square$

So there are more real numbers than rational numbers. This seems a bit paradoxical, since the rationals pack densely along the number line: between any two rationals there is another rational. But the reals are, in a sense, much denser.

Also, the set of all real numbers can be matched up with a finite interval. There is a simple trick to do this. The function $y = \arctan x$ maps every real number to a number in the interval $(-\pi/2, \pi/2)$, and we can go back by the inverse function $x = \tan y$.

You might think that the number of points in a square would be larger again than the number of points on the line. But this is not so:

Theorem 3.6 *The cardinality of the set of points in the unit square is the same as that of the set of points in the unit interval $[0, 1]$ on the line.*

Proof This is not a completely accurate proof of the theorem, since there is a small problem caused by numbers ending with 9s or 0s. But the basic idea is OK.

We have to match up the points in the square with the points on the line. A point in the square is represented by two coordinates (x, y) . We interleave the digits in their decimal expansions to get the expansion of a point in the unit interval.

For example, if

$$(x, y) = (0.3549697764 \dots, 0.8082031410 \dots),$$

then

$$z = 0.38504892609371746140 \dots$$

□

In a similar way, the number of points in 3-dimensional space is the same as the number of points on the real line, or on the unit interval.

3.4 Larger and larger sets

Is it true, then, that every infinite set either is countable or can be matched up with the real numbers? Cantor discovered that this is not so. Whatever set we take, its power set (the set of all subsets of A) is larger:

Proposition 3.7 *For any set A , there is no matching between A and its power set $\mathcal{P}(A)$.*

Proof Suppose that we had such a matching. Then for every element $a \in A$, there is a subset S_a of A matched to it, so that every subset of A occurs as S_a for some a . We have to show that this is impossible, by producing a set which is missed out.

We follow Bertrand Russell's idea (see the supplementary material for Chapter 2). Let B consist of all those elements a of A which do not lie in the subset they label:

$$B = \{a \in A : a \notin S_a\}.$$

Now, if we have a complete matching, then the subset B must occur in the matching, that is, $B = S_b$ for some element $b \in B$. But this is impossible. For, if $b \in S_b$, then by definition $b \notin B$, while if $b \notin S_b$, then $b \in B$; so the sets B and S_b differ at the element b (one contains it, the other doesn't), and cannot be equal. So the proposed matching is not complete. □

See another version of this proof [here](#) on Theorem of the Day.

Cantor also showed that $\mathbb{R}$ can be matched with $\mathcal{P}(\mathbb{N})$, giving another proof that $\mathbb{R}$ cannot be matched with $\mathbb{N}$.

Cantor then began to wonder whether there is any set intermediate in cardinality between $\mathbb{N}$ and $\mathbb{R}$, but was unable to decide this question. The question became known as the *Continuum Hypothesis*, since “continuum” is an old word for the set of real numbers.

This problem resisted all attempts at solution until finally Paul Cohen showed that it is “independent”, that is, it cannot be decided on the basis of the usual assumptions of mathematics: we are free to assume it is true or false as we choose, without running into any inconsistency.

But these are deep waters into which we cannot venture here.

For another look at the ideas we have discussed in this chapter, I recommend to you the short video [How Big is Infinity?](#)

3.5 Study skills: Sentences or symbols?

If you look at proofs in the lecture notes, you will see that they mostly consist of words. You may be more used to producing mathematics which just consists of a page of symbols with a few $=$ signs or $\Rightarrow$ signs linking them up. Indeed, you may think, “I decided to study mathematics because I have had enough of writing essays.”

If you think like that, I want to change your mind.

Remember that a proof has to be convincing. A person with reasonable intelligence and patience should be able to read the argument and be persuaded that the conclusion is correct. This is unlikely to happen if you write just a page of symbols.

The other problem is that, even if your chains of $=$ or $\Rightarrow$ signs are correct, you may need to explain why they are correct, and how each line follows from the one before.

And finally, when you have your degree and are working at a job, your boss is unlikely to be impressed by a page of symbols.

I will take here a proof in the notes, and give the same proof again in a much more condensed style. You have to decide which style is easier to understand.

This is the first part of Proposition 2.9 on page 6 of Notes 2.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

The proof I gave there ran as follows:

Before we begin, remember the important remark in the previous section. We are trying to prove that two sets are equal, so we have to

show that every element of the first belongs also to the second, and also the other way round.

(a) An element belongs to $A \cup (B \cap C)$ if it belongs either to A or to both B and C . If it is in A , then it is in both $A \cup B$ and $A \cup C$, and so it is in their intersection. On the other hand, if it is in both B and C , then it is in both $A \cup B$ and $A \cup C$, and so in their intersection. In either case, it is in $(A \cup B) \cap (A \cup C)$.

The other way round, suppose that X belongs to $(A \cup B) \cap (A \cup C)$. Then x is in both $A \cup B$ and $A \cup C$. If $x \in A$, then certainly $x \in A \cup (B \cap C)$. On the other hand, if $x \notin A$, then we must have $x \in B$ and $x \in C$; so $x \in B \cap C$, and then $x \in A \cup (B \cap C)$. So the assertion holds in either case.

Now here is the same proof in telegraph style.

$$\begin{aligned}
 x \in A \cup (B \cap C) &\Leftrightarrow x \in A \text{ or } x \in (B \cap C) \\
 &\Leftrightarrow x \in A \text{ or } (x \in B \text{ and } x \in C) \\
 &\Leftrightarrow (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C) \\
 &\Leftrightarrow (x \in A \cup B) \text{ and } (x \in A \cup C) \\
 &\Leftrightarrow x \in (A \cup B) \cap (A \cup C).
 \end{aligned}$$

It is shorter than the other, but raises a few problems:

- There are no pointers to where the argument is going.
- The symbol $\Leftrightarrow$ has a technical meaning. It means that the statement on either side of it follows from the statement on the other side. If you write out an argument in this way, it is very tempting just to check that the statements follow left-to-right, without going back to check whether they really do follow right-to-left.
- The main thing, really, is that at several points in this argument you probably found yourself wondering, “Why?”

It is time for you to go and read Kevin Houston’s second technique for thinking like a mathematician: “Write in whole sentences”.

The symbols = and $\Leftrightarrow$

The symbols = and $\Leftrightarrow$ are a particular trap. It is very important to use them correctly. It is all too common for people to use one or other of these just to mean something vague like “and then ...”.

Consider the following argument:

$$\begin{aligned}x^2 - 6x + 8 = 0 &= (x - 2)(x - 4) = 0 \\&= x = 2 \text{ or } x = 4.\end{aligned}$$

The two equals signs lining up in the middle should not be there. The first is correct but misleading: $x^2 - 6x + 8$ is equal to $(x - 2)(x - 4)$, whether or not it is equal to zero. The second is wrong: whoever wrote this has actually said

$$\dots 0 = x = 2 \dots$$

which is certainly not what they meant.

Think about the following evaluation of $(2 + 3) \times 6$:

$$2 + 3 = 5 \times 6 = 30.$$

This may be an attempt to write down what your calculator tells you, but it is not a correct use of the $=$ sign.

The main reason people write like this is not thinking about what is being said. Read what you write and check whether it makes sense! You also should think about whether some explanation is needed. We have seen examples of long strings of formulae connected by $=$ signs, where the argument is correct but the reason for each $=$ sign needs to be explained.

Remember that $=$ means that the two *formulae* on either side of it have the same value, while $\Rightarrow$ and $\Leftrightarrow$ are logical relation between the two *statements* on either side. Thus, both of the following are incorrect:

$$\begin{aligned}x - 2 = 0 &= x = 2 \\2 + 3 &\Rightarrow 5 \times 6.\end{aligned}$$

3.6 Supplement: Galileo Galilei

Galileo (1564–1642) was a famous Italian scientist of the seventeenth century. He wrote on many branches of science and mathematics. He is best known as the champion of the view put forward by Copernicus, that the earth orbits the sun, rather than the sun going round the earth, as had been believed in Europe in the Middle Ages.

Galileo is important to us because, in his book *Dialogue on Two New Sciences*, he explained very clearly the idea that an infinite set can have the same cardinality as (that is, can be matched up with) a subset of itself. His example was the natural

numbers and the perfect squares:

1	2	3	4	5	...
↓	↓	↓	↓	↓	
1	4	9	16	25	...

You can read Galileo's statement of this paradox in this [Wikipedia article](#).

Galileo did not invent the telescope (as sometimes claimed), but he did improve it considerably, and used it to make important observations of the solar system: he discovered the phases of Venus, the four largest moons of Jupiter, and the rings of Saturn.

Like all our heroes, Galileo had feet of clay. He found himself in trouble with the Catholic Church because, while he was permitted to say that the heliocentric universe could be considered as a hypothesis, he insisted that it was true, even after having been warned not to do so. Also, sometimes he was completely wrong (he developed a theory of the tides which predicted a single high tide at midday every day). Also, he worked out the laws of uniformly accelerated motion: in fact this had been done a couple of hundred years earlier by the French mathematician Nicole Oresme. And finally, Galileo continued to insist that the planets moved in circular orbits, even though Kepler had shown by painstaking observation and calculation that the orbits were in fact ellipses.

Galileo died in the year that Newton was born.

3.7 Supplement: Hilbert's hotel

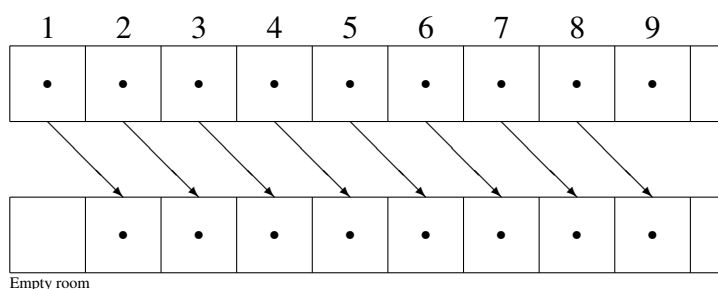
The mathematician David Hilbert was a strong supporter of Cantor's set theory. He devised *Hilbert's hotel* to explain how an infinite set can be of the same cardinality as a proper subset of itself.

The picture, by [Neill Cameron](#), shows a corridor in Hilbert's hotel. The guests arriving are the Rev Thomas Kirkman and some of his schoolgirls; you will learn their story if you study Combinatorics.

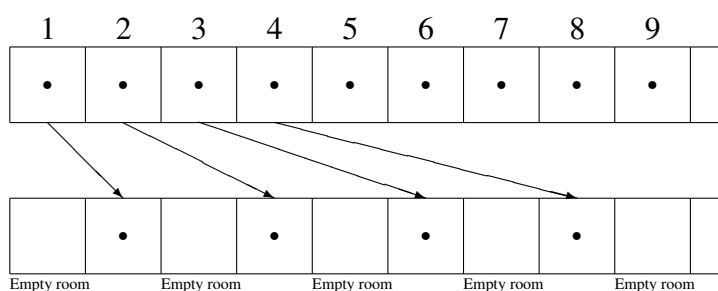


Imagine an infinite hotel, with rooms numbered 1, 2, 3, One night, when every room is occupied, a new guest arrives. In a finite hotel, you would have to

send the guest away. But in an infinite hotel, the manager simply moves the guest from room 1 into room 2, the guest from room 2 into room 3, and so on. This works because the hotel is infinite. Room 1 is now free, so the new guest can be given that room.



Now suppose a very large coach pulls up, with infinitely many more new guests wanting to stay for the night. A problem? No; the resourceful manager moves the guest from room 1 into room 2, room 2 into room 4, room 3 into room 6, and so on; in general, the guest in room n moves into room $2n$. Then all the even-numbered rooms are occupied, and the new guests can be put into the odd-numbered rooms.



Exercise Suppose that the hotel is empty, and infinitely many coaches pull up in the car park, each with infinitely many passengers needing rooms for the night. What does the manager do?

Here is a picture of me in Hilbert's hotel. It is a big hotel; you can see how the mass of the hotel has caused the structure of space-time to become curved!



Exercises

3.1 In each of the following cases, is the set in question finite or infinite? If it is infinite, is it countable or uncountable?

- (a) The set of different essays that can be stored on a 320GB hard disk.
- (b) The set of natural numbers which are perfect cubes.
- (c) The set of continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$.

3.2 A biologist has to write down all sequences of length 4 consisting of the letters A and B to design an experiment. She finds AAAB, BABB, AABA, BBBB, AABB, BAAB, BAAA, BBBA, ABBA, ABBB, ABAB, ABAA, BBAB, AAAA.

- (a) What would you say to the biologist?
- (b) Suggest a systematic way of writing down all such sequences.

3.3 The library of Babel¹ consists of interconnecting hexagonal rooms. Each room contains twenty shelves, with thirty-five books of uniform format on each shelf. A book has four hundred and ten pages, with forty lines to a page, and eighty characters on a line, taken from an alphabet of twenty-five orthographical symbols (twenty-two letters, comma, period and space). Assuming that one copy of every possible book is kept in the library, how many rooms are there?

3.4 (Puzzle) Four ants, A, B, C, D, are at the corners of a square of side length 1. At time $t = 0$, they begin to crawl at the same speed towards each other so that A always crawls towards B, B crawls towards C, C crawls towards D, and D crawls towards A. Eventually they meet in the middle of the square. How far have they crawled?

¹Jorge Luis Borges, *Labyrinths* (1964).

3.5 (Project)

- (a) Show that the union of two countably infinite sets is countably infinite.
- (b) Show that the union of a countably infinite number of countably infinite sets is countably infinite.

Chapter 4

Functions and relations

If sets are the static part of mathematics, containing the objects we reason about, then functions are the dynamic part, moving from one set to another. This chapter will explain what functions are and some extra properties they can have. The explanation, in terms of “black boxes”, doesn’t look very mathematical, but gives a clear idea what is going on.

4.1 Functions

Until a couple of hundred years ago, mathematicians thought of a function as something expressed by a formula, such as $x^3 - 7x + 6$. Such functions can be added and multiplied, almost as if they are numbers, and can also be differentiated and integrated.

The study of ways of representing functions such as Fourier series threw up some oddities, and at first it was not clear whether they should be regarded as functions at all. We have already met examples of functions which have a “split” definition, where the value is given by one expression in some places and by a different expression in others. What about this one?

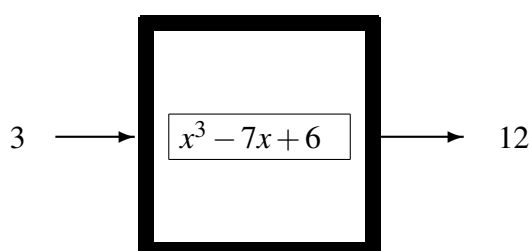
$$f(x) = \begin{cases} x & \text{if } x \text{ is rational,} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

There is a problem with this function. What is $f(x)$ when $x = 3.14$? If you trust me, you will say $3.14 = 314/100$ is rational, and so $f(x) = 3.14$. But you may think that when I wrote 3.14 I really meant π but was too lazy to give you the full decimal expansion, in which case $f(x) = 0$. Moreover, there is no question of differentiating this function, since it jumps up and down between 0 and x on every scale, no matter how small.

Eventually it was decided to remove all conditions about addition and differentiation, and give the following definition:

Definition 4.1 Let A and B be sets. A *function* from A to B is a black box such that, when an element $a \in A$ is put in, an element $b \in B$ comes out. If the name of the function is f , we write $f : A \rightarrow B$, and write the output given by the input a as $f(a)$.

In other words, we don't think of a function as an expression, but simply a way of transforming elements of A into elements of B . For example, the function $x^3 - 7x + 6$ earlier is a black box such that, when we insert the number 3, the output is the number 12. We put a label on the box giving the name of the function.



Remember, $f : A \rightarrow B$ means: elements of A in; elements of B out; and f is the name of the function. If $f(a) = b$, we say that f maps a to b .

What does it mean for two functions to be equal? If f and g are both functions with input set A and output set B , we say that $f = g$ if $f(a) = g(a)$ for every $a \in A$; in other words, when given the same input, they produce the same output. We don't care how the mechanism inside the black box actually works to determine the output!

Definition 4.2 Let $f : A \rightarrow B$ be a function. The set A of allowable inputs to the function is the *domain* of the function, and the set B containing the possible outputs is the *codomain* of the function.

There is a bit of vagueness here. We only require that the set B contains all possible outputs to the function; there is no guarantee that every element of B is a possible output. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$. The codomain is the set of real numbers; but only real numbers which are positive and zero are possible outputs. So we make another definition:

Definition 4.3 Let $f : A \rightarrow B$ be a function. The *range* of f is the set of all outputs from f which can occur when arbitrary elements of A are input.

So, in our example, the range of the function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$ is the set of non-negative real numbers, that is, $\{y \in \mathbb{R} : y \geq 0\}$.

Remember that the difference between the codomain and the range is:

- the codomain is, along with the domain, part of the specification of the function. If I sell you a function $F : A \rightarrow B$, I promise that if you input any element of A , the machine will output an element of B , while if you input anything not in A , the machine will jam and not work. I do not promise that every element of B actually occurs as output; but if it ever produces an output which is not in B , then you are entitled to ask for your money back!
- the range is the set of all the elements of B which occur as output from the function. It depends on the actual working of the function, and we may not know what it is without a lot more effort.

4.2 Names of functions

I said above that $x^3 - 7x + 6$ is the name of a function. This is not strictly true. This is the function which maps the number x to the number $x^3 - 7x + 6$. Equally, it maps the number y to the number $y^3 - 7y + 6$; so it could equally well be called the function $y^3 - 7y + 6$.

A better notation involves the symbol $\mapsto$, used to mean what the function does to one particular input. Thus, whatever we call this function, it has the property that $3 \mapsto 12$. (Read this “3 maps to 12”.)

Now, we could specify the function unambiguously by calling it “the function $x \mapsto x^3 - 7x + 6$ ”, meaning the function which maps the number x to the number $x^3 - 7x + 6$. But this is a bit cumbersome! Mostly we just give arbitrary functions a short name like f . We could write

$$\begin{aligned} f &: \mathbb{R} \rightarrow \mathbb{R} \\ f &: x \mapsto x^3 - 7x + 6 \end{aligned}$$

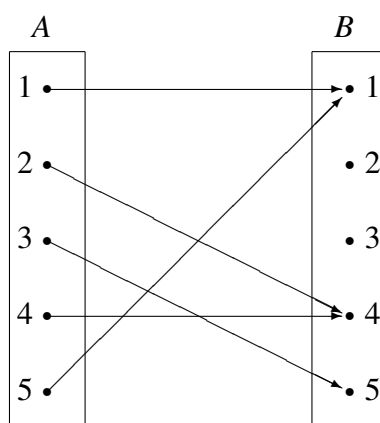
Some functions have standard names already. We can talk about “the function $\cos$ ” or “the function $\log$ ”, meaning the function $x \mapsto \cos x$ or the function $x \mapsto \log x$. But what about the function $x \mapsto x^2$? We could call it “the function squared” or “the squaring function”. Saying “the function 2 ” is not very sensible: how would you read this? Mostly we just accept some imprecision and say “the function x^2 ”.

Example Here is an example. Let $A = B = \{1, 2, 3, 4, 5\}$, and let F be the function which maps 1 to 1, 2 to 4, 3 to 5, 4 to 4, and 5 to 1. Then F is a function with domain A and codomain B (these sets happen to be equal), with $F(1) = 1$, $F(2) = 4$, and so on. Its range is $\{1, 4, 5\}$. (In this particular case, it happens that F is given by a fairly simple formula: $F(x) = 6x - x^2 - 4$.)

We can represent this function in a table. (You will recognise these tables which are used for probability mass functions in Introduction to Probability.)

x	1	2	3	4	5
$F(x)$	1	4	5	4	1

We can also represent it in a diagram showing the domain and codomain and with arrows to represent the effect of F , as follows. (Unlike our earlier pictures, this represents what the black box *does*, not the box itself.)



4.3 Injective, surjective, bijective

These are three important words which describe special types of functions.

Definition 4.4 Let $f : A \rightarrow B$ be a function with domain A and codomain B .

- (a) We say that f is *injective*, or *one-to-one*, if different inputs to f produce different outputs; that is, $a_1 \neq a_2$ implies that $f(a_1) \neq f(a_2)$. [We can write this in the equivalent form: $f(a_1) = f(a_2)$ implies that $a_1 = a_2$.]
- (b) We say that f is *surjective*, or *onto*, if every element in B occurs as an output of f ; that is, the range of f is equal to its codomain.
- (c) We say that f is *bijective* if it is both injective and surjective.

The function $f : x \mapsto x^3 - 7x + 6$ from $\mathbb{R}$ to $\mathbb{R}$ is not injective, since $1 \neq 2$ but $f(1) = f(2) = 0$. This function is surjective; this is a bit harder to prove, but if you draw the graph of the function you should be able to convince yourself that it is so.

The function $f : x \mapsto 2^x$ from $\mathbb{R}$ to $\mathbb{R}$ is injective, since if $x_1 < x_2$ then $2^{x_1} < 2^{x_2}$. It is not surjective, since it only takes positive values.

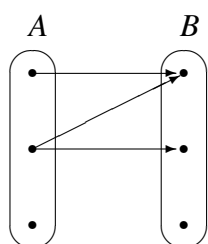
The function $f : x \mapsto x^3$ from $\mathbb{R}$ to $\mathbb{R}$ is both injective and surjective, and so is bijective.

Note that we need to know what the domain and codomain of a function are before we can decide whether it is injective or surjective.

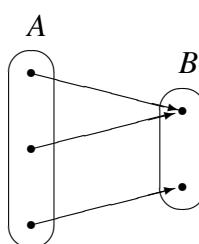
The pictures below show examples of something which is not a function, an injective function which is not surjective, and *vice versa*, and a bijective function. The first diagram fails to be a function for two reasons:

- (a) the second input gives two different outputs;
- (b) the third input gives no output at all.

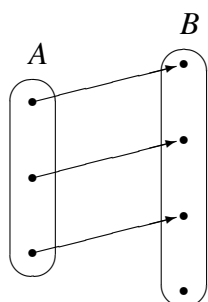
Remember that every input to a function must give just one output. So, in this kind of diagram, every point of the domain has just one arrow leaving it. But a point of the codomain might have none, one, or more than one arrow entering it.



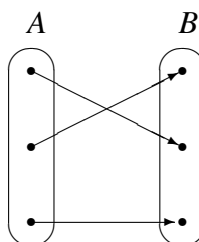
Not a function



Surjective, not injective



Injective, not surjective



Bijective

4.4 Bijective functions match up two sets

We have seen bijective functions before!

Suppose that $f : A \rightarrow B$ is a bijective function. Then to every element a of A , there is an element $f(a)$ of B ; and to every element $b \in B$, there is a unique

element a of A corresponding to it. (The fact that f is surjective guarantees that there is an element $a \in A$ with $f(a) = b$; and the fact that it is injective guarantees that there cannot be more than one such element, for if $f(a_1) = b = f(a_2)$ with $a_1 \neq a_2$, then f would fail the definition of an injective function.)

So we have:

Proposition 4.5 *Let A be a set.*

- (a) $|A| = n$ if and only if there is a bijective function $f : A \rightarrow \{1, 2, \dots, n\}$.
- (b) A is countably infinite if and only if there is a bijective function $f : A \rightarrow \mathbb{N}$.

Another important property of bijective functions is that they have inverse functions which undo their effects.

Definition 4.6 Let $f : A \rightarrow B$ and $g : B \rightarrow A$ be functions. We say that g is an *inverse* of f if the following two conditions hold:

- (a) $g(f(a)) = a$ for all $a \in A$;
- (b) $f(g(b)) = b$ for all $b \in B$.

This means that, if we connect the black boxes for f and g (in either order) so that the output of the first becomes the input to the second, then the output from the second box is the same as the input to the first.

Proposition 4.7 *Let $f : A \rightarrow B$ be a function. Then f has an inverse if and only if it is bijective. If so, then the inverse is unique.*

Proof The “if and only if” in the theorem means that we have to do the proof both ways round.

So suppose first that f has an inverse $g : B \rightarrow A$. Then

- (a) f is injective. For suppose that $f(a_1) = f(a_2)$. Then, by the definition of inverse,

$$a_1 = g(f(a_1)) = g(f(a_2)) = a_2.$$

- (b) f is surjective. For suppose we are given any element b of the codomain of f . Let $a = g(b)$. Then a is an element of the domain of f , and

$$f(a) = f(g(b)) = b.$$

Second, suppose that f is bijective. We have to construct a black box which gives a function g which is an inverse to f . The function g has domain B and codomain A . Now input an element $b \in B$ to the black box for g . We know that, since f is surjective, there is an element $a \in A$ for which $f(a) = b$; and, since f is injective, there is only one such element a . So we let a be the output of the black box; that is, $g(b) = a$. This defines a function. Since, by definition, we have $f(a) = b$, we see that $f(g(b)) = b$. Moreover, for any $a \in A$, let $f(a) = b$; then $g(b) = a$, so $g(f(a)) = a$. Thus, g is an inverse for f .

Finally, we have to show the last part of the proposition, namely, that the inverse g is unique. This means that, if g_1 and g_2 are two functions which are both inverses to f , then $g_1 = g_2$. Remember that equality of functions means that they produce the same output for any input. So take an element $b \in B$. As we said in the last paragraph, there is a unique element $a \in A$ such that $f(a) = b$; then

$$g_1(b) = g_1(f(a)) = a = g_2(f(a)) = g_2(b),$$

so the outputs of g_1 and g_2 are equal, as claimed. $\square$

The inverse function corresponds to “running the black box in reverse”.

4.5 Counting

Suppose that A and B are finite sets, with (say) $|A| = m$ and $|B| = n$. How many functions are there from A to B ? How many of these are injective, and how many are surjective?

We can represent a function on a finite set by a table giving the output for each possible input to the function. Here is a very small example: the tables representing all functions from the set $A = \{a, b\}$ to the set $B = \{c, d\}$. Remember that, to specify a function $f : A \rightarrow B$, we simply have to say what happens when the input is a , and what happens when the input is b .

$$\begin{array}{c|cc} x & a & b \\ \hline f(x) & c & c \end{array} \quad \begin{array}{c|cc} x & a & b \\ \hline f(x) & c & d \end{array} \quad \begin{array}{c|cc} x & a & b \\ \hline f(x) & d & c \end{array} \quad \begin{array}{c|cc} x & a & b \\ \hline f(x) & d & d \end{array}$$

Proposition 4.8 *Let $|A| = m$ and $|B| = n$.*

- (a) *The number of functions $f : A \rightarrow B$ is n^m .*
- (b) *The number of injective functions $f : A \rightarrow B$ is $n(n-1) \cdots (n-m+1)$.*

Proof (a) The table representing the function f has m columns after the vertical line: we write each possible input to f in the first row, and below it the corresponding output. Now there are n possibilities for the output for each input, namely any element of B ; so the number of tables is

$$n \cdot n \cdots (m \text{ factors}) = n^m.$$

(b) If the function is to be injective, then all the entries in the second row of the table must be different. So there are n possibilities for the first entry, $n - 1$ possibilities for the second entry (since it must be different from the first), and so on, up to $n - (m - 1) = n - m + 1$ possibilities for the m th entry. $\square$

Note that, if $m > n$, then one of the factors in the product for the number of injective functions is $n - n = 0$, so the number of injections is zero. This makes sense; if the set B is smaller than the set A , there cannot be an injective map from A to B . For such a map would take the m elements of A to m distinct elements of B , and there are not enough elements of B to go round.

I didn't give you a formula for the number of surjective functions. This is a bit more complicated – see the supplementary material – but you will learn about it if you study Combinatorics in your degree.

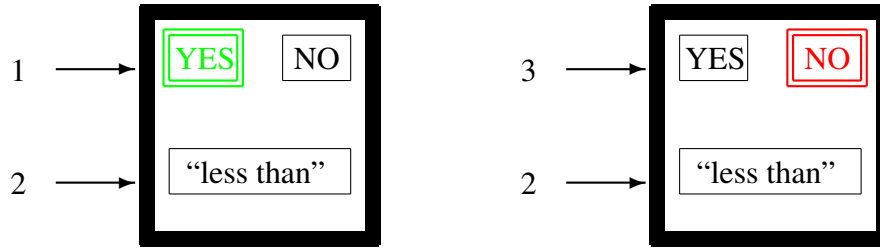
4.6 Relations

A *relation* on a set A is a different sort of black box.

Definition 4.9 Let A be a set. A *relation* on A is a black box which takes as input two elements of A (which may be the same or different, but come in a definite order). Rather than output an element, the black box responds “yes” or “no”. The set A is called the *domain* of the relation.

An example of a relation on a set is the order relation “less than” on the real numbers. If we input two real numbers a and b to the black box for this relation, it will respond “yes” if $a < b$, and “no” otherwise. We can think of the black box as having two lights, one of which illuminates to give us the answer. In the picture, I have put the first input above the second.

Informally, a relation is something which holds between some pairs of elements of the set and fails to hold between others – think of the relation “friend” between people in your class.



The black box is demonstrating that 1 is less than 2 but 3 is not less than 2. We write the relation “less than” as the symbol $<$, which we place between the two inputs. Thus, the black box is showing that $1 < 2$ is true but $3 < 2$ is false.

A general relation should have a non-committal name like R ; we write aRb to mean that the relation holds when a is the first input and b is the second.

Now we come to a very important type of relation. First, here are some properties which may hold for certain relations.

Definition 4.10 A relation R , with domain A , is said to be

- (a) *reflexive* if aRa for all $a \in A$;
- (b) *symmetric* if, whenever aRb , then also bRa ;
- (c) *transitive* if, whenever aRb and bRc , then also aRc .

For example, the relation $<$ on the real numbers is transitive (this says that if $a < b$ and $b < c$ then $a < c$) but not reflexive or symmetric (since $1 < 1$ is false, and $1 < 2$ is true but $2 < 1$ is false). The relation $\leq$ on the real numbers is reflexive and transitive but not symmetric. The relation $=$ on the real numbers is reflexive, symmetric and transitive.

Definition 4.11 An *equivalence relation* is a relation which is reflexive, symmetric and transitive.

In the next section we describe the important role that equivalence relations play in mathematics.

4.7 The Equivalence Relation Theorem

Definition 4.12 Let A be a set. A *partition* $\mathcal{P}$ of A is a collection $\{B_1, B_2, \dots\}$ of subsets of A with the properties

- (a) every set B_i in the partition is non-empty;

- (b) the intersection of any two sets in the partition is the empty set;
- (c) the union of all the sets in the partition is A .

Said otherwise, condition (c) says that every element of A is contained in at least one of the sets of $\mathcal{P}$; condition (b) says that no element can be contained in more than one of these sets. So a partition “chops up” the set A into non-empty pieces $B_1, B_2, \dots$, with no overlap.

For example, $\{\{1, 2, 6\}, \{3, 5\}, \{4\}\}$ is a partition of the set of the first six natural numbers. Three collections which are not partitions of $\{1, \dots, 6\}$ are:

- $\{\{1, 2, 6\}, \{3, 5\}, \{4\}, \emptyset\}$: one of its parts is empty;
- $\{\{1, 2, 6\}, \{3, 5\}, \{4, 5\}\}$: two parts have an element in common;
- $\{1, 6\}, \{3, 5\}, \{4\}$: the element 2 is not in any part.

Now we come to a very important theorem, the Equivalence Relation Theorem. Informally it says that the job of an equivalence relation is to chop up a set into non-empty disjoint parts, in other words, to give us a partition of its domain. Before stating the theorem, we need a preliminary definition.

Definition 4.13 Let R be a relation on a set A . For each $a \in A$, let

$$R(a) = \{b \in A : aRb\},$$

in other words, $R(a)$ is the set of all elements related to a , the set of all second inputs which, when they follow the first input a , produce the response “yes”. If R is an equivalence relation, the sets $R(a)$ are called the *equivalence classes* of R .

Theorem 4.14 (Equivalence Relation Theorem) *Let R be an equivalence relation on a domain A . Then the equivalence classes of R form a partition of A . In symbols,*

$$\{R(a) : a \in A\}$$

is a partition of A .

Proof We have to show that the collection of sets $R(a)$ satisfies the three conditions in the definition of a partition.

(a) Each set $R(a)$ is non-empty. For, by the fact that R is reflexive, we have aRa , so $a \in R(a)$ by definition.

(b) Two unequal sets $R(a)$ and $R(b)$ are disjoint. Said otherwise, we have to show that, if $R(a)$ and $R(b)$ are not disjoint, then they are equal. So suppose that $R(a)$ and $R(b)$ have an element c in common. We have to show that $R(a) = R(b)$.

Remember that equality of sets means that every element of the first lies in the second, and *vice versa*.

So suppose that $x \in R(a)$. Then aRx . Now we have aRc , since $c \in R(a)$, so cRa by symmetry; also bRc , since $c \in R(b)$. Putting the three conditions

$$bRc, cRa, aRx$$

together by applying transitivity twice, we conclude that bRx , so that $x \in R(b)$.

The proof the other way around (that if $x \in R(a)$ then $x \in R(b)$) is an exercise for you.

(c) Every element of A is in at least one of the sets $R(a)$; for, simply, the element $a \in A$ belongs to $R(a)$. $\square$

The converse of this theorem is also true: every equivalence relation comes from a partition. This is not so important here, but you can find a proof in the supplementary material. This means that, as we said above, equivalence relations and partitions do exactly the same job, and are effectively the same thing.

Consider our partition of $\{1, 2, \dots, 6\}$ above. It is the set of equivalence classes of the equivalence relation which is true for the pairs

$$(1, 1), (1, 2), (1, 6), (2, 1), (2, 2), (2, 6), (3, 3), (3, 5), (4, 4), (5, 3), (5, 5), (6, 1), (6, 2), (6, 6)$$

and is false in all other cases.

We see that an equivalence relation R can be specified in either of two ways:

- (a) list all the pairs (a, b) for which aRb ;
- (b) give the corresponding partition.

The second is likely to be more compact and useful than the first.

4.8 An example

Here is a small example of using an equivalence relation to construct a partition.

Suppose that $A = \{1, 2, 3, 4, 5\}$, and we have a relation which holds for the pairs $(1, 1), (1, 3), (2, 2), (2, 5), (3, 1), (3, 3), (4, 4), (5, 2), (5, 5)$ and fails for all other pairs. (That is, if we put 1 as the first input and 1 as the second, the box answers “yes”, while 1 as first input and 2 as the second gives the answer “no”, and so on. So $1R1$ is true but $1R2$ is false.)

You can check, somewhat laboriously, that R is an equivalence relation, that is, it is reflexive, symmetric and transitive:

- Since $1R1, 2R2, 3R3, 4R4$, and $5R5$ all hold, the relation is reflexive.

- To check it's symmetric, a typical case is: $1 R 3$, so we need to check that $3 R 1$ holds – yes, it does.
- To check it's transitive, a typical case is: $1 R 3$ and $3 R 1$, so we need to check that $1 R 1$ holds – yes, it does.

Now to construct the equivalence classes:

- Take a box and label it $R(1)$. In this box we put all elements x for which $1 R x$ holds. By inspection we see that 1 and 3 get put into the box (since $1 R 1$ and $1 R 3$ hold), but 2, 4 and 5 do not.
- Now take another box and label it $R(2)$. We find that 2 and 5 go into this box, while the other elements do not.
- Now take another box and label it $R(3)$. We find that 1 and 3 go into this box, but 2, 4 and 5 do not. But now the contents of the box $R(3)$ are exactly the same as the contents of the box $R(1)$. Since two sets with exactly the same elements are equal, we don't need box $R(3)$.
- The box labelled $R(4)$ gets to contain just 4.
- Finally, the box $R(5)$ turns out to have the same contents as the box $R(2)$, so again we don't need it.

So finally, we have three boxes, which give us the following partition P of the set $\{1, 2, 3, 4, 5\}$:

$$P = \{\{1, 3\}, \{2, 5\}, \{4\}\}.$$

Let us just stop and see why this works. We saw that the box $R(1)$ contained 1 and 3. What must $R(3)$ contain?

- $3 R 3$ holds (by the reflexive law), so 3 is in $R(3)$.
- $1 R 3$ holds (since 3 is in $R(1)$), and so $3 R 1$ holds (by the symmetric law), so 1 is in $R(3)$.
- Could we have 4 in $R(3)$? This would only happen if $3 R 4$. But we know that $1 R 3$ (since $3 \in R(1)$), so by the transitive law, we would have $1 R 4$, and 4 would be in $R(1)$, which is not so. Similarly, nothing else goes in $R(3)$ except what is in $R(1)$.

4.9 Study skills: Read the question!

When you are trying to solve a homework or exam problem, the first thing is to make sure that you know exactly what the question is asking you. This means that you should *read the question carefully*. It is not enough to skim the question looking for keywords, and jump to a guess about what answer is needed based on these.

If there is a mathematical word you don't understand, you probably have a definition of it in your notes. You should look up this definition before going any further.

In tutorials, especially large group tutorials, the tutors will not be prepared to answer your questions until you have tried to use your own resources.

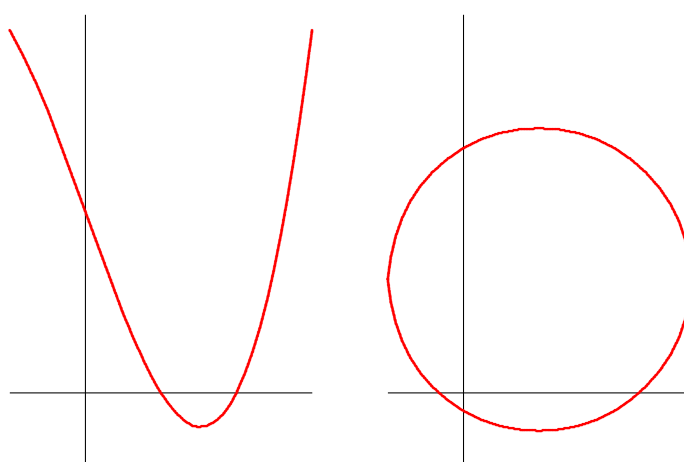
It very often happens that later parts of a question refer to earlier parts. So, if you don't understand part of a question, then trying to go on to the next part may just make your confusion worse.

On the other hand, in the exam, if you have tried and failed to solve part of a question, then it is OK to go on to the next part, assuming what you were asked to do in the previous part.

4.10 Supplement: The graph of a function

How is the notion of a function (as the output of a black box) related to what you are already familiar with, something which has a graph?

The picture on the left shows the graph of a function; the one on the right shows a curve which is not the graph of a function.



Can you see the difference?

Definition 4.15 Let f be a function whose domain and codomain are subsets of the set $\mathbb{R}$ of real numbers. The *graph* of f is the set of all points (x, y) in the plane such that $y = f(x)$.

Thus, the graph of the function must have the property that, for each number a in the domain of the function, there is a unique number b such that (a, b) is a point of the graph. In other words, the vertical line $x = a$ meets the graph in just one point (a, b) , where $b = f(a)$. The curve on the right fails to be the graph of a function because there are vertical lines which meet it in two points.

You see that the graph of the function gives us complete information about the function. A black box computing the function might work by evaluating some mathematical formula. On the other hand, it might work by simply taking the input a , finding the point (a, b) where the vertical line $x = a$ meets the graph, and measuring and outputting the value b .

4.11 Supplement: Is a black box a mathematical object?

I described a function as a black box, with the three essential features:

- the domain and range are part of the specification;
- each input produces a unique output;
- we don't have to know what goes on inside.

You might still feel that black boxes are not really mathematical objects.

There is a more formal way of describing a function. This follows on from the discussion of the graph of a function in the preceding section. The graph tells us everything about the function, and does satisfy the same two features as the black box. So we can proceed as follows:

Definition 4.16 Let A and B be sets.

- (a) Let $a \in A$ and $b \in B$. An *ordered pair* (a, b) is a gadget which has a “first element” and a “second element”, such that the first element is a and the second element is b . (Just like the coordinates of a point in the plane, where a and b are real numbers.
- (b) A *function* $F : A \rightarrow B$ is a set of ordered pairs (a, b) , where $a \in A$ and $b \in B$, with the property that, for each choice of an element $a \in A$, there is precisely one $b \in B$ such that (a, b) belongs to F . Here b is the output of the function when the input is a .

Thus, the function from $\{a, b, c\}$ to $\{x, y\}$ which maps a and c to x and b to y would be represented as the set

$$\{(a, x), (b, y), (c, x)\}$$

of ordered pairs. You see that this is almost identical to specifying the function by writing $a \mapsto x, b \mapsto y, c \mapsto x$.

It is now routine to translate the notions of “injective”, “surjective” and “bijective” into the language of ordered pairs.

The statements $b = F(a)$ and $(a, b) \in F$ mean exactly the same; but you may feel more comfortable with the first.

In the same way, we can say:

Definition 4.17 A *relation* on a set A is a set R consisting of ordered pairs (a_1, a_2) , where $a_1, a_2 \in A$.

Thus, $a_1 R a_2$ and $(a_1, a_2) \in R$ mean exactly the same.

Again we can translate concepts about relations into this language. For example, R is reflexive if $(a, a) \in R$ for every element $a \in A$.

4.12 Supplement: Functions and subsets

We saw that, if $|A| = m$ and $|B| = n$, then the number of functions from A to B is n^m . We also saw that, if $|A| = m$, then the number of subsets of A (the cardinality of the *power set* $\mathcal{P}(A)$) is 2^m . Is there any connection? Not surprisingly, there is; this gives us a way to represent a subset of A by a function.

Definition 4.18 Let X be a subset of A . The *characteristic function* of X is the function $I_X : A \rightarrow \{0, 1\}$ defined by

$$I_X(x) = \begin{cases} 1 & \text{if } x \in X, \\ 0 & \text{if } x \notin X. \end{cases}$$

So this function is a black box which, when you input an element of the set A , tells you whether the element in question belongs to the subset X or not, by outputting 1 if the element is in X and 0 otherwise. You can think of 1 and 0 as code for “YES” and “NO”.

Any subset of A can be represented by its characteristic function. On the other hand, any function $F : A \rightarrow \{0, 1\}$ is the characteristic function of some subset. For let F be such a function, and define

$$X_F = \{x \in A : F(x) = 1\},$$

the set of all inputs which produce the output 1. Then F is the characteristic function of the set X_F .

For example, if F is the constant function with value 0, it is the characteristic function of the empty set; and if it is the constant function with value 1, then it is the characteristic function of the set A .

We have matched up subsets of A (elements of $\mathcal{P}(A)$) with functions from A to $\{0, 1\}$, where a set is matched with its characteristic function. So the number of subsets is equal to the number of functions from A to the 2-element set $\{0, 1\}$, as we noticed.

Characteristic functions provide a very useful way to represent a function by a set; you are likely to meet them in analysis, probability, and various other branches of mathematics.

4.13 Supplement: Counting surjections

Here is the formula for the number of surjective functions between arbitrary finite sets.

Theorem 4.19 *Let A and B be finite sets, with $|A| = m$ and $|B| = n$. Then the number of surjective functions $f : A \rightarrow B$ is*

$$\sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^m.$$

Here is a proof in two cases, when $n = 2$ and when $n = 3$.

Case $n = 2$ We are mapping from the set A to the set $\{x, y\}$. The only ways that such a function can fail to be surjective are either that it maps everything to x , or that it maps everything to y . So of the 2^m functions from A to B , there are $2^m - 2$ surjective functions. Now

$$2^m - 2 = \binom{2}{0} 2^m - \binom{2}{1} 1^m + \binom{2}{2} 0^m,$$

so it agrees with the formula given.

Case $n = 3$ In this case, let $B = \{x, y, z\}$. Now a function fails to be surjective if at least one of $\{x, y, z\}$ is not in its range. Let Ω be the set of all functions; we know that $|\Omega| = 3^m$.

Let X be the set of functions whose range does *not* contain x , and similarly for Y and Z . A function is *not* surjective if (and only if) it lies in $X \cup Y \cup Z$. We

can calculate the cardinality of this union by using the Principle of Inclusion and Exclusion from the preceding chapter of the notes.

Now X is the set of functions mapping A to $\{y, z\}$, so $|X| = 2^m$. Similarly $|Y| = |Z| = 2^m$.

What about $X \cap Y$? This is made up of functions which don't take either of the values x or y , so map everything to z . There is only one such function, so $|X \cap Y| = 1$. Similarly, $|X \cap Z| = |Y \cap Z| = 1$.

Finally, $X \cap Y \cap Z$ consists of functions which don't take any of the three values. But this is impossible: there is no such function.

Now the Principle of Inclusion and Exclusion tells us that

$$\begin{aligned} |X \cup Y \cup Z| &= |X| + |Y| + |Z| - |X \cap Y| - |X \cap Z| - |Y \cap Z| + |X \cap Y \cap Z| \\ &= 3 \cdot 2^m - 3, \end{aligned}$$

so the number of functions which *are* surjections is obtained by subtracting this from the total number 3^m of functions, giving

$$3^m - 3 \cdot 2^m + 3 \cdot 1^m - 0^m = \sum_{k=0}^3 (-1)^k \binom{3}{k} (3-k)^m,$$

as the formula asserts.

Remark The Principle of Inclusion and Exclusion has a form which applies to any number of sets. This form can be used to prove the formula for the number of surjections between sets of any finite cardinalities.

4.14 Supplement: Infinite sets

How many functions are there between infinite sets? We won't give a complete answer here; we just prove the following.

Theorem 4.20 *Let B be a set containing more than one element. Then the number of functions from $\mathbb{N}$ to B is uncountable.*

Proof Choose two elements of B , say x and y . It is clear that there are infinitely many functions from $\mathbb{N}$ to B . We have to show that the set of such functions is not countable, that is, cannot be matched to $\mathbb{N}$ by a bijection.

Suppose that such a matching exists. This means that we can label the functions as $f_1, f_2, \dots$, where the indices run through the natural numbers. Now we use Cantor's diagonal method. Construct a function g by the rule

$$g(n) = \begin{cases} x & \text{if } f_n(n) \neq x, \\ y & \text{if } f_n(n) = x. \end{cases}$$

Now we have constructed g so that it disagrees with every function f_n : in fact, $g(n) \neq f_n(n)$. So our enumeration of the functions could not have been complete.

□

This should come as no surprise after the section where we matched up subsets with characteristic functions. The number of subsets of $\mathbb{N}$ is equal to the number of functions from $\mathbb{N}$ to $\{0, 1\}$.

4.15 Supplement: Equivalence relations and partitions

We proved in the notes that an equivalence relation gives us a partition, whose parts are the equivalence classes.

It goes the other way too. Given a partition, we can find an equivalence relation whose equivalence classes are the parts of the partition. The idea is very simple: two elements of the domain are related if they lie in the same part of the partition.

So let A be a set, and $\mathcal{P} = \{B_1, B_2, \dots\}$ a partition of A . Now construct a black box which answers the question “Are the two inputs in the same part of the partition?” In other words, given inputs a and b , it responds “yes” if there exists a part B_i such that $a, b \in B_i$, and “no” if no such part exists.

We claim that the relation defined by this black box is an equivalence relation: that is, it is reflexive, symmetric and transitive.

Reflexive: Given $a \in A$, by the definition of a partition, there is a part B_i containing a . Then both a and a belong to B_i , so if both inputs are equal to a , the black box responds “yes”.

Symmetric: Suppose that the black box responds “yes” on inputs a and b . This means that some part B_i of the partition contains a and b . But then B_i contains b and a , and so the black box responds “yes” to inputs b and a .

Transitive: Suppose that a, b, c are three elements, and the black box responds “yes” to inputs (a, b) and (b, c) . Let B_i be the part of the partition containing b . The “yes” response to (a, b) means that a also lies in B_i , and similarly the “yes” response to (b, c) means that c also lies in B_i . Then a and c are both in B_i , so the box responds “yes” to the inputs (a, c) .

So the relation is an equivalence relation, as claimed.

Exercises

4.1 Each of the following purports to define a function from the set $\{a, b, c, d\}$ to the set $\{x, y, z\}$. Which of them are functions? Of those which are functions, which are injective? which are surjective, and which are both?

(a) $a \mapsto x, b \mapsto x, c \mapsto y, d \mapsto y$.

(b) $a \mapsto x, a \mapsto y, a \mapsto z, b \mapsto y$.

(c) $a \mapsto x, b \mapsto y, c \mapsto x, d \mapsto x$.

(d) $a \mapsto z, b \mapsto y, c \mapsto x, d \mapsto y$.

4.2 Which of the following relations R on sets X are (i) reflexive, (ii) symmetric, (iii) transitive? Give proofs or counterexamples.

(a) $X = \mathbb{N}$, $a R b$ if and only if a divides b .

(b) $X = \mathbb{R}$, $a R b$ if and only if $a > b$.

(c) $X = \mathbb{C}$, $a R b$ if and only if $|a| = |b|$.

(d) $X = \mathbb{Z}$, $a R b$ if and only if $a + b$ is a multiple of 5.

(e) X is the set of countries of Europe, $a R b$ if and only if a and b have a common border.

In the preceding question, in each case where R is an equivalence relation, describe the equivalence classes of R .

4.3 Consider the following argument:

False proposition *If a relation is symmetric and transitive then it is reflexive.*

Proof Let R be a symmetric and transitive relation. Take elements x, y satisfying $x R y$. Then $y R x$ (since R is symmetric), and so $x R x$ (since R is transitive; put $z = x$ in the transitive law). So R is reflexive.

(a) Say what is wrong with this argument.

(b) Give a counterexample to the false proposition.

4.4 One of the following is not a function; which one, and why? In the other cases, is the function one-to-one? is it onto?

- (a) The function $F : \mathbb{C} \rightarrow \mathbb{R}$ given by $F(z) = \arg(z)$. (If you don't know what this function is, see Chapter 8.)
- (b) The function $F : \mathbb{R} \rightarrow \mathbb{R}$ given by $F(x) = x^3 - x$.
- (c) The function $F : \mathbb{R} \rightarrow \mathbb{R}$ given by $F(x) = e^x$.

4.5 (Puzzle) You have two fuses, each of which burns for exactly one minute. However, the fuses are not of uniform thickness, and do not burn at a uniform speed. How can you use the two fuses to measure a time of 45 seconds?

4.6 (Project) The square grid shown is to be filled with the numbers 1, 2, 3 and 4 in such a way that each of these numbers occurs once in each row and once in each column.

1	2		
		3	4
3		1	

- (a) Complete the grid, in accordance with the rule given.
- (b) Give a convincing argument to show that there is only one way to complete the grid. (As mathematicians say, “the solution is unique”.)
- (c) Let $f(n)$ be the smallest number of entries in an $n \times n$ grid which can be completed in only one way. Show that $f(2) = 1$ and $f(3) = 3$. [For $f(3) = 3$, for example, you have to give an example of a grid with three entries, and show that no grid with only two entries has a unique solution.] What can you say about larger values of n ?

Chapter 5

The natural numbers

In this chapter you will learn about the natural numbers: what they are; induction; division, divisibility, greatest common divisor. The Study skill concerns using definitions. The supplementary material contains more about induction and Euclid's algorithm; a proof of the Fundamental Theorem of Arithmetic; and Pascal's triangle.

In this part of the course, we will talk about the various number systems of mathematics: why we need them, how we build them, and how to prove things about them.

5.1 What are the natural numbers?

It is often the case that we are more interested in what an object does than in precisely what it is. In this section, I will assume that you know what the natural numbers are, and will concentrate on their properties. Recall that we are taking the set $\mathbb{N}$ of natural numbers to be $\{1, 2, 3, \dots\}$.

We usually visualise the natural numbers as points spaced out along a line, extending to infinity on the right:



The most basic property of the natural numbers is closely tied up with our use of them for counting. We can separate this into three parts, with one extra important property:

- (a) There is a first (or smallest) natural number 1.
- (b) Each natural number has a successor; so the successor of 1000000 is 1000001.
- (c) Apart from 1, every natural number has an immediate predecessor.

- (d) Any natural number can be reached by starting at 1 and “counting up”, passing from each number to its successor.

The last property is crucial and we will come back to it. When my children were small, they had a counting rhyme which went:

One, two, missed a few,
Ninety-nine, a hundred.

That presumably means that they know they could count up to 100 if they had to, but couldn't be bothered! Mathematicians do the same when they write $1, 2, \dots, 99, 100$.

Based on this idea, we can define the other properties of natural numbers:

Addition: To work out $a + b$, we start at a and count on b steps. In particular, $a + 1$ is the successor of a . So

$$a + b = ((\dots(a + 1) + 1) + \dots + 1),$$

where there are b ones in the sum.

Multiplication: Multiplication is “repeated addition”. To work out $a \times b$, we add a to a , and then add a to that, and continue until there are b occurrences of a in the sum: that is,

$$a \times b = ((\dots(a + a) + a) + \dots + a),$$

where there are b a s in the sum.

Order: a is smaller than b (written $a < b$ as usual) if we can start from a and count up to b . In other words, $a < b$ if there is a natural number d such that $a + d = b$.

As usual, we often write $a \times b$ as ab . Putting two symbols next to each other is called *juxtaposition*; so we can say in fancy language “multiplication is represented by juxtaposition”. Of course care is required; it is not true that $7 \times 8 = 78$. In cases like this, we sometimes use a dot in place of the multiplication sign: $7 \cdot 8$.

These operations and this relation have many properties. Here are a few of them.

Commutative laws: For any natural numbers a and b ,

$$a + b = b + a, \quad a \times b = b \times a.$$

Associative laws: For any natural numbers a, b, c ,

$$(a + b) + c = a + (b + c), \quad (a \times b) \times c = a \times (b \times c).$$

Distributive law: For any natural numbers a, b, c ,

$$(a + b) \times c = (a \times c) + (b \times c).$$

Cancellation laws: For any natural numbers a, b, c , if $a + c = b + c$, then $a = b$; and if $ac = bc$, then $a = b$.

Order laws: For any natural numbers a, b, c , if $a < b$ then $a + c < b + c$ and $a \times c < b \times c$.

These “laws” just codify the rules that anyone who does arithmetic knows very well. If you add a column of figures, you can add from top to bottom and from bottom to top, and get the same answer (at least, as long as you don’t make a mistake); this follows from the commutative and associative laws for addition. The distributive law allows us to “expand brackets”. So there is nothing unfamiliar here.

It is possible to write down proofs of these laws using our descriptions of the operations. Surprisingly, the commutative laws are the hardest. Here is a proof of the associative law for addition, as an example.

Proposition 5.1 *For any natural numbers a, b, c , we have $a + (b + c) = (a + b) + c$.*

Proof According to our description, $a + (b + c)$ means the number we reach if we start at a and count on $b + c$ steps. Now $b + c$ is the number we reach if we start at b and count on c steps. So counting up to $b + c$ means counting b steps and then counting on c steps. This means that starting at a and counting on $b + c$ steps, we can first count on b steps (reaching $a + b$) and then count on another c steps (reaching $(a + b) + c$). Since the result must be the same, we have $a + (b + c) = (a + b) + c$. $\square$

I am not going to go through proofs of all the laws. If you want to put the theory of the natural numbers on a really secure foundation, you really need a better definition, which can be done using set theory. I gave some hints of this in the supplementary material for the last chapter.

I will, however, give you a proof of the order laws, to show how they follow from the other laws.

Proposition 5.2 *For any natural numbers a, b, c , if $a < b$, then $a + c < b + c$ and $ac < bc$.*

Proof Remember that $a < b$ means that there exists a natural number d such that $a + d = b$. Then

$$b + c = (a + d) + c = a + (d + c) = a + (c + d) = (a + c) + d,$$

where we used the associative law twice and the commutative law once; so $a + c < b + c$. (We have to add the number d to $a + c$ to get $b + c$.)

Similarly,

$$bc = (a + d)c = ac + dc,$$

using the distributive law; so we have to add dc to ac to get bc , whence $ac < bc$. $\square$

5.2 Induction

We come now to the most important property of the natural numbers: induction. In the next section, we will see how to use it to prove things.

Here is the basic statement of the induction principle.

Theorem 5.3 (Principle of Induction) *Let S be a subset of the set $\mathbb{N}$ of natural numbers. Suppose that*

(a) $1 \in S$;

(b) *for any natural number n , if $n \in S$, then $n + 1 \in S$.*

Then $S = \mathbb{N}$.

Proof I have called this a theorem, because it really depends on the details of how we construct the natural numbers. The proof below shows that it follows from the property mentioned in the last section, namely, that we can count up to any natural number.

Clearly every element of S belongs to $\mathbb{N}$. So to prove that the sets S and $\mathbb{N}$ are equal, we only have to do the other half of the job, show that any natural number n belongs to S .

Now count up to n . The first number 1 belongs to S , and when we proceed from any number a to its successor $a + 1$, we preserve the property of belonging to S . So at the end of the count, we find that $n \in S$. $\square$

Since this result is important, I will explain it another way. Imagine the natural numbers laid out along a line, with the members of S painted red. Then, by assumption, 1 is red. Because 1 is red, the second assumption shows that 2 is red, and then 3 is red, and so on; so every natural number is red.

To take the children's rhyme again: To show that 100 is red: we know that 1 is red, and so 2 is red, and so on. Missing a few, 99 is red, and so 100 is red.

The Principle of Induction can be stated in many other ways. Here are a couple.

Theorem 5.4 (a) Suppose that T is a subset of $\mathbb{N}$ with the properties that, for any number n , if all the numbers smaller than n belong to T , then n belongs to T . Then $T = \mathbb{N}$.

(b) Any non-empty subset of $\mathbb{N}$ has a smallest element.

Proof (a) Suppose that T is a subset of $\mathbb{N}$ which satisfies the hypothesis. We construct a new set S as follows: S is the set of natural numbers with the property that all their predecessors belong to T ; in other words,

$$S = \{n \in \mathbb{N} : \text{if } m < n, \text{ then } m \in T\}.$$

Now consider the number 1. There are no smaller natural numbers at all; so, in a vacuous sense, all natural numbers smaller than 1 belong to T . Hence $1 \in S$.

Suppose that $n \in S$. By assumption, all the numbers smaller than n belong to T . Then, by assumption, n belongs to T . But this means that n and all smaller numbers (i.e. all the numbers smaller than $n + 1$) belong to T ; so by definition, $n + 1$ belongs to S .

We conclude from the Principle of Induction (Theorem 5.3) that $S = \mathbb{N}$.

Finally we want to conclude that $T = \mathbb{N}$; so take any natural number n . By what we just proved, $n + 1 \in S$. So all smaller numbers belong to T , and in particular, $n \in T$, as required.

(b) Let U be a subset of $\mathbb{N}$, and suppose that U has no smallest element. Arguing by contradiction, we have to show that $U = \emptyset$. If T is the complement of U , that is, if $T = \mathbb{N} \setminus U$, then we have to prove that $T = \mathbb{N}$.

Take any natural number n , and suppose that all the numbers smaller than n belong to T . Could we have $n \in U$? If so, it would be the smallest element of U , which contradicts our assumption. So necessarily $n \in T$ (it must belong to one or the other).

By part (a), we see that $T = \mathbb{N}$, as required. $\square$

Theorem 5.4(a) is sometimes called the *Principle of Strong Induction*, since we apparently have to assume less: a number n is only put into the set if all its predecessors lie in the set, whereas in the Principle of Induction only the immediate predecessor is needed. But in fact, as our argument shows, the two forms of the Principle are both true, and we can use either one as needed.

5.3 Proof by induction

The Principle of Induction is a very powerful tool for proving things about the natural numbers. It works like this.

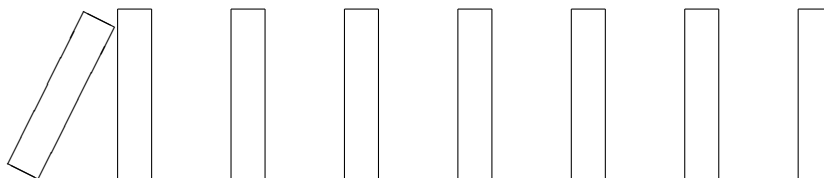
Theorem 5.5 (Proof by Induction) *Suppose we have some property P of natural numbers, and we can verify the following two facts:*

- (a) $P(1)$ holds;
- (b) for any natural number n , if $P(n)$ holds, then $P(n+1)$ holds.

Then $P(n)$ is true for every natural number n .

Proof Simply let S be the set of natural numbers n such that $P(n)$ is true: $S = \{n \in \mathbb{N} : P(n)\}$. The hypotheses tell us that $1 \in S$ and that $n \in S$ implies $n+1 \in S$ for every natural number n . Then Theorem 5.3 shows that $S = \mathbb{N}$, as required. $\square$

Proof by induction can be thought of in the following way. Suppose that we have a line of dominos, as shown in the diagram.



If we push over the first domino, what will happen? It will knock over the second, which will knock over the third, and so on; eventually all the dominos will fall. This is like induction. The inductive step is the fact that each domino knocks over the next one, and starting the induction is giving the first domino a push.

One can also formulate the Principle of Strong Induction as a proof technique. I leave the argument to you.

Theorem 5.6 (Proof by Strong Induction) *Suppose we have some property P of natural numbers, and we can verify that, if all natural numbers smaller than n satisfy P , then $P(n)$ holds. Then we conclude that $P(n)$ is true for all natural numbers n .*

Here are several examples.

Example 1 We saw earlier Gauss's ingenious argument (which he found while he was still at school) that the sum of the numbers from 1 to n is $n(n+1)/2$:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

Here is a proof by induction. The advantage of an induction proof is that you do not have to be as clever as Gauss to spot the trick to do it; you have two well-defined tasks to perform.

Let us call the displayed equation $P(n)$.

First, $P(1)$ is true. For, when $n = 1$, the left-hand side is $\sum_{k=1}^1 k$; the sum has only one term, namely 1, and so the value is 1. The right-hand side is $1 \cdot 2/2 = 1$. So the property holds.

Second, let us suppose that $P(n)$ holds, and prove that $P(n+1)$ also holds.

The left-hand side of $P(n+1)$ is $\sum_{k=1}^{n+1} k$, the sum of all the numbers from 1 to $n+1$.

We can write this as the sum of all the numbers from 1 to n , with $n+1$ added to it, and then use the assumed truth of $P(n)$ to put in the value for the sum:

$$\begin{aligned} \sum_{k=1}^{n+1} k &= \sum_{k=1}^n k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n(n+1)}{2} + \frac{2(n+1)}{2} \\ &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

The last expression is exactly the right-hand side of $P(n+1)$. So, assuming $P(n)$, we have proved $P(n+1)$.

By Theorem 5.5, $P(n)$ is true for all natural numbers n .

Example 2 For a second example, let us revisit an issue that came up in the first section of the notes, where we discussed Euclid's proof that there are infinitely many primes. We figured out that we had to prove the following statement:

Lemma 5.7 *Every natural number greater than 1 has a prime factor.*

Proof We will give a proof by Strong Induction.

Let $P(n)$ be the statement "if $n > 1$, then n has a prime factor".

To prove this by Strong Induction, we suppose that $P(m)$ holds for all numbers $m < n$, and have to deduce $P(n)$. So consider the number n . There are three cases:

- (a) $n = 1$. In this case there is nothing to prove, since $P(n)$ is vacuously true for $n = 1$.
- (b) n is prime. Then $P(n)$ holds, since n is a prime factor of itself.
- (c) $n > 1$, but n is not prime. In this case, $n = ab$, where $1 < a < n$ and $1 < b < n$. We are assuming that $P(m)$ holds for all $m < n$. So in particular, $P(a)$ holds, and a has a prime factor, say p . Then clearly p is a prime factor of $n = ab$, so $P(n)$ holds.

By Strong Induction, $P(n)$ is true for all natural numbers n ; so every natural number greater than 1 has a prime factor. $\square$

More examples of proofs by induction can be found in the Supplementary Material or in the course texts.

5.4 Division and divisibility

For this section, I am going to change the definition of natural numbers slightly, to allow 0 to be a natural number. You will see the reason for this soon.

Division is the reverse of multiplication; but it cannot always be performed exactly in the natural numbers. You can divide 6 by 2, since $6 = 2 \cdot 3$; but you cannot divide 7 by 2.

Definition 5.8 Let m and n be natural numbers. We say that m *divides* n , or that n *is divisible by* m , if there is a natural number k such that $n = mk$. We write $m \mid n$ to mean that m divides n .

Note that this is a *relation*: if we put the inputs 2 and 6 into the black box for divisibility, it responds “yes”, but if we put 2 and 7 in, it responds “no”. *Do not confuse $m \mid n$ with the fraction m/n , which is a number.*

Allowing zero to be a natural number here confuses things a bit:

- (a) for any natural number m , $m \mid 0$ (this is because $0 = m \cdot 0$);
- (b) $0 \mid n$ holds if and only if $n = 0$ (since $0 \cdot k = 0$ for any k).

Although exact division is not always possible, there is a substitute. This is called the *division algorithm*, since to do the calculations you use the method for long division you learned in primary school. An *algorithm* is a constructive method or recipe for producing some specified result.

Theorem 5.9 (The Division Algorithm) *Let a and b be natural numbers, with $b \neq 0$. Then there exist unique natural numbers q and r such that*

$$(a) \ a = bq + r;$$

$$(b) \ 0 \leq r \leq b - 1.$$

The numbers a, b, q, r are called the *dividend, divisor, quotient* and *remainder* respectively.

Proof First we show that we can find q and r satisfying the two conditions.

Let S be the set of positive integers n such that $bn > a$. Certainly S is non-empty since, for example, $b(a + 1) > ba \geq a$. So, by Theorem 5.5(b), the set S has a smallest element. Let n be the smallest element. Since $n \geq 1$, we can write $n = q + 1$, where $q \geq 0$.

Now $bq \leq a$ (since q is smaller than n , which was the smallest number for which $bn > a$; so certainly $a = bq + r$ for some number r , where $r \geq 0$).

Finally,

$$bq + r = a < bn = b(q + 1) = bq + b,$$

so $r < b$, which implies that $r \leq b - 1$.

Now we have to prove that the quotient and remainder are unique. Accordingly, suppose that we have two quotient-remainder pairs: that is, $a = bq_1 + r_1 = bq_2 + r_2$, where $0 \leq r_1 \leq b - 1$ and $0 \leq r_2 \leq b - 1$. If $r_1 = r_2$, then $bq_1 = bq_2$, and so $q_1 = q_2$. So suppose that r_1 and r_2 are unequal. One of them is larger; suppose that $r_1 < r_2$.

Then

$$bq_1 = a - r_1 > a - r_2 = bq_2,$$

so $q_1 > q_2$. This means that $q_1 = q_2 + c$ for some natural number $c > 0$. Then

$$bq_2 + r_2 = bq_1 + r_1 = b(q_2 + c) + r_1 = bq_2 + bc + r_1,$$

so that $r_2 = bc + r_1$. But then $r_2 \geq bc \geq b$, contrary to the assumption that $r_2 \leq b - 1$. $\square$

So we can say: $b \mid a$ holds if and only if, when we divide a by b , the remainder is zero. This is why we added 0 to the natural numbers in this section.

In this proof we used the cancellation laws for natural numbers: if $a + c = b + c$ then $a = b$; and if $ac = bc$, then $a = b$. We stated them for the natural numbers not including zero; and indeed the second law fails if $c = 0$. So we have to check our proof to see that we didn't cancel 0.

5.5 Greatest common divisor

We continue with the convention that 0 is a natural number here. Again this complicates things a bit.

Definition 5.10 Let a and b be two natural numbers. The *greatest common divisor* or *highest common factor* of a and b , written $\gcd(a, b)$, is a natural number d such that

- (a) $d \mid a$ and $d \mid b$;
- (b) if e is any natural number such that $e \mid a$ and $e \mid b$, then $e \mid d$.

It is not obvious that a number satisfying these conditions even exists. But we can say for sure that there cannot be more than one! For suppose that d_1 and d_2 are both greatest common divisors of a and b . Then $d_1 \mid a$ and $d_1 \mid b$; since d_2 is a greatest common divisor, it follows from part (b) of the definition that $d_1 \mid d_2$. Similarly, $d_2 \mid d_1$. Hence $d_1 = d_2$.

It happens that $\gcd(0, 0) = 0$. Let us check this. We saw above that every natural number divides 0. So $0 \mid 0$ and $0 \mid 0$; and, if $e \mid 0$ and $e \mid 0$, then $e \mid 0$. So by definition 0 is the greatest common divisor! [I hope this argument doesn't persuade you to switch from mathematics to Mongolian history!]

If a and b are not both zero, say $a \neq 0$, then the greatest common divisor cannot be larger than a , and it is (as the name suggests) the largest natural number which divides both a and b . (For if $d \neq 0$ and $e \mid d$, then $e \leq d$, since $d = ef$ for some natural number f .) If we had not included 0, we could have defined the greatest common divisor of two natural numbers to be the largest natural number dividing both of them; the definition is simpler, and there is no problem about uniqueness. But there is a reason for our strange choice, which will appear.

We are going to describe *Euclid's algorithm* for finding the greatest common divisor of two natural numbers.

Lemma 5.11 (a) For every natural number a , $\gcd(a, 0) = a$.

(b) For any two natural numbers a and b , if $a = bq + r$, then $\gcd(a, b) = \gcd(b, r)$.

Proof (a) is an exercise for you.

(b) We use the general principle: if two pairs of integers have exactly the same divisors, then they have the same greatest common divisor. For, assuming that neither pair is $(0, 0)$, the greatest common divisor is simply the largest number which divides both the numbers in the pair.

So we need to show that, if $a = bq + r$, then (a, b) and (b, r) have the same set of divisors.

Suppose that $d \mid a$ and $d \mid b$. Then also $d \mid bq$, so $d \mid a - bq = r$. (In detail: $d \mid a$ and $d \mid b$, so $a = kd$ and $b = ld$ for some natural numbers k and l . Then $r = a - bq = kd - ldq = d(k - lq)$, so $d \mid r$.)

In the other direction, if $d \mid b$ and $d \mid r$, then $d \mid bq + r = a$. $\square$

Euclid's Algorithm To find $\gcd(a, b)$ for two natural numbers a and b :

- (a) if $b = 0$, then $\gcd(a, b) = a$.
- (b) if $b \neq 0$, apply the Division Algorithm to find q and r with $a = bq + r$ and $0 \leq r < b$. Then calculate $\gcd(b, r)$; the result is equal to $\gcd(a, b)$.

Why does it work? We know that, if $a = bq + r$, then $\gcd(a, b) = \gcd(b, r)$. Also, $r < b$. If $r = 0$, then the answer is a ; otherwise we apply the algorithm again. The remainders get smaller at each step; we know that they can't go on getting smaller for ever, so after a finite number of steps the remainder reaches zero.

It is easier to explain with an example. What is $\gcd(87, 33)$?

$$\begin{aligned} 87 &= 33 \cdot 2 + 21, \\ 33 &= 21 \cdot 1 + 12, \\ 21 &= 12 \cdot 1 + 9, \\ 12 &= 9 \cdot 1 + 3, \\ 9 &= 3 \cdot 3 + 0. \end{aligned}$$

So

$$\gcd(87, 33) = \gcd(33, 21) = \gcd(21, 12) = \gcd(12, 9) = \gcd(9, 3) = \gcd(3, 0) = 3.$$

The rule is:

Continue dividing the previous divisor by the remainder until the remainder is zero. The last divisor used is the greatest common divisor.

It is possible, though complicated, to give a proof that the algorithm works correctly; but hopefully this is clear to you.

In the supplementary material we will see that Euclid's algorithm has another use as well.

5.6 Study skills: Definitions

Definitions are the dictionary entries of mathematics. They tell us precisely what the words mean. Without them, mathematics could not be an exact subject. Statements of theorems or problems very often contain unexplained terms; we need definitions to tell us how to understand the statements.

A lot of the vocabulary of mathematics consists of everyday words used in a different sense. Each of the following words has a specialised use in mathematics that cannot be guessed from its everyday meaning: *ring*, *field*, *tree*, *root*, *linear*, *complete*, *differentiate*. Indeed, “field” has two quite different meanings, one in algebra and one in applied mathematics!

The best way to understand a definition is to use it. The coursework problems are designed to make use of definitions from the lecture notes and help you to understand them.

There are two parts to understanding a definition. These are not the same thing, but they do reinforce each other.

- You must know precisely what the definition says. You may state it in your own words, but make sure that you have not changed the sense of the definition.
- You must be able to apply the definition.

Here is an example from Chapter 2 of the notes.

Definition Let A be a set. The *power set* of A , written $\mathcal{P}(A)$, is the set whose members are all the subsets of A .

You should be able to state this, or something with the same meaning, on demand. In exams, there will often be marks given for accurate statement of a definition; you are throwing away marks if you cannot give it.

But using the definition is a bit different. The first thing you have to do is to understand the term *subset*. Earlier in the notes we find:

Definition Let A and B be sets. Then A is a *subset* of B , written $A \subseteq B$, if every member of A is also a member of B .

The best test of your understanding of a definition is to apply it to an extreme or unusual case. Suppose you were asked: *What is the power set of the empty set?* You could argue like this. The empty set is a subset of itself, so the power set contains at least one element. There cannot be any others, since a non-empty set A cannot be a subset of the empty set. (According to the definition of subset,

this would require that every member of A would have to lie in the empty set; but nothing lies in the empty set.) So the answer to the question is:

$$\mathcal{P}(\emptyset) = \{\emptyset\},$$

a set with one element.

Please make sure that you understand the difference between $\emptyset$ (a set with no members) and $\{\emptyset\}$ (a set with one member).

5.7 Supplement: More proofs by induction

Here are two proofs by induction. They both rely on a property of binomial coefficients that we saw in Chapter 2: If $0 < k < n$, then

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k}.$$

Adding up binomial coefficients Prove that

$$\sum_{i=k}^n \binom{i}{k} = \binom{n+1}{k+1}.$$

For this we will use a variant of Proof by Induction, as follows:

Suppose that P is a property of natural numbers. Suppose that

- (a) $P(k)$ is true;
- (b) for $n \geq k$, if $P(n)$ holds, then $P(n+1)$ holds.

Then $P(n)$ is true for all $n \geq k$.

Can you see why this is true?

The reason we need it is that our definition of $\binom{i}{k}$, as the number of k -element subsets of a set of cardinality i , doesn't make much sense if $i < k$, so we have to start the induction at $n = k$. There is a fudge: we could define $\binom{i}{k}$ to be 0 if $i < k$, but then the proof becomes a bit more complicated.

So to the proof. Is $P(k)$ true? $P(k)$ asserts

$$\sum_{i=k}^k \binom{i}{k} = \binom{k+1}{k+1}.$$

The sum on the left has a single term, with $i = k$, and its value is $\binom{k}{k} = 1$; on the right we have $\binom{k+1}{k+1} = 1$. So $P(k)$ is true.

Suppose that $P(n)$ is true. To get from the left-hand side of $P(n)$ to that of $P(n+1)$, we have to add just one further term, namely $\binom{n+1}{k}$. So, using the assumption $P(n)$, we have

$$\begin{aligned}\sum_{i=k}^{n+1} \binom{i}{k} &= \binom{n+1}{k+1} + \binom{n+1}{k} \\ &= \binom{n+2}{k+1},\end{aligned}$$

using our result about binomial coefficients (with $n+2$ replacing n , and $k+1$ replacing k).

So the result is true, by induction.

Proof of the Binomial Theorem:

$$(x+y)^n = \sum_{k=0}^n x^{n-k} y^k.$$

Let $P(n)$ be the statement of the Binomial Theorem.

Then $P(1)$ asserts that

$$(x+y)^1 = \binom{1}{0} x^1 y^0 + \binom{1}{1} x^0 y^1,$$

which is true since the binomial coefficients are 1.

Suppose that $P(n)$ holds. Then

$$\begin{aligned}(x+y)^{n+1} &= (x+y)(x+y)^n \\ &= (x+y) \sum_{k=0}^n x^{n-k} y^k.\end{aligned}$$

On the right, every term contains a power of x and a power of y where the exponents add up to $n+1$. The coefficient of x^{n+1} is $\binom{n}{0} = 1$, and the coefficient of y^{n+1} is $\binom{n}{0} = 1$. Any other term, say $x^{n+1-k} y^k$, comes from two places, when we expand the bracket:

- the term in $x^{n-k}y^k$, multiplied by x (coefficient $\binom{n}{k}$); and
- the term in $x^{n-k+1}y^{k-1}$, multiplied by y (coefficient $\binom{n}{k-1}$.)

So, overall, the coefficient of $x^{n+1-k}y^k$ is

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k},$$

just as $P(n+1)$ requires it to be. So we have proved $P(n+1)$ and finished the inductive step.

So the theorem is proved.

Subsets of $\{1, 2, \dots, n\}$ Here is a proof by induction that the number of subsets of $\{1, 2, \dots, n\}$ is 2^n . Again, let $P(n)$ denote this assertion.

Then $P(1)$ is true, since the set $\{1\}$ has two subsets, namely $\emptyset$ and $\{1\}$.

Suppose that $P(n)$ is true. To count the subsets of $\{1, \dots, n+1\}$, we divide them into two classes, those containing $n+1$ and those not containing $n+1$. To take the second class first, a subset of $\{1, \dots, n+1\}$ not containing $n+1$ is nothing but a subset of $\{1, \dots, n\}$; by $P(n)$, there are 2^n of these.

A subset containing $n+1$ has the form $S \cup \{n+1\}$, where S is a subset of $\{1, \dots, n\}$; so there are 2^n of these also.

Thus the total number of subsets is $2^n + 2^n = 2^{n+1}$, so $P(n+1)$ holds. This finishes the proof of the inductive step, and hence (by induction) the theorem.

5.8 Supplement: More on Euclid's Algorithm

In this section, we will anticipate the next chapter of the notes, in which we construct the integers (positive and negative). Euclid's algorithm has the following consequence:

Theorem 5.12 *Let a and b be natural numbers with $\gcd(a, b) = d$. Then there are integers u and v such that $au + bv = d$.*

I will not prove this theorem; I will demonstrate how it works, using the example of Euclid's algorithm from the notes. But first let me make a remark. Suppose that a, b, d are natural numbers such that

- (a) $d \mid a$ and $d \mid b$;

(b) $d = ax + by$ for some integers x, y .

Then $d = \gcd(a, b)$.

For suppose that $e \mid a$ and $e \mid b$. Then $e \mid ax$ and $e \mid by$, so $e \mid ax + by = d$.

Now recall our example:

$$\begin{aligned} 87 &= 33 \cdot 2 + 21, \\ 33 &= 21 \cdot 1 + 12, \\ 21 &= 12 \cdot 1 + 9, \\ 12 &= 9 \cdot 1 + 3, \\ 9 &= 3 \cdot 3 + 0. \end{aligned}$$

We conclude that $\gcd(87, 33) = 3$.

Now work up from the second last equation, expressing 3 in terms of what is above:

$$\begin{aligned} 3 &= 12 - 9 \cdot 1 \\ &= 12 - (21 - 12 \cdot 1) \cdot 1 = 12 \cdot 2 - 21 \cdot 1 \\ &= (33 - 21 \cdot 1) \cdot 2 - 21 \cdot 1 = 33 \cdot 2 - 21 \cdot 3 \\ &= 33 \cdot 2 - (87 - 33 \cdot 2) \cdot 3 = 33 \cdot 8 - 87 \cdot 3, \end{aligned}$$

so $\gcd(87, 33) = 3 = 87x + 33y$, where $x = -3$ and $y = 8$.

Again, I hope it is clear that this procedure will always work.

5.9 Supplement: The Fundamental Theorem of Arithmetic

Theorem 5.13 (Fundamental Theorem of Arithmetic) *Every natural number greater than 1 can be factorised into primes; the factorisation is unique up to the order of the factors.*

Proof We already saw in the notes (Lemma 5.7) that every natural number greater than 1 has a prime factor. This is obviously the first step in proving that there is a factorisation into prime factors. (We leave the uniqueness until later.) So we will prove by Strong Induction the statement

$$P(n): n \text{ can be factorised into prime factors.}$$

So let us assume that every natural number m satisfying $1 < m < n$ can be factorised into prime factors. Now there are two possibilities for n :

5.9. SUPPLEMENT: THE FUNDAMENTAL THEOREM OF ARITHMETIC 91

- (a) n is prime. In this case we have our factorisation, with just one factor.
- (b) n is not prime. By Lemma 5.7, n has a prime factor p ; and so $n = pm$, where $1 < m < n$. (If $m = 1$, then $p = n$, contrary to the case assumption; and if $m = n$, then $p = 1$, contrary to the definition of a prime.) By the induction hypothesis, $m = q_1 q_2 \cdots q_r$, where $q_1, q_2, \dots, q_r$ are primes; then $n = p q_1 q_2 \cdots q_r$, where $p, q_1, \dots, q_r$ are primes. So the inductive step is complete.

The proof that the factorisation is unique, that is, that two factorisations of n into prime factors can differ only in the order of the factors, goes like this. We suppose that we have two factorisations of n into primes, say

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s.$$

We have to show that $r = s$ and that the q s are just the p s possibly in a different order. So we look first at p_1 . If we could show that one of the primes on the right, say q_i , was equal to p_1 , then we could match p_1 with q_i and cancel this factor. Then we could continue the process, matching p_2 with one of the remaining q s, and so on, until we had matched up the two factorisations. So the crucial thing is to show:

If p is prime and p divides $a_1 a_2 \cdots a_s$, then p divides a_i for some i .

Note that we allow the a s to be arbitrary natural numbers here. If we then specialise to the case where they are primes, if $p \mid q_i$ and q_i is prime, then $p = q_i$ (since the only factors of q_i are q_i and 1, and $p \neq 1$).

We do this first for $s = 2$.

Lemma 5.14 *Suppose that p is prime and $p \mid ab$ for some natural numbers a, b . Then either $p \mid a$, or $p \mid b$.*

Proof If p divides a , then our conclusion holds. So we may suppose that p does not divide a , and must conclude that then it divides b .

If p does not divide a , then the greatest common divisor of p and a is 1 (since 1 and p are the only divisors of p). By our extended version of Euclid's algorithm, we know that there are integers u and v such that

$$up + va = 1.$$

Multiplying by b , we see that $upb + vab = b$. Now p obviously divides upb ; and by assumption, p divides ab , so p divides vab . Hence p divides the sum of these two numbers, which is $upb + vab = b$, as required. $\square$

Now it is easy to extend this to the product of more than two factors. Suppose that p is prime, and p divides $a_1 a_2 \cdots a_s$. Then either $p \mid a_1$ or $p \mid a_2 \cdots a_s$. In the first case, we have succeeded in showing that p divides one of the a s. In the second case, either $p \mid a_2$ or $p \mid a_3 \cdots a_s$. Continuing, we eventually conclude that p divides a_i for some i . $\square$

5.10 Supplement: Pascal's Triangle

The binomial coefficients are conveniently displayed in a triangular array called *Pascal's Triangle*, where the n th row consists of the numbers

$$\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \dots, \binom{n}{n},$$

as shown below. The row numbers, and numbers along each row, start from 0.

$$\begin{array}{cccccccc} & & & & 1 & & & \\ & & & & & 1 & & 1 \\ & & & 1 & & 2 & & 1 \\ & & 1 & & 3 & & 3 & & 1 \\ & 1 & & 4 & & 6 & & 4 & & 1 \\ 1 & & 5 & & 10 & & 10 & & 5 & & 1 \end{array}$$

Despite its name, Pascal's Triangle was not invented by Pascal. The figure on the next page shows the triangle as given in Chu Shi-Chieh's *Ssu Yuan Yü Chien*, dated 1303.

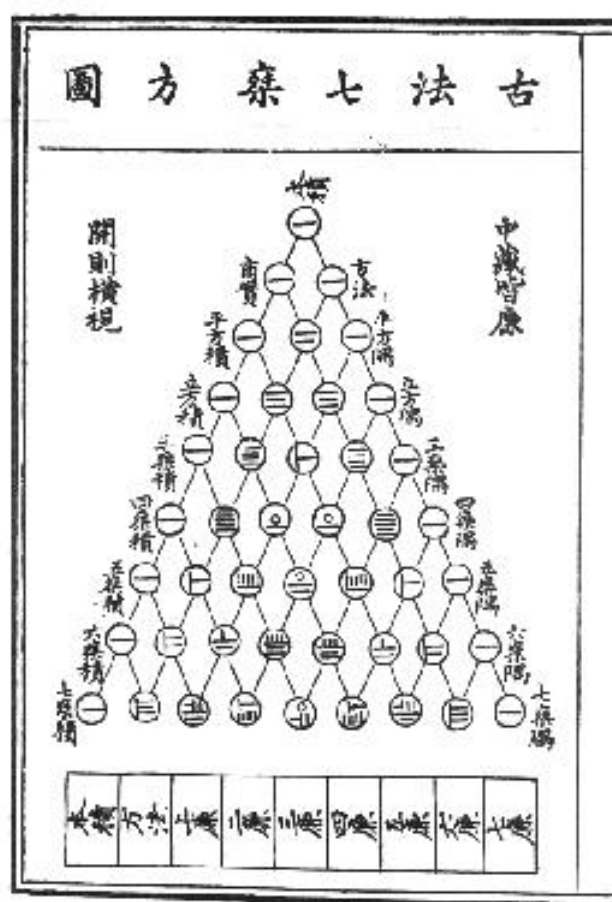
The easiest way to generate the table is to use the relation given at the start of this week's supplementary material:

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k}.$$

This shows that, when the binomial coefficients are written out in the triangular array, each one is the sum of the two to its left and right in the row above. Thus, $\binom{5}{2} = 10$ is found by adding the two values shown:

$$\begin{array}{ccc} 4 & & 6 \\ & \searrow & \swarrow \\ & 10 & \end{array}$$

So, to generate the triangle, we write 1s down the left and right borders and then use this rule to fill in the rest.



The numbers in a row of Pascal's triangle are the coefficients in the expansion given by the Binomial Theorem:

$$(x + y)^5 = x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5.$$

The binomial coefficients and Pascal's triangle have many other remarkable properties. Here are two to try for yourself.

- Take Pascal's triangle: replace the even numbers by 0 and the odd numbers by 1. What pattern do you get? How can the triangle with rows $0, \dots, 8$ be obtained from the triangle with rows $0, \dots, 4$? Does this pattern continue?
- Work out the sums of the numbers in sloping lines, of which a typical line is

$$\binom{5}{0} + \binom{4}{1} + \binom{3}{2} = 1 + 4 + 3 = 8.$$

What rules do these sums obey?

5.11 Supplement: Representing numbers

You are very familiar with the usual way of writing numbers in the decimal system using the digits $0, 1, \dots, 9$. This is called a *place value system*, since the value of a digit depends on its place in the number: thus, in the number 345, the number 3 represents three hundreds, 4 represents four tens, and 5 five units. In this way, any natural number, however large, can be represented.

Other number systems have been used; not all have this property. Here is an extreme example, from a short story “Funes, his memory” by the Argentine writer Jorge Luis Borges. Ireneo Funes has been crippled by a fall from a horse; the fall has made his memory extraordinarily efficient (or, perhaps, has made it impossible for him to forget anything).

He told me that in 1886 he had invented a numbering system original with himself, and that within a very few days he had passed the twenty-four thousand mark ... His original motivation, I think, was his irritation that the thirty-three Uruguayan patriots should require two figures and three words rather than a single figure, a single word. He then applied this mad principle to the other numbers. Instead of 7013, he would say, for instance, “Máximo Pérez”; instead of 7014, “the railroad”; other numbers were “Luis Melián Lafinur”, “Olimar”, “sulphur”, “clubs”, “the whale”, “gas”, “a stewpot”, “Napoleon” ... I tried to explain to Funes that his rhapsody of unconnected words was exactly the opposite of a number *system*. I told him that when one said “365” one said “three hundreds, six tens, and five ones,” a breakdown impossible with the numbers “Nigger Timoteo” or “a ponchoful of meat”. Funes either could not or would not understand me.

Roman numerals, still familiar to us on public clocks and in film credits, are more systematic. The units digits are represented as follows:

1	=	I	
2	=	1 + 1	= II
3	=	1 + 1 + 1	= III
4	=	1 less than 5	= IV
5	=	V	
6	=	5 + 1	= VI
7	=	5 + 1 + 1	= VII
8	=	5 + 1 + 1 + 1	= VIII
9	=	1 less than 10	= IX

0 is represented by nothing. Then the tens digits are represented on the same pattern with X for 10 and L for 50; and the hundreds digits similarly, with C for

100 and *D* for 500. Finally, *M* represents 1000. Thus, all numbers from 1 to 3999 can be represented with only seven different symbols. For example, 2012 is *MMXII*.

However, the apparent advantage over our notation (which needs ten different symbols) is an illusion, for two reasons:

- (a) If you have ever tried adding or multiplying Roman numbers, you will realise that it is rather complicated!
- (b) To go further, we need more symbols. In a sense, the Roman system has the same drawback as Funes' "system"; to represent all numbers we would need infinitely many symbols, even though we don't have to invent new ones quite as often!

Exercises

5.1 Prove that

$$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2$$

for all natural numbers n . (You may want to find a formula for the right-hand side and then prove by induction that it is equal to the left-hand side.)

5.2 Here is a *definition* of the **factorial function**, which maps the natural numbers to the natural numbers. (Despite the curious notation, this really is a function!)

- $0! = 1$.
- If $n > 0$, then $n! = n \times (n - 1)!$.

- (a) Prove by induction that this really does define a function; that is, prove that, for any natural number n , the rule given enables us to calculate $n!$.
- (b) Prove by induction that, for $n \geq 1$,

$$2^{n-1} \leq n! \leq n^n.$$

5.3 How many zeros occur at the end of the number $100!$?

5.4 A natural number n is *even* if it has the form $n = 2k$ for some natural number k , and is *odd* if it has the form $2l + 1$ for some natural number l . Prove carefully that

- (a) every natural number is either even or odd;
- (b) no natural number is both even and odd.

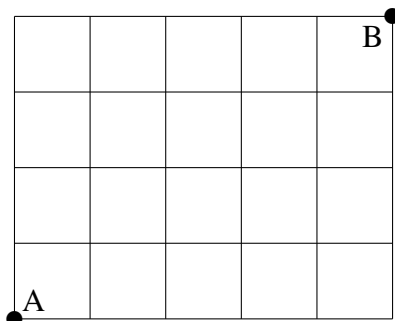
5.5 Let m and n be positive integers. Recall the *greatest common divisor* of m and n , written $\gcd(m, n)$: it is the largest positive integer which divides both m and n . Now define the *least common multiple* of m and n , written $\text{lcm}(m, n)$, to be the smallest positive integer which is divisible by both m and n .

- (a) Prove that $\gcd(m, n) \cdot \text{lcm}(m, n) = mn$.
- (b) Prove that $\gcd(m, n) \leq \text{lcm}(m, n)$, with equality if and only if $m = n$.
- (c) In an essay entitled “Love is Dead”, John Betjeman described (and disparaged) the common man of his day in these words:

He is the Lowest Common Multiple, not even the Highest Common Factor.

(Here “lowest common multiple” and “highest common factor” are alternative names for least common multiple and greatest common divisor.) Why does Betjeman’s metaphor fail to do what he intends?

5.6 (Puzzle) The street plan of a certain city is a grid, as shown.



You are at point A, and need to go to point B. How many routes can you take if you travel only northwards or eastwards?

Generalise this result to a $m \times n$ grid.

5.7 (Project) You are a bank robber and want to crack a safe with a dial combination lock. The dial has the numbers 0, 1, 2, ..., 9 on it. As you rotate the dial, the last four numbers dialled make up the combination. So, if you dialed the sequence 1253452, you would test the combinations 1253, 2534, 5345 and 3452.

Can you find a sequence of numbers that will take you through every possible combination once and once only? How long should such a sequence be?

Chapter 6

Integers and rational numbers

In this chapter, we will see constructions of the integers and the rational numbers; and we will see that our number system still has gaps (equations we can't solve) and needs to be extended further. The Study guide suggests that you collect a stock of examples. In the supplementary material we see that two famous numbers (the square root of 2, and the base of natural logarithms) are irrational.

6.1 Why do we need to extend the number system?

There are good practical reasons for needing a larger number system than the natural numbers.

Natural numbers are ideal for counting – that is what they are for! But as we become more sophisticated, we need new kinds of numbers.

First, once we have a banking system, we need to face the fact that a customer might actually owe the bank money. I had £100 in the bank, but as a result of a sudden emergency, I had to withdraw £150, leaving me £50 in debt to the bank. I would say “I am in the red”, because this balance would be written in red in the bank's ledgers. But it would be better if, instead of two kinds of numbers (red and black), there was only one kind, which could be positive or negative.

The other extension comes from the need for numbers in measurement. If I draw a square with side of length 1, and measure the diagonal, it comes out (as near as I can make it) to 1.414, while the circumference of a circle of diameter 1 turns out to be about 3.142. I don't know whether these values are exact, but I can't even express them with just natural numbers (the nearest natural numbers would be 1 and 3 respectively).

From a mathematician's point of view, there is another reason: solving equations. Much of mathematics is concerned with exactly this. If we start with the natural numbers, we can solve the equation $3 + x = 5$ (the solution is $x = 2$, but we

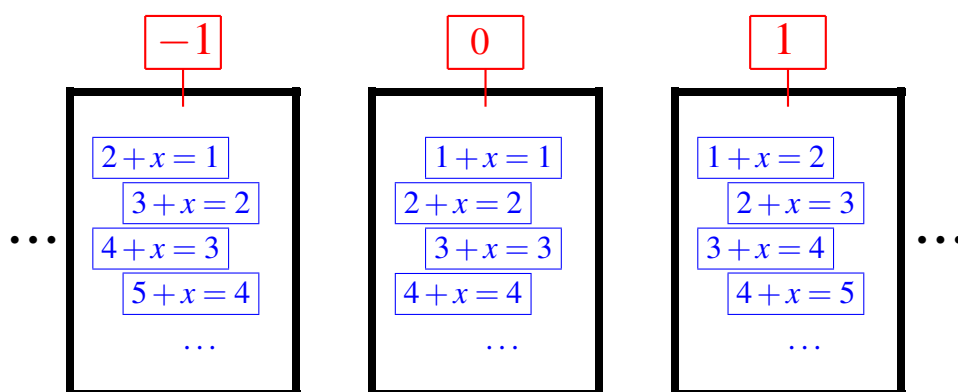
can't solve the equation $5 + x = 3$. We need negative numbers for this. Similarly, we can solve the equation $3x = 6$, but not the equation $3x = 7$, unless we introduce fractions.

6.2 The integers

We want to enlarge the natural numbers so that equations of the form $b + x = a$ always have a solution, no matter what a and b are.

The simplest solution is the most profligate. For every two natural numbers a and b , add a new number $m(a, b)$ so that $x = m(a, b)$ is a solution to this equation. The reason why we don't want to do this is that there will be far too much duplication. For example, the equations $5 + x = 3$ and $7 + x = 5$ should have the same solution. So we want to take the set of equations of this form, and sort them into bags, each of which gives us a new number: equations in the same bag will have the same solution. Then we will label each bag with the integer which is the common solution to all the equations in that bag, and regard the bags as being the integers. When we use an integer in a sum, we just read the label on the bag; we don't have to worry about the contents.

For example, all the equations $2 + x = 1$, $3 + x = 2$, $4 + x = 3$, $\dots$, will correspond to the integer -1 , as the picture shows.



Remember that partitions come from equivalence relations: we want to define an equivalence relation on the set of all the equations whose equivalence classes do what we want.

So we do a bit of rough work. If the equations $b + x = a$ and $d + x = c$ are to have the same solution, then

$$b + c = b + (d + x) = (b + x) + d = a + d,$$

where we used a bit of manipulation involving the commutative and associative laws in there. (More precisely, we want to construct the integers in such a way that these laws continue to hold; this will force our choices.)

So we make a definition and prove a lemma about it.

Definition 6.1 Let S be the set of all equations of the form $b + x = a$, for natural numbers a and b . Define a relation R on the domain S by the rule

$$(b + x = a) R (d + x = c) \text{ if and only if } a + d = b + c.$$

Lemma 6.2 R is an equivalence relation on S .

Proof We have three rules to check.

Reflexive: $a + b = b + a$ (by the commutative law), so $(b + x = a) R (b + x = a)$.

Symmetric: Suppose that $(b + x = a) R (d + x = c)$. Then $a + d = b + c$. Hence $c + b = d + a$ (by the commutative law again), so $(d + x = c) R (b + x = a)$.

Transitive: Suppose that $(b + x = a) R (d + x = c)$ and $(d + x = c) R (f + x = e)$. Then $a + d = b + c$ and $c + f = d + e$. Hence

$$a + d + f = b + c + f = b + d + e,$$

and the cancellation law gives $a + f = b + e$, so $(b + x = a) R (f + x = e)$. □

Now watch closely. The next bit is the real test of a mathematician. If you can understand what I am going to say next, you will have no trouble with anything else you meet in your degree.

We have defined an equivalence relation on the set of equations of the form $b + x = a$ so that two equations are equivalent if they have the same solution. So there is one solution for each equivalence class. Now, being brave, we say, take each equivalence class to be one of our new numbers:

Definition 6.3 An *integer* is an equivalence class of the relation R on the domain S just defined.

To repeat: each equivalence class corresponds to a single integer; we take the integers to be the equivalence classes. We will now change our notation a bit and write the equivalence class $R((b + x = a))$ (consisting of all the equations related to this one) as $[a, b]$.

Anyway, we have now defined the integers. We need to be able to add, multiply and order them. Remembering that the equivalence class $[a, b]$ corresponds to solutions of the equation $b + x = a$, in other words, $x = a - b$, we can figure out what the rules must be, by a short calculation:

- (a) $(a - b) + (c - d) = (a + c) - (b + d)$,
- (b) $(a - b) \times (c - d) = (ac + bd) - (ad + bc)$,
- (c) $(a - b) < (c - d)$ if and only if $a + d < b + c$.

So

Definition 6.4 We define addition and multiplication of integers by the rules

$$\begin{aligned} [a, b] + [c, d] &= [a + c, b + d], \\ [a, b] \times [c, d] &= [ac + bd, ad + bc], \\ [a, b] < [c, d] &\Leftrightarrow a + d < b + c. \end{aligned}$$

Now it is just a case of laboriously verifying that the integers we have constructed satisfy properties like those of the natural numbers: the commutative, associative, distributive, and order laws, and the cancellation laws (except that we can't cancel 0 from multiplication: that is, if $0a = 0b$, we are not allowed to conclude that $a = b$).

But two more final jobs are also very important:

Proposition 6.5 *Inside the integers, we can find a “copy” of the natural numbers. That is, the integer $[a + n, a]$ (which contains all pairs $(x + n, x)$) behaves just like the natural number n :*

- (a) $[a + n, a] + [b + m, b] = [c + m + n, c]$;
- (b) $[a + n, a] \times [b + m, b] = [c + mn, c]$;
- (c) $[a + n, a] < [b + m, b]$ if and only if $m < n$.

Proposition 6.6 *If e and f are any two integers, then the equation $f + x = e$ has a unique integer solution.*

Proof If $e = [a, b]$ and $f = [c, d]$, then the solution x is $[a + d, b + c]$. For

$$[c, d] + [a + d, b + c] = [c + a + d, d + b + c] = [a, b].$$

This is because $(c + a + d, d + b + c)R(a, b)$, since $(c + a + d) + b = (d + b + c) + a$.
□

This has been a long and complicated ride. So let us summarise what we have done.

Starting from the set $\mathbb{N}$ of natural numbers, we have constructed a new set of objects called *integers* such that

- (a) their addition, multiplication, and order relations satisfy “all the usual rules”;
- (b) within them, we can find a subset whose addition, multiplication and order are exactly the same as those for the natural numbers;
- (c) equations of the form $f + x = e$ have unique solutions in the new system.

What does this mean on practice?

What it certainly *doesn't* mean is that I want you to stop writing -23 and write $[1, 24]$ or $[100, 123]$ or something else instead. The integer -23 is equal to $1 - 24$, or to $100 - 123$, as it always was!

The things we have constructed are the usual integers; but we have put them on a firm mathematical footing, based on our knowledge of the natural numbers.

We have already taken this point of view in Chapter 3, where we saw that the set of integers is countably infinite.

6.3 The rational numbers

If I were to write out this section in full, it would look very similar to the last one, with only some differences in detail. Instead, I will just sketch the differences.

We want to solve equations $bx = a$, where a and b are integers and $b \neq 0$. The solution will eventually be the rational number a/b . Now $a/b = c/d$ if and only if $ad = bc$. So we define

- (a) S is the set of equations $bx = a$ of integers with $b \neq 0$.
- (b) The relation R holds between equations $bx = a$ and $dx = c$ whenever $ad = bc$.

Again, R is an equivalence relation; we define a *rational number* to be an equivalence class of this relation.

Now the rules for adding and multiplying fractions, namely $(a/b) + (c/d) = (ad + bc)/bd$ and $(a/b) \times (c/d) = ac/bd$, give us the rules for adding and multiplying our new numbers. If $[[a, b]]$ is the equivalence class containing the equation $bx = a$, then we put

$$\begin{aligned} [[a, b]] + [[c, d]] &= [[ad + bc, bd]], \\ [[a, b]] \times [[c, d]] &= [[ac, bd]]. \end{aligned}$$

There is a slight twist to the order because the signs give us some trouble; I won't give the details.

We write the rational number $[[a, b]]$ as a/b , just as you would expect.

What we end up with is a new set of objects called *rational numbers* such that

- (a) their addition, multiplication, and order relations satisfy “all the usual rules”;
- (b) within them, we can find a subset whose addition, multiplication and order are exactly the same as those for the integers;
- (c) equations of the form $fx = e$ with $f \neq 0$ have unique solutions in the new system.

Again there is no need to change our practice about how to write these things: we still write $22/7$ rather than $[[22, 7]]$.

And once again, we have used this already, when we showed that the set of rational numbers is countably infinite.

6.4 We’re not finished yet!

Mathematicians realised quite early on that this was not the end of the story for solving equations.

We saw earlier that, empirically, if the side of a square is 1, then the diagonal is about 1.42. But “about” is not good enough: what is it exactly?

Pythagoras’ Theorem tells us that, if we take one of the two right-angled triangles into which the diagonal divides the square, then the square on the hypotenuse is equal to the sum of the squares on the other two sides. That is, the length of the hypotenuse is a number whose square is 2.

Your calculator will tell you that $1.414^2 = 1.999396$; close to 2 but not exactly 2. If you try more places of decimals, you will get closer and closer to 2, but you will never get it exactly. This is because all the numbers you can type into your calculator are rational numbers (and rather special rational numbers at that, of the form $1999396/1000000$, say, where the denominator is a power of 10. Another famous theorem of Pythagoras tells us that there is no rational number whose square is equal to 2.

We need one preliminary result. A natural number of the form $2k$ (for some natural number k) is *even*; one of the form $2k + 1$ is *odd*. Every natural number is of one of these types: for if we apply the division algorithm to divide n by 2, we get $n = 2k + r$, where $0 \leq r \leq 1$; that is, $r = 0$ or $r = 1$.

Lemma 6.7 *The square of an even number is even, and the square of an odd number is odd.*

Proof If n is even, say $n = 2k$, then $n^2 = 4k^2 = 2(2k^2)$; so n^2 is even. If n is odd, say $n = 2k + 1$, then $n^2 = 4k^2 + 2k + 1 = 2(2k^2 + k) + 1$; so n^2 is odd. $\square$

Theorem 6.8 *There is no rational number x such that $x^2 = 2$.*

Proof First, an observation about rational numbers. Let m/n be a rational number, and suppose that m and n have a common factor d . This means that $m = dp$ and $n = dq$ for some integers p and q ; so $m/n = p/q$. (To tie this in with what you learned in the last section, notice that $mq = dpq = np$, so the pairs (m, n) and (p, q) represent the same rational number.) Informally we say that a common factor can be cancelled from the numerator and denominator of a rational number without changing its value.

So we can assume that all common factors have been cancelled, so that $\gcd(m, n) = 1$.

Now suppose that $x = m/n$ satisfies $x^2 = 2$. That is, $(m/n)^2 = 2$, so that $m^2 = 2n^2$. This means that m^2 is even, so m is even (because the square of an odd number would be odd). Say $m = 2k$ for some natural number k .

Then we have $4k^2 = m^2 = 2n^2$, so we can cancel a factor of 2 and obtain $2k^2 = n^2$. Thus, n^2 is even, and so as before n is even.

But now m and n are both even, so 2 is a common factor, contradicting our assumption that all the common factors have been cancelled.

This contradiction shows that no such rational number x can exist. $\square$

This provided a big problem for Pythagoras and his students. It is obvious that the length of the diagonal of a square has to be some number; but, if rational numbers are all that we have, there is no such number!

As well as polynomial equations like $x^2 - 2 = 0$ that we can't solve in the rational numbers, there are other types of equations, for example

- (a) $\tan x = 1$, which has a solution $x = \pi/4$, one-eighth of the circumference of a circle of unit radius;
- (b) $\log x = 1$, which has a solution $x = e$, the base of natural logarithms.

These numbers are both irrational. The proof for π is rather difficult. The proof for e is in the supplementary material, using a result from calculus.

We see that we need to enlarge once again our stock of numbers. We do this in the next section.

6.5 Study skills: Make a stock of examples

Later in the module we will discuss the uses of examples. But already you can see that they will be useful. You might remember from an earlier chapter that, although it often happens that, if p is prime, then the Mersenne number $2^p - 1$ is also prime, this is not always the case; it fails for $p = 11$, since $2^{11} - 1 = 2047 =$

23×89 . If you remember this example, then you can immediately answer the question whether this general statement is true.

In calculus it is particularly important to have a stock of examples at your disposal: a continuous function which is not differentiable, a convergent sequence of continuous functions whose limit is not continuous, and so on. But the principle applies in general to any branch of mathematics.

Is a sum of rational numbers always rational? Yes, if it is a finite sum; but no, in general. Here is an example:

$$\sqrt{2} = 1.41421\dots = \frac{1}{1} + \frac{4}{10} + \frac{1}{100} + \frac{4}{1000} + \frac{2}{10000} + \frac{1}{100000} + \dots$$

Here is a famous example which appears in many different parts of mathematics:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}.$$

If p is a prime number greater than 5, and you calculate $1/p$ as a decimal, it will recur, with period which divides $p-1$. Sometimes it is equal (for example, $1/7 = 0.142857142857\dots$); sometimes not (for example, $1/11 = 0.090909\dots$). You may find a use for these examples one day.

Keep your own little stock, and add to it anything you find interesting.

6.6 Supplement: Another proof of Pythagoras' Theorem

Pythagoras' Theorem tells us that $\sqrt{2}$ is not a rational number. Here is a different proof, possibly closer to the one that Pythagoras found.

Instead of $\sqrt{2}$, we will instead show that $1 + \sqrt{2}$ is irrational. This is obviously equivalent.

We will as usual argue by contradiction, and suppose that $1 + \sqrt{2} = m/n$ for some natural numbers m and n . Now the diagonal of a unit square clearly has length between 1 and 2, so $1 + \sqrt{2}$ lies between 2 and 3. This means that, when we apply the division algorithm to m and n , we obtain

$$m = 2n + r, \quad \text{where } 0 \leq r \leq n-1.$$

And r cannot be zero, since this would mean that $m = 2n$, so $1 + \sqrt{2} = m/n = 2$, which is obviously wrong.

Now

$$\begin{aligned}
 \frac{n}{r} &= \frac{n}{m-2n} \\
 &= \frac{n}{(1+\sqrt{2})n-2n} \\
 &= \frac{1}{\sqrt{2}-1} \\
 &= \sqrt{2}+1 \\
 &= \frac{m}{n}.
 \end{aligned}$$

In the fourth line, we used the fact that $(\sqrt{2}+1)(\sqrt{2}-1) = 2-1 = 1$.

This means that, whatever fraction m/n we find for $1+\sqrt{2}$, we can find a fraction n/r taking the same value, but with smaller numerator and denominator. This is impossible!

Another way of saying it is that the equation $m = 2n + r$ is the first step of Euclid's algorithm for finding the greatest common divisor of m and n . But the ratio n/r is the same as the ratio m/n . This will remain true at every step of the algorithm; so the algorithm will never terminate.

This argument can also be expressed geometrically, more in line with the way that Pythagoras (and Euclid) actually thought. You can find this in the [Number Theory notes](#), on page 29.

6.7 Supplement: The irrationality of e

The number e , which is the basis of natural logarithms, is known to be irrational. The proof follows, using an expression for e which you will meet in calculus.

The expression is:

Proposition 6.9

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

This course is not about infinite sums, so I will have to refer you to the Calculus course for this. We will need another sum, the *geometric series*:

Proposition 6.10 If $-1 < r < 1$, then

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}.$$

Given this, here is the proof that e is irrational. As you might expect by now, it is a proof by contradiction.

Let us suppose that e is equal to the rational number p/q . Then $q!e = p(q-1)!$, since the q in the denominator cancels with the factor q in $q!$, and the remaining factors $1, 2, \dots, q-1$ have product $(q-1)!$. In particular, we see that $q!e$ is an integer.

But from the infinite series, we have

$$q!e = q! \sum_{n=0}^{\infty} \frac{1}{n!}.$$

We break this sum into two parts; first the terms from 0 to q , and then the remaining terms.

For the first part,

$$q! \sum_{n=0}^q \frac{1}{n!} = \sum_{n=0}^q \frac{q!}{n!}.$$

Since $n \leq q$, the fraction $q!/n!$ is an integer: it is equal to $(n+1) \cdots q$, since the factors $1, \dots, n$ cancel). So the sum of all these terms is also an integer.

For the second part, we have again to deal with fractions $q!/n!$, this time with $n > q$. Now the factors in the numerator cancel into the denominator, and we have

$$\frac{q!}{n!} = \frac{1}{(q+1)(q+2) \cdots n} \leq \frac{1}{(q+1)^{n-q}},$$

since we have replaced some of the factors in the denominator by smaller ones. So the second part of the sum is

$$\sum_{n=q+1}^{\infty} \frac{q!}{n!} < \sum_{n=q+1}^{\infty} \frac{1}{(q+1)^{n-q}}.$$

The last sum is

$$\frac{1}{q+1} + \frac{1}{(q+1)^2} + \cdots$$

This is a geometric series, whose sum is

$$\frac{1/(q+1)}{1 - 1/(q+1)} = \frac{1}{q} \leq 1.$$

So the integer $q!e$ is the sum of two terms, of which the first one is an integer and the second is a positive number smaller than 1. This is impossible.

The contradiction shows that e is irrational.

Exercises

6.1 Here is an extract from the Introduction to Probability lecture notes.

We say that the events A and B are *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \times \mathbb{P}(B).$$

You may assume that events are independent in the following situations:

- (i) they are clearly physically unrelated (eg depend on different coin tosses),
- (ii) you calculate their probabilities and find that $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$,
- (iii) the question tells you that the events are independent!

- (a) What is this? Is it a definition, a theorem, a proof, an example, or just some chit-chat?
- (b) Your boss asks you to summarise it in two or three brief bullet points, without using any symbols. What would you say?
- (c) Why is the P in the formula written in a different typeface from the other symbols?

6.2 Let a, b, c, d be positive integers, and suppose that $a/b < c/d$. Show that

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

6.3 *Time flies like an arrow, but fruit flies like a banana.*

I like bananas; does that make me a fruit fly?

6.4 A selection of questions from Essential Mathematics. Accurate working is important!

- (a) Evaluate

$$\frac{5}{30} - \frac{1}{7} \times \left(\left(4 - \frac{18}{27} \right) / \frac{8}{3} + \left(\frac{3}{8} \right)^2 \times \left(\frac{7}{4} \times \frac{7}{9} + \frac{5}{12} \right) \right).$$

- (b) Simplify

$$\left(\frac{-ab^2c^3}{cb^3} \right)^3 \left(\frac{a^5}{-b^7c} \right)^{-2}.$$

(c) Compute the remainder when $-x^4 + 3x^2 + 2x - 1$ is divided by $x^2 + 2$.

(d) Add and simplify

$$\frac{1}{y^2 - 2y - 15} + \frac{3}{y^2 - 10y + 25}.$$

(e) Compute $f\left(\frac{-1}{a^3}\right)$, where

$$f(x) = \frac{x^2 - x - 1}{x - 1}.$$

(f) Simplify

$$\frac{1}{\sqrt{5}} \frac{\sqrt{30} - \sqrt{12}\sqrt{15}}{(\sqrt{2} - \sqrt{3})^2}$$

to the form $m + n\sqrt{d}$, where m , n , and d are integers.

(g) Simplify

$$\sqrt{x^2 - x^3} - \sqrt{4 - 4x}.$$

(h) Solve the equation

$$4x - \frac{x^2 - 9}{x + 3} = \frac{6}{3x - 1}.$$

6.5 (Puzzle) You are given a card, with a statement on each side, as shown.

The statement on the other side of this card is true

The statement on the other side of this card is false

Which of the two statements is true, and which is false?

6.6 (Project) Find a counterexample to the statement:

If p is prime, then $2^p - 1$ is prime.

You should explain what a counterexample to this statement is.

Use MAPLE to find as many counterexamples as you can.

Is the converse of the statement true?

Chapter 7

Real numbers

We construct real numbers as infinite decimals in this chapter, and look at their most important property, the Principle of the Supremum. We also look at representing numbers with other bases. The Study skills section looks at the purpose of coursework. In the supplementary material, we ask whether Achilles catches the tortoise, and observe that all positive numbers have square roots.

We turn to the real numbers; we will contain a number system containing $\sqrt{2}$, π , e , and most of the other important numbers of mathematics.

It is possible to build the real numbers from the rational numbers, but the process is more complicated than constructing the integers from the natural numbers. In fact there are two quite different ways of doing it, called *Dedekind cuts* and *Cauchy sequences*. To simplify matters, I will not use either.

As in previous sections, I assume you already have some idea what real numbers are. I will simply build on that knowledge.

7.1 Infinite decimals

We make a preliminary definition; we will have to explore it a bit before we understand it.

Definition 7.1 A *real number* is a number represented by an infinite decimal; that is, we allow infinitely many digits after the decimal point, but only finitely many before it.

For example,

$$\begin{aligned}\frac{1}{2} &= 0.50000000000000000000\dots \\ \frac{1}{3} &= 0.33333333333333333333\dots\end{aligned}$$

$$\begin{aligned}
\frac{1}{7} &= 0.14285714285714285714\dots \\
\sqrt{2} &= 1.41421356237309504880\dots \\
\pi &= 3.14159265358979323846\dots \\
e &= 2.71828182845904523536\dots
\end{aligned}$$

The part before the decimal point represents the integer part. If the digits after the decimal point are $a_1, a_2, a_3, \dots$, then the fractional part is

$$\sum_{n=1}^{\infty} \frac{a_n}{10^n} = \frac{a_1}{10} + \frac{a_2}{100} + \frac{a_3}{1000} + \dots$$

You will meet in Calculus the way in which we give a meaning to an infinite sum like this. That doesn't matter now, since we are *defining* real numbers to be expressions like this.

We also allow negative real numbers, by simply putting a $-$ sign in front.

If the digits are all zero from some point onwards, we don't bother writing them. So, for example, $1/2 = 0.5$. Such a decimal is said to be *terminating*.

What can the digits after the decimal point be? Anything at all, with one small proviso. Since

$$0.999999999999999999\dots = 1$$

(see the Supplementary material), and similarly (for example)

$$0.234569999999999999\dots = 0.23457,$$

we disallow decimals which are all 9 from some point onwards. With this proviso, different decimals represent different numbers.

We already used this representation of real numbers when we proved that the set $\mathbb{R}$ of real numbers is uncountable.

7.2 Decimals for rational numbers

We would like our new number system to include what we already have, the rational numbers. So what do the decimal expansions of rational numbers look like?

A decimal is said to be *periodic* if, from some point on, a sequence of k digits repeats forever. The smallest such number k is called the *period*. A terminating decimal is periodic (since after some point 0 repeats forever), though we often treat this case separately. Formally:

Definition 7.2 A decimal whose digits after the decimal point are $a_1, a_2, a_3, \dots$ is said to be periodic if there exist numbers m and k such that, for all $n \geq m$, we have $a_{n+k} = a_n$.

Proposition 7.3 *A terminating or periodic decimal represents a rational number. Conversely, any rational number is represented by a terminating or periodic decimal.*

Proof In the forward direction, we need the formula for the sum of a *geometric series*: if $-1 < r < 1$, then

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}.$$

Now consider a periodic decimal. Assume for convenience that it is positive. The numbers before the decimal point represent an integer N . After the decimal point, we have $a_{n+k} = a_n$ for $n \geq m$. So we break the decimal into two parts:

- (a) The non-periodic part (up to a_{m-1}). This is a terminating decimal, and its value is $M/10^{m-1}$, where M is the integer represented by the digits $a_1a_2 \dots a_{m-1}$.
- (b) The periodic part (from a_m onwards). Let L be the number represented by the recurring digits $a_ma_{m+1} \dots a_{m+k-1}$. Then the first block of digits represents $L/10^{m+k-1}$. The next block consists of the digits from a_{m+k} to a_{m+2k-1} , and represents $L/10^{m+2k-1}$. So the periodic part adds up to

$$\begin{aligned} \frac{L}{10^{m+k-1}} + \frac{L}{10^{m+2k-1}} + \dots &= \frac{L}{10^{m+k-1}} \left(1 + \frac{1}{10^k} + \frac{1}{10^{2k}} + \dots \right) \\ &= \frac{L}{10^{m+k-1}} \frac{10^k}{10^k - 1} \\ &= \frac{L}{10^{m-1}(10^k - 1)}. \end{aligned}$$

So each of the three parts is a rational number, and their sum is rational.

For example,

$$0.142857142857 \dots = \frac{142857}{999999} = \frac{1}{7}.$$

For the converse, consider any rational number m/n . Again, we lose nothing by assuming that it is positive.

We find the integer part by the division algorithm: $m = nq + r$, so $m/n = q + r/n$, where q is an integer and r/n a fraction less than one. So we only have to deal with the fractional part.

Now you calculate r/n by long division. If you think about the long division algorithm, you realise that at each stage, you divide n into a number smaller than $10n$; the quotient gives you the next digit, and for the remainder you bring down a digit and repeat the operation. After some time, every digit brought down is a 0.

Now there are only a finite number of possibilities for the number being divided by n ; so after a while you will see one that you saw before, and the calculation will simply repeat from this point on.

For example, consider the calculation of $1/7$.

$$\begin{array}{r}
 7 \) \ \underline{1.0000000...} \quad (\ 0.142857 \\
 \underline{7} \\
 30 \\
 \underline{28} \\
 20 \\
 \underline{14} \\
 60 \\
 \underline{56} \\
 40 \\
 \underline{35} \\
 50 \\
 \underline{49} \\
 10
 \end{array}$$

Since we have seen 10 before, the decimal recurs from this point on. □

Our conclusion is: every rational number is a real number; so once again we have extended our number system.

7.3 The Principle of the Supremum

So far, this is all technical detail. The important thing, which is what makes the real numbers so powerful and useful, is a very important property they have, essentially that there are “no gaps” in the real number line. (This contrasts with what happens in the rationals, where there is a gap where the square root of 2 should go.)

We start with a definition.

Definition 7.4 (a) Let S be a subset of $\mathbb{R}$. We say that S is *bounded above* if there is a number $a \in \mathbb{R}$ such that $s \leq a$ for all $s \in S$. Such a number a is called an *upper bound* for S .

(b) Let S be a subset of $\mathbb{R}$. We say that a number a is the *supremum*, or *least upper bound*, of S , if

(i) a is an upper bound for S ;

- (ii) for any positive number ε , no matter how small, the number $a - \varepsilon$ is not an upper bound for S . In other words, any number smaller than a fails to be an upper bound for S .

Note: Mathematicians typically use ε (the Greek letter epsilon) for a small positive number. The mathematician Paul Erdős, whom we met in Chapter 1, used to refer to his friends' children as "epsilons".

For example, suppose that S is the set

$$\{x \in \mathbb{R} : 0 < x < 1\}$$

(the set of all real numbers strictly between 0 and 1). Then S is bounded above: there is no shortage of upper bounds, we can take 2, or 1.5, for example. I claim that the supremum of S is 1. Why? Clearly 1 is an upper bound for S (since all elements $x \in S$ satisfy $x < 1$, by definition). On the other hand, suppose that a is any number less than 1. We can suppose a is positive, since 0 is clearly not an upper bound. Now a is not an upper bound, since we can find elements of S which are larger than a : for example, the number $(1 + a)/2$, halfway between a and 1, belongs to S but is greater than a .

In this case, the supremum of S is not a member of S . There are other cases where the supremum does belong to S . For example, if S is the set

$$\{x \in \mathbb{R} : 0 < x \leq 1\},$$

then 1 is the supremum, but belongs to S . In fact, if the supremum of S is an element of S , then it is the greatest element (can you see why?)

I have been talking about "the supremum": this is OK, since there cannot be more than one. If a_1 and a_2 were both suprema, and $a_1 < a_2$, then $a_1 = a_2 - \varepsilon$ for some positive ε ; so by definition, a_1 cannot be an upper bound for S , and certainly cannot be the supremum, contradicting our assumption.

Does every set have a supremum? No, a set may have no upper bound at all (like the set of all real numbers), or it may have arbitrarily small upper bounds (like the empty set). These are the only exceptions. This important result is called the *Principle of the Supremum*.

Theorem 7.5 (Principle of the Supremum) *Let S be a non-empty set of real numbers which is bounded above. Then S has a supremum.*

Proof This will be a "proof by example". The method used here works in general. I will take the set S to be

$$S = \{x \in \mathbb{R} : x^2 < 2\}$$

and finding the supremum of S is effectively the same as finding the square root of 2.

First we note that S is bounded: in fact every element of S is less than 2. But not every element is less than 1, since $1.4 \in S$. So we will start our construction with 1. (In general, take any integer upper bound for S , and successively decrease it by 1 until for the first time it is no longer an upper bound; stop there.)

Now we calculate the first decimal place. We consider 1.1, 1.2, $\dots$. By squaring these, we find that 1.1, $\dots$, 1.4 are not upper bounds for S , but 1.5 is. So the next digit in the decimal is 4.

To calculate the second decimal place, we do a similar calculation. We check that 1.41 is not a lower bound but 1.42 is. So the next digit is 1.

By continuing this process, we calculate a decimal expansion representing a number a . We claim that a is the supremum. We have to show two things:

- (a) a is an upper bound;
- (b) No smaller number is an upper bound.

(a) Suppose that a is not an upper bound. That means that there is some element $x \in S$ with $x > a$. Now $x^2 < 2$, so $a^2 < 2$. This means that at some point we could have chosen a larger digit and still had an element which was not an upper bound, contradicting the way we constructed a .

(b) The second part is easy. Any number smaller than a will have a smaller digit at some point, and we chose a so that any such number fails to be an upper bound. $\square$

We write the supremum of the set S as $\sup(S)$.

What about lower bounds: We can make corresponding definitions for them:

Definition 7.6 (a) Let S be a subset of $\mathbb{R}$. We say that S is *bounded below* if there is a number $a \in \mathbb{R}$ such that $s \geq a$ for all $s \in S$. Such a number a is called a *lower bound* for S .

(b) Let S be a subset of $\mathbb{R}$. We say that a number a is the *infimum*, or *greatest lower bound*, of S , if

- (i) a is a lower bound for S ;
- (ii) for any positive number ε , no matter how small, the number $a + \varepsilon$ is not a lower bound for S . In other words, any number bigger than a fails to be a lower bound for S .

We have a similar theorem:

Theorem 7.7 (Principle of the Infimum) *Let S be a non-empty set of real numbers which is bounded below. Then S has an infimum.*

You can probably imagine that we could go through a similar proof to establish this. But in fact we can make use of what we already did:

Proof Let U be the set of lower bounds for S . Then U is non-empty, and bounded above (by any element of S), and so U has a supremum. Now it is an easy exercise to show that the supremum of U is the infimum of S . $\square$

We denote the infimum of S by $\inf(S)$.

7.4 Some useful functions

Here are some useful functions for dealing with real numbers.

The floor function If you put a positive real number into the black box for the floor function, it strips away the decimal part and returns just the integer part. The floor of the number x is written $\lfloor x \rfloor$. The function is sometimes called the “round down” function. The precise definition is

The *floor* of x is the largest integer not exceeding x .

So, for example, $\lfloor 2.4 \rfloor = 2$ and $\lfloor -2.4 \rfloor = -3$.



Suppose that you are in a tower block, where the distance between floors is 1 (in some units). If your height above the ground is x , then the height of the floor on which you are standing is $\lfloor x \rfloor$. (In Europe, this will be the number of the floor you are on; but in North America, you have to add one to the height to get the floor number.)

If x is an integer, then $\lfloor x \rfloor = x$.

The ceiling This function rounds up, and gives you the height of the ceiling above your head. The definition is

The *ceiling* of x is the smallest integer which is not smaller than x .

It is written $\lceil x \rceil$. Thus, $\lceil 2.4 \rceil = 3$ and $\lceil -2.4 \rceil = -2$.

The modulus The black box for this function simply strips away the sign of the real number you enter, and returns the positive value (or 0, if you input 0). The modulus of x is written $|x|$. The definition is

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Maximum and minimum The *maximum* of two numbers is simply the larger of the two. It is written $\max(x, y)$, and the definition is:

$$\max(x, y) = \begin{cases} x & \text{if } x \geq y, \\ y & \text{otherwise.} \end{cases}$$

Similarly the minimum is the smaller:

$$\min(x, y) = \begin{cases} x & \text{if } x \leq y, \\ y & \text{otherwise.} \end{cases}$$

So $\max(2.4, -5.6) = 2.4$, $\min(2.4, -5.6) = -5.6$, $\max(2.4, 2.4) = \min(2.4, 2.4) = 2.4$.

These can be extended to more than two numbers. The maximum of any finite set of numbers is the largest number in the set, and the minimum is the smallest. They can be calculated by comparing two at a time; for example,

$$\max(x, y, z) = \max(\max(x, y), z).$$

An infinite set of numbers doesn't always have a maximum or minimum, as we saw: for example, the set $\{x \in \mathbb{R} : 0 < x < 1\}$ has neither. However, if it is bounded above, then it has a supremum, and often we can use this instead.

A quick question for you: is it true that $|x| = \max(x, -x)$ for any real number x ?

7.5 Other bases

We have used infinite decimals to represent real numbers, so that

$$0.a_1a_2a_3\dots = \sum_{n=1}^{\infty} a_n 10^{-n}.$$

Why do we use 10 in this formula?

In fact, any positive integer greater than 1 would do equally well as the *base* for our representation. Let b be an integer, with $b > 1$. Suppose that we have symbols $s_0, s_1, s_2, \dots, s_{b-1}$ to represent the digits $0, 1, 2, \dots, b-1$. Then we can represent any real number using these symbols.

First, we let the sequence $s_{i_m}s_{i_{m-1}} \dots s_{i_1}s_{i_0}$ represent the natural number

$$i_mb^m + i_{m-1}b^{m-1} + \dots + i_1b + i_0.$$

We assume that the first digit i_m is not zero (except in the special case where we are representing the number 0, which is done by the digit s_0).

For example, if we use the usual digits $0, 1, \dots, 6$ in base 7, then the number represented by 135 is $1 \times 7^2 + 3 \times 7 + 5$, which is 75 (in the usual base 10).

To represent a positive real number, we write the integer part in front of the decimal point, as above, and then an infinite sequence of digits after the decimal point, where

$$.s_{j_1}s_{j_2}s_{j_3} \dots = \sum_{n=1}^{\infty} j_nb^{-n}.$$

There is an ambiguity, like the problem with decimals ending with infinitely many 9s. In this case, a “decimal” ending with infinitely many s_{b-1} s is equal to what we obtain by increasing the digit before them by 1 and adding an infinite string of s_0 s (which we usually omit).

A couple of questions might occur to you at this point.

How can we be sure that the real numbers represented in a different base are “the same” as the ones we are familiar with?

We can transform one representation into the other. Take a base b “decimal”, say

$$0.s_{j_1}s_{j_2}s_{j_3} \dots = \sum_{n=1}^{\infty} j_nb^{-n}.$$

Now follow the usual procedure to convert this into base 10.

Why do we use base 10?

Clearly this has something to do with the fact that we have ten fingers. We count on our fingers; and we call the symbols $0, 1, 2, \dots, 9$ *digits*, a word that also means “fingers”.

But there is also an element of historical accident about it. The system invented in India, which was adopted by the Islamic world, then Europe, and then worldwide, happened to use base 10. Other cultures have used other bases: for example, 5, or 20, or 60.

Also, we can remember and recognise 10 digits without too much effort. The Babylonians, who used base 60, were forced to use a compound symbol for each of their 60 digits.

Computers naturally work in base 2, where there are just two digits, which we usually take to be 0 and 1. A computer is essentially a collection of interconnected switches, and a switch can be in one of two positions, off or on, represented by the two digits.

This is very simple, but the disadvantage is that the representations of integers are more than three times as long as those in base 10. For example, 2012 (in base 10) is 11111011100 (in base 2). To make their code more readable by humans, computer scientists often use base 8 (octal) or base 16 (hexadecimal). The octal digits are $0, \dots, 7$, and the hexadecimal digits are $0, \dots, 9, a, b, c, d, e, f$.

It is very easy to convert between binary (base 2) and either of these bases. For example, to convert 11111011100 into hexadecimal:

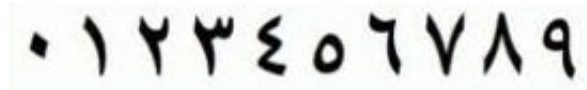
- (a) Break the string into blocks of four from the right, padding the start with 0s if necessary: 0111, 1101, 1100.
- (b) Calculate the number represented by each block, and replace it by the corresponding digit: 7, 13, 12 become 7, d , c .

So the answer is $7dc$. Indeed, $7 \times 16^2 + 13 \times 16 + 12 = 2012$.

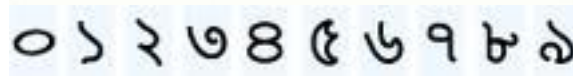
Could we use different symbols for the digits?

Yes, and indeed many people do. Here are a couple of examples.

Arabic:



Bengali:



7.6 Study skills: The purpose of coursework

You learn better by doing. In particular, you learn to understand the abstract things you hear about in lectures by using them to solve problems. This is the main purpose of coursework.

You will read in the handbook that coursework questions are designed to be like exam questions. What the handbook doesn't tell you is that there is a big difference between these things. Coursework gives you practice to help you understand the material; exams test your understanding. For example, some coursework

questions will include drill in the use of a particular technique; an exam question might have just a single example.

There is another big difference, which we will come to later when we talk about taking exams.

Although you do not get numerical marks for your coursework solutions, you should not think that we don't regard them as important. We expect you to attack each question seriously; it is not enough just to look at the question and think "I know how to do that", and turn to the next question. Write down your solution, and compare it to mine later. It probably won't be the same; there are many different ways of solving a maths problem. But check carefully, to make sure that you haven't missed something important. And remember that it is not enough to come up with the right answer, or write down a string of symbols connected by $=$ or $\Leftrightarrow$; you must explain what you are doing. Your solution must be clear, grammatical and logical, as well as mathematically correct.

The purpose of tutorials is not for tutors to tell you how to solve the problems. They are there to answer your questions. But if your question is about the meaning of a word which is defined in the lecture notes, expect the tutor to say, "Look it up in the notes".

You can also ask your questions about material in the lectures which you didn't understand.

7.7 Supplement: Achilles and the tortoise

Achilles, one of the Greek heroes of the Trojan war, has a race with a tortoise. Because Achilles runs ten times as fast as the tortoise, he agrees to give it a head start of, say, nine metres. The two contestants start at the same moment. Does Achilles catch the tortoise?



Image from Science2.0

Let us suppose that Achilles runs ten metres in a second, while the tortoise runs one metre. Common sense says that, after one second, Achilles has run ten metres and the tortoise one, so Achilles has made up the nine-metre deficit and caught the tortoise.

However, consider the following argument.

- (a) After 0.9 seconds, Achilles has run 9 metres and reached the point from which the tortoise started. But the tortoise has crawled 0.9 metres in this time, and is still ahead.
- (b) After another 0.09 seconds, Achilles has run 0.9 metres and again reached the point where the tortoise was. But the tortoise has crawled 0.09 metres in this time, and is still ahead.
- (c) After another 0.009 seconds, Achilles has run 0.09 metres and again reached the point where the tortoise was. But the tortoise has crawled 0.009 metres in this time, and is still ahead.
- (d) And so on.

The conclusion is that Achilles never catches the tortoise, since when he reaches the last point the tortoise was, the tortoise has moved on a bit.

This is a very famous paradox dating from classical times. Personally, I can't really see the point of it, but philosophers have taken it very seriously. It is not enough simply to say "Achilles catches the tortoise after one second"; you have to say what is wrong with the argument given.

From our point of view, the time Achilles takes to catch the tortoise is

$$0.9 + 0.09 + 0.009 + \dots = 0.999\dots$$

and, of course, $0.999\dots = 1$, so the two analyses of the problem agree.

But if you search the Internet, you will find that there are many people in the world who do not believe that $0.999\dots = 1$. How would you answer them?

Here is a quick argument. Divide both sides by 3:

$$\begin{aligned} 0.999\dots/3 &= 0.333\dots, \\ 1/3 &= 0.333\dots \end{aligned}$$

If two numbers divided by 3 give the same answer, they must be equal!

Do you find that convincing?

Ultimately the problem is to do with what we mean by the sum of an infinite series.

7.8 Supplement: Square roots

In the system of rational numbers, there is no square root of 2. Let's see that we have at least fixed this problem by extending the rationals to the real numbers:

Proposition 7.8 *Every non-negative real number has a real square root.*

Proof There is an algorithm for finding the square root of a number in decimal notation, which churns out one digit at a time.

I learned this algorithm in primary school. To be more precise: I attended a small country school where a single teacher took years 4–8. The teacher took a dislike to me, for some reason; so one day when I was off sick, he taught the class square roots, and when I returned, set a test on them. I had to figure it out on the spot. I still rate this as one of my best mathematical achievements.

The basic idea is very simple, although there are refinements to make the calculations easy enough for a primary school pupil. I will illustrate by finding the square root of $1/7$.

We know that

$$\frac{1}{7} = 0.142857142857 \dots$$

Let x be its square root.

- $0.3^2 = 0.09$ and $0.4^2 = 0.16$, so $x = 0.3 \dots$,
- $0.37^2 = 0.1369$ and $0.38^2 = 0.1444$, so $x = 0.37 \dots$,
- $0.377^2 = 0.142129$ and $0.378^2 = 0.142884$, so $x = 0.377 \dots$,

and so on. The numbers are found by trial and error. In the last step I showed, you can square 0.370, 0.371, 0.372, $\dots$, and repeat until the answer is bigger than the target; then the previous step tells you what the next digit is.

Clearly, starting with any decimal number, this procedure will generate the decimal digits of its square root. $\square$

7.9 Supplement: A problem

There is a difficulty with the representation of numbers by decimals, which I have not told you about. This difficulty is the reason why, in more advanced treatments, you will meet different constructions of the real numbers, by *Dedekind cuts* or *Cauchy sequences*. The problem comes when we try to add or multiply infinite decimals, since we can't "start at the right".

Problem What is the first digit after the decimal place in

$$0.414213562 \dots + 0.585786437 \dots ?$$

Adding the two numbers given, we obtain $0.999999999 \dots$. But we don't know whether, further down the line, we will come upon two digits adding up to more than 10, in which case the answer would be $1.000000000 \dots$.

Exercises

7.1 Non-mathematicians often have a lot of trouble with the assertion

$$0.999\dots = 1.$$

- (a) Is the assertion true or false?
- (b) Someone taking the opposite view to yours accosts you at a party. What argument would you use to convince them that you are correct?

7.2 (a) Draw the graph of the function $f(x) = |x|$, identifying the points where it takes the value 0.

- (b) Same question for $f(x) = |1 - |x||$.
- (c) Same question for $f(x) = |1 - |1 - |x|||$.
- (d) Now generalise!

7.3 (a) Show that, if p is a prime number, then $\sqrt{p}$ is irrational.

- (b) Show that $\sqrt[3]{2}$ is irrational.
- (c) Can you generalise these two facts?

7.4 (Puzzle) The *logarithm to base 10* of a number x , written $\log_{10} x$, is the number y such that $10^y = x$. Show that there do not exist natural numbers m and n such that

$$\log_{10} 2 = \frac{m}{n}.$$

7.5 (Project) The *maximum* of two numbers x and y is defined by the rule

$$\max(x, y) = \begin{cases} x & \text{if } x \geq y, \\ y & \text{otherwise.} \end{cases}$$

- (a) Give a definition of the *minimum* $\min(x, y)$ of two numbers x and y .
- (b) Prove that $\max(x, y) = \max(y, x)$.
- (c) Prove that $\min(\max(x, y), y) = y$.
- (d) Show that $\min(x, y) < a$ if and only if either $x < a$ or $y < a$.

- (e) Littlewood, in his book *Littlewood's Miscellany*, gives the following curious piece of legal history:

There was a Rent Act after 1914, and the definition of when a house was subject to it was as follows. The 'standard rent' was defined to be the rent in 1914, unless this was less than the rateable value, in which case it was to be the rateable value. The house is subject to the Act if and only if either the standard rent or the rateable value is less than £105.

Letting R be the standard rent, R_0 the rent in 1914, and V the rateable value, show that

- (i) $R = \max(R_0, V)$;
- (ii) the house is subject to the Act if and only if $\min(R, V) < 105$.

Deduce that the house is subject to the Act if and only if its rateable value is less than £105.

- (f) Prove that $\max(x, y) + \min(x, y) = x + y$.
- (g) There are many other theorems concerning the maximum and minimum functions. Make one up yourself and give a proof of it.

Chapter 8

Complex numbers

In this chapter, we reach the last of our important number systems, the complex numbers. You will learn

- (a) the construction of the complex numbers;
- (b) their geometric representation in the Argand diagram;
- (c) the Fundamental Theorem of Algebra.

The Study skill points out that examples do not make a proof. In the supplementary material we show that every quadratic equation over the complex numbers can be solved, and discuss Euler's famous formula $e^{i\pi} = -1$.

The final extension arises because there are still equations we can't solve, such as $x^2 = -1$ (which has no real solution) or $x^3 = 2$ (which has only one, though for various reasons we would like it to have three). It turns out that the first equation is the crucial one.

8.1 The square root of minus one

Definition 8.1 A *complex number* is a number of the form $a + bi$, where a and b are real numbers, and i is a mysterious symbol which will have the property that $i^2 = -1$. The rules for addition and multiplication are

$$\begin{aligned}(a + bi) + (c + di) &= (a + c) + (b + d)i, \\ (a + bi)(c + di) &= (ac - bd) + (ad + bc)i.\end{aligned}$$

The rule for multiplication comes out by just expanding in the usual way and using $i^2 = -1$:

$$(a + bi)(c + di) = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i.$$

Subtraction and division (except by zero) also work for complex numbers. You can work out the rule for subtraction. How do we divide? You can check that the rule above gives

$$(a + bi)(a - bi) = a^2 + b^2,$$

which is a positive number unless $a = b = 0$. So, to divide by $a + bi$, we multiply by

$$\left(\frac{a}{a^2 + b^2} \right) - \left(\frac{b}{a^2 + b^2} \right) i.$$

Thus, in the complex numbers, we can add, subtract, multiply, and divide (except by zero), and the laws we met earlier all apply here too.

Complex numbers are not called complex because they are complicated: a modern advertising executive would certainly have come up with a different name! They are called “complex” because each complex number is built of two parts, each of which is simpler (being a real number).

Here, for example, is a proof of the distributive law. Let $z_1 = a_1 + b_1i$, $z_2 = a_2 + b_2i$, and $z_3 = a_3 + b_3i$. Now

$$\begin{aligned} z_1(z_2 + z_3) &= (a_1 + b_1i)((a_2 + a_3) + (b_2 + b_3)i) \\ &= (a_1(a_2 + a_3) - b_1(b_2 + b_3)) + a_1(b_2 + b_3) + b_1(a_2 + a_3)i, \end{aligned}$$

and

$$\begin{aligned} z_1z_2 + z_1z_3 &= ((a_1a_2 - b_1b_2) + (a_1b_2 + a_2b_1)i) + ((a_1a_3 - b_1b_3) + (a_1b_3 + a_3b_1)i) \\ &= (a_1a_2 - b_1b_2 + a_1a_3 - b_1b_3) + (a_1b_2 + a_2b_1 + a_1b_3 + a_3b_1)i, \end{aligned}$$

and a little bit of rearranging shows that the two expressions are the same.

Definition 8.2 If $z = a + bi$ is a complex number (where a and b are real), we say that a and b are the *real part* and *imaginary part* of z respectively. The complex number $a - bi$ is called the *complex conjugate* of z , and is written as $\bar{z}$.

So the rules for addition and subtraction can be put like this:

To add or subtract complex numbers, we add or subtract their real parts and their imaginary parts.

The rule for multiplication looks more complicated as we have written it out. There is another representation of complex numbers which makes it look simpler.

Definition 8.3 Let $z = a + bi$ be a non-zero complex number. The *modulus* and *argument* of z by

$$\begin{aligned} |z| &= \sqrt{a^2 + b^2}, \\ \arg(z) = \theta &\text{ where } \cos \theta = a/|z| \text{ and } \sin \theta = b/|z|. \end{aligned}$$

In other words, if $|z| = r$ and $\arg(z) = \theta$, then

$$z = r(\cos \theta + i \sin \theta).$$

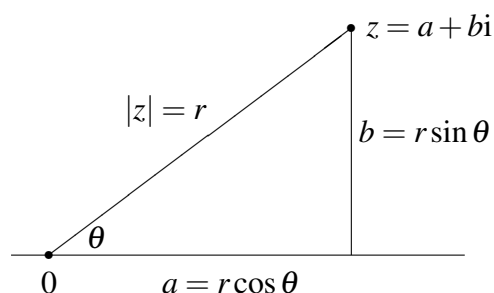
Now the rules for multiplication and division are:

To multiply two complex numbers, multiply their moduli and add their arguments. To divide two complex numbers, divide their moduli and subtract their arguments.

Remark The definition of modulus of a complex number agrees with the definition in the last chapter of the modulus of a real number. The argument of a positive real number is 0; that of a negative real number is π (or 180°).

8.2 The complex plane, or Argand diagram

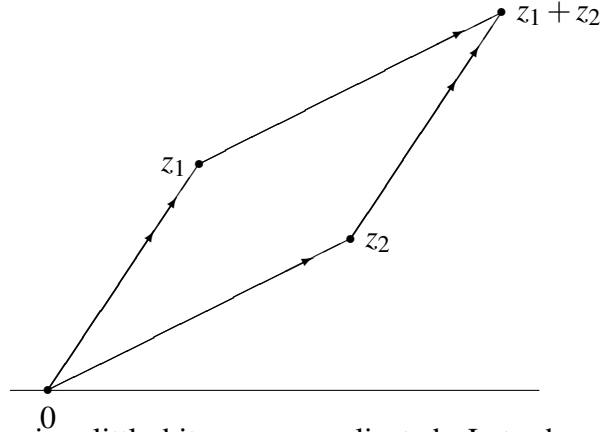
The complex numbers can be represented geometrically, by points in the Euclidean plane (which is usually referred to as the *Argand diagram* or the *complex plane* for this purpose). The complex number $z = a + bi$ is represented as the point with coordinates (a, b) . Then $|z|$ is the length of the line from the origin to the point z , and $\arg(z)$ is the angle between this line and the x -axis. The next diagram shows this.



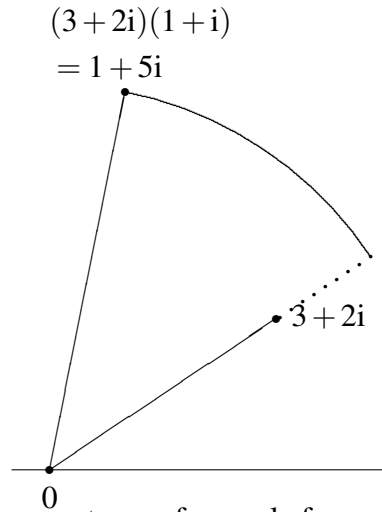
In terms of the complex plane, we can give a geometric description of addition and multiplication of complex numbers. The addition rule is the *parallelogram rule* which you will meet again for vectors in Geometry I, next semester. It states:

Draw lines from the origin to the points representing the two complex numbers z_1, z_2 to be added. Construct the parallelogram having these lines as two of its sides. Then the point opposite the origin represents $z_1 + z_2$.

This is illustrated in the next diagram.



Multiplication is a little bit more complicated. Let z be a complex number with modulus r and argument θ , so that $z = r(\cos \theta + i \sin \theta)$. Then the way to multiply an arbitrary complex number by z is a combination of a stretch and a rotation: first we expand the plane so that the distance of each point from the origin is multiplied by r ; then we rotate the plane through an angle θ . In the next diagram, we are multiplying by $1 + i = \sqrt{2}(\cos(\pi/4) + i \sin(\pi/4))$; the dots represent the stretching out by a factor of $\sqrt{2}$, and the circular arc represents the rotation by $\pi/4$.



Now let's check the correctness of our rule for multiplying complex numbers. Remember that the rule is: to multiply two complex numbers, we multiply the moduli and add the arguments. To see that this is correct, suppose that z_1 and z_2 are two complex numbers; let their moduli be r_1 and r_2 , and their arguments $\theta_1 + \theta_2$. Then

$$\begin{aligned} z_1 &= r_1(\cos \theta_1 + i \sin \theta_1), \\ z_2 &= r_2(\cos \theta_2 + i \sin \theta_2). \end{aligned}$$

Then

$$\begin{aligned} z_1 z_2 &= r_1 r_2 (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= r_1 r_2 ((\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + (\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2)i) \\ &= r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)), \end{aligned}$$

which is what we wanted to show.

From this we can prove *De Moivre's Theorem*:

Theorem 8.4 (De Moivre's Theorem) *For any natural number n , we have*

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

Proof The proof is by induction. Starting the induction is easy since $(\cos \theta + i \sin \theta)^0 = 1$ and $\cos 0 + i \sin 0 = 1$.

For the inductive step, suppose that the result is true for n , that is,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

Then

$$\begin{aligned} (\cos \theta + i \sin \theta)^{n+1} &= (\cos \theta + i \sin \theta)^n \cdot (\cos \theta + i \sin \theta) \\ &= (\cos n\theta + i \sin n\theta)(\cos \theta + i \sin \theta) \\ &= \cos(n+1)\theta + i \sin(n+1)\theta, \end{aligned}$$

which is the result for $n+1$. So the proof by induction is complete.

Note that, in the second line of the chain of equations, we have used the inductive hypothesis, and in the third line, we have used the rule for multiplying complex numbers. $\square$

The argument is clear if we express it geometrically. To multiply by the complex number $(\cos \theta + i \sin \theta)^n$, we rotate n times through an angle θ , which is the same as rotating through an angle $n\theta$.

De Moivre's Theorem is useful in deriving trigonometrical formulae. For example,

$$\begin{aligned} \cos 3\theta + i \sin 3\theta &= (\cos \theta + i \sin \theta)^3 \\ &= (\cos^3 \theta - 3 \cos \theta \sin^2 \theta) + (3 \cos^2 \theta \sin \theta - \sin^3 \theta)i, \end{aligned}$$

so

$$\begin{aligned} \cos 3\theta &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta, \\ \sin 3\theta &= 3 \cos^2 \theta \sin \theta - \sin^3 \theta. \end{aligned}$$

These can be converted into the more familiar forms $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ and $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ by using the equation $\cos^2 \theta + \sin^2 \theta = 1$.

8.3 The Fundamental Theorem of Algebra

We enlarged the real numbers to the complex numbers by adding a square root of minus one, a solution of the equation $x^2 + 1 = 0$. It turns out that we have, incidentally, provided solutions for a huge class of equations.

Definition 8.5 (a) A *polynomial* (over some number system) is an expression of the form

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where $a_n, a_{n-1}, \dots, a_1, a_0$ belong to the number system under consideration, and the *leading coefficient* a_n is not equal to zero. The number n is the *degree* of the polynomial.

(b) A *root* of the above polynomial is a number r (in the appropriate number system) such that

$$a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = 0.$$

For example, $r = 1$ and $r = 2$ are roots of the polynomial $x^3 - 7x + 6$.

Note: Polynomials have a kind of double existence. On the one hand, they are just algebraic expressions; the x in the polynomial doesn't have any real existence, it is just a placeholder. On the other hand, a polynomial defines a function on the number system in question: putting a value into the black box labelled by the polynomial simply evaluates the polynomial with x put equal to this value, as we saw in Chapter 4.

We enlarged our number systems successively to find roots of various polynomials: $bx - a$ (to build $\mathbb{Q}$), $x^2 - 2$ (to build $\mathbb{R}$), $x^2 + 1$ (to build $\mathbb{C}$). Maybe we have to do more complicated constructions to find roots of other polynomials? No, we have reached the end:

Theorem 8.6 (Fundamental Theorem of Algebra) *Any polynomial over $\mathbb{C}$ of degree at least 1 has a root in $\mathbb{C}$.*

The first rigorous proof of this theorem was given by Gauss. Indeed, he gave many different proofs of the theorem: ten, according to Michael Atiyah. He clearly felt very proud of this!

There is another account of it [here](#) on Theorem of the Day.

In the supplementary material, I will show you something much weaker: at least we can solve all quadratic equations.

8.4 Study skills: Examples don't make a proof

How would you prove the following theorem?

Theorem All odd numbers are prime.

You should know by now that the argument

- 3 is prime,
- 5 is prime,
- 7 is prime,
- and so on . . .

is not a proof. If an argument contains the words “and so on” or a row of dots, you should immediately suspect that there is a proof by induction going on somewhere. But there is no way to prove this theorem by induction: the fact that 7 is prime tells you nothing about whether 9 is prime.

For more examples of how to prove that all odd numbers are prime, look at [this web page](#).

In particular, note the confused undergraduate's proof:

Let p be any prime number larger than 2. Then p is not divisible by 2, so p is odd. □

What is the mistake here? We discussed this in an earlier study guide.

It is a very common mistake to think that a few examples make a proof (as above). Take care!

We saw earlier the definition of *Mersenne numbers*: these are numbers of the form $2^p - 1$, where p is prime. Now

- $2^2 - 1 = 3$ is prime,
- $2^3 - 1 = 7$ is prime,
- $2^5 - 1 = 31$ is prime,
- $2^7 - 1 = 127$ is prime,
- but $2^{11} - 1 = 2047 = 23 \times 89$.

If you have a good, general example of the thing that has to be proved, then with a bit of luck you can turn it into a general proof. Indeed, this is what mathematicians did before they came up with the idea of proof in the modern sense. Here, for example, is how they would have solved a quadratic equation in ancient Egypt.

Example Solve the quadratic $x^2 + 16 = 10x$.

- (a) Calculate the square of half the coefficient of x : $5^2 = 25$.
- (b) Subtract the constant term from this: $25 - 16 = 9$.
- (c) Take the square root of this: $\sqrt{9} = 3$.
- (d) Add and subtract this from half the coefficient of x : $5 + 3 = 8$, $5 - 3 = 2$.
- (e) The solutions are $x = 8$ and $x = 2$. Substitute them in and you will find that this works.

I hope you can write down a general proof that the method works.

8.5 Supplement: Solving quadratic equations

Proposition 8.7 *Every quadratic equation over the complex numbers has a solution. In other words, every polynomial of degree 2 has a root.*

Everyone knows the formula for the solution of the quadratic equation $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

You probably learned this formula in the case where a, b, c are real numbers. Does it work over $\mathbb{C}$?

The easiest way to show that it does is simply to substitute the two values of x into the quadratic equation and simplify. I will just take the positive sign here; the argument for the negative sign is similar. If $x = (-b + \sqrt{b^2 - 4ac})/2a$, then

$$\begin{aligned} ax^2 + bx + c &= \frac{b^2 - 2b\sqrt{b^2 - 4ac} + (b^2 - 4ac)}{4a} + \frac{b(-b + \sqrt{b^2 - 4ac})}{2a} + c \\ &= \frac{b^2 + b^2 - 2b^2 - 4ac + 4ac + (-2b + 2b)\sqrt{b^2 - 4ac}}{4a} \\ &= 0. \end{aligned}$$

In the second line, we have simply put everything over the common denominator $4a$. The manipulations that we do in simplifying the expression use the various laws (commutative, associative, distributive) that hold in the complex numbers just as they do in the real numbers. So the answer is, yes, the formula is valid.

The only problem is: does the square root $\sqrt{b^2 - 4ac}$ exist?

Yes it does:

Lemma 8.8 *Every complex number z has a square root.*

There are two ways to see this.

The easier way is to use the modulus-argument form. We can suppose that $z \neq 0$ (since 0 certainly has a square root), so we can write

$$z = r(\cos \theta + i \sin \theta).$$

Then, if we put $w = \sqrt{r}(\cos \theta/2 + i \sin \theta/2)$, we see that $w^2 = z$, and we have found our square root.

But how do we know that $\sqrt{r}$ exists? Well, r is a positive real number, and we showed in the supplementary material for the last chapter of the notes that every positive real number has a square root.

The second method works just with the real and imaginary parts. We are given a complex number $z = a + bi$, and we want to find $w = x + yi$ such that

$$(x + yi)^2 = a + bi,$$

in other words,

$$\begin{aligned} x^2 - y^2 &= a, \\ 2xy &= b. \end{aligned}$$

Now a, b, x, y are real numbers, so this is just a question about real numbers.

We can assume that $b \neq 0$. For if $b = 0$, we are trying to find the square root of a real number a . If $a \geq 0$, we know that this exists; and if a is negative, say $a = -c$, then $c > 0$ and we can take $(\sqrt{c})i$ to be the square root. Now, if $b \neq 0$, the second equation shows that $x \neq 0$, so $y = b/2x$.

Substituting in the first equation, we get

$$x^2 - \left(\frac{b}{2x}\right)^2 = a,$$

so (clearing the denominators),

$$4x^4 - 4ax^2 - b^2 = 0.$$

Putting $u = x^2$, we have a quadratic equation for u , namely $4u^2 - 4au - b = 0$, which has the solution

$$u = \frac{a \pm \sqrt{a^2 + b^2}}{4}.$$

Now $a^2 + b^2 > 0$, so $\sqrt{a^2 + b^2}$ exists as a real number. Moreover, $a^2 + b^2 > a^2$, so $\sqrt{a^2 + b^2} > |a|$; so, if we choose the positive sign to describe u , that is,

$$u = \frac{a + \sqrt{a^2 + b^2}}{32},$$

then $u > 0$, and so we can find $x = \sqrt{u}$. Knowing x , we can find $y = b/2x$, which is also OK since $x \neq 0$.

This is a special property of the complex numbers. Look at [this account of the quadratic formula](#) on Theorem of the Day. The problem is to make a frame for Euler's formula, which we will discuss next, to display in a gallery of beautiful mathematics. You will see that quadratic equations over the real numbers (even those arising in practice) don't always have solutions.

8.6 Supplement: $e^{i\pi} = -1$

The formula $e^{i\pi} = -1$, due to Euler, is one of the most famous in all of mathematics. It connects the negative unit -1 , the imaginary unit i , and the two most famous mathematical constants π and e .

So of course you would like to see a proof.

Well, I can't prove it. After all, you have never had a definition of raising a real number to an imaginary power; and without that, we can't even start!

In fact the formula follows from a more general formula,

$$e^{ix} = \cos x + i \sin x.$$

Substitute $x = \pi$, and observe that $\cos \pi = -1$ and $\sin \pi = 0$.

This formula can be justified on the basis of consistency. But this argument can be very powerful.

First, it has sensible conclusions. Just before De Moivre's Theorem in the notes, we did a calculation to show that

$$(\cos x_1 + i \sin x_1)(\cos x_2 + i \sin x_2) = \cos(x_1 + x_2) + i \sin(x_1 + x_2).$$

On the other hand, the laws of exponentiation would say that

$$e^{ix_1} \cdot e^{ix_2} = e^{i(x_1 + x_2)}.$$

(Of course, we don't know that the laws of exponentiation hold when the exponent is a complex number; but we would like this to happen.) So at least the proposed formula doesn't disagree with what we know.

The second argument is stronger, in my view. One of the most important things about complex numbers (which we won't discuss in this module, but you

will meet if you study Complex Variables) is that we can extend ordinary calculus to them. The rewards are very rich. Quantum mechanics (the theory which underlies all our consumer electronics, among many other things) rests on the fact that we can do calculus over the complex numbers, and the usual rules apply. The most famous equation in quantum mechanics is *Schrödinger's equation*, which mixes derivatives with complex numbers. Among many other applications is fluid mechanics, where the calculus of complex numbers helps us study the flow of air over aeroplane wings.

Enough of the commercial: how will calculus over the complex numbers help us?

It turns out that, to make calculus work properly, we have to assume that the power series we use to express functions over the real numbers continue to apply over the complex numbers.

Three very famous power series, which you definitely need to know, are

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots, \\ \cos x &= \sum_{m=0}^{\infty} (-1)^m \frac{x^{2m}}{(2m)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \\ \sin x &= \sum_{m=0}^{\infty} (-1)^m \frac{x^{2m+1}}{(2m+1)!} = x - \frac{x^3}{3!} + \cdots. \end{aligned}$$

So our *assumption* will be that these series continue to hold true when we let x take a complex value. In fact, we can regard this assumption as a *definition* of what e^x , $\cos x$ and $\sin x$ mean when x is a complex number.

Now we are going to calculate e^{ix} , by substituting ix for x in the series. The powers i^n cycle through the values $1, i, -1, -i$ as n runs through the indices $0, 1, 2, 3, \dots$. So the even-numbered terms will be real and the odd-numbered terms imaginary, and we can separate the real and imaginary parts:

$$\begin{aligned} e^{ix} &= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) + \left(x - \frac{x^3}{3!} + \cdots\right)i \\ &= \cos x + i \sin x, \end{aligned}$$

as required.

Exercises

8.1 Use De Moivre's Theorem to express $\cos 4\theta$ as a polynomial in $\cos \theta$, and to express $\sin 4\theta$ as a polynomial in $\sin \theta$.

8.2 Suppose that the Euclidean plane is rotated in the anticlockwise direction through an angle θ . Prove (using complex numbers or otherwise) that the point (x, y) in the plane moves to the point $(x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$.

8.3 The Fibonacci numbers F_n , for $n \geq 1$, are defined by

$$F_1 = 1, \quad F_2 = 2, \quad F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 3.$$

Write down the *last digits* of the first few Fibonacci numbers. Does the sequence of last digits ever repeat? If so, after how many steps?

Could you have predicted that the sequence will repeat without calculating the number of steps?

8.4 (Puzzle) You have two mugs, one containing coffee and the other containing milk, in exactly equal quantities. You transfer three spoonsful of milk from the first mug to the second, and stir thoroughly. Then you transfer three spoonsful of the milky coffee in the second mug to the first, and stir thoroughly.

Is there more milk in the coffee than coffee in the milk, or emphvice versa?

8.5 (Project) In Calculus, you will meet the notion of a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$. You don't need to know the definition to do this question, but you will need to know one property: if the graph of a continuous function contains a point on either side of a line, then between these two points it contains a point on the line.

- (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Suppose that there exist real numbers x and y with $x < y$ such that $f(x) = y$ and $f(y) = x$. Show that there exists a real number z with $x < z < y$ such that $f(z) = z$.
- (b) A monk sets out from his monastery at 9am to journey to a hermitage on a mountain, where he arrives at 6pm. (You should not assume that he travels at a constant speed.) He meditates all night, and leaves at 9am the next morning to travel back to the monastery along the same route that he took the day before. He arrives at 6pm. Show that there is a point on the route which the monk passes at the same time on both days. You should state carefully any assumptions you make when you translate this description into mathematics.
- (c) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Suppose that there exist real numbers x, y, z with $x < y < z$ such that $f(x) = y$, $f(y) = z$ and $f(z) = x$. Show that there exists a real number w such that $f(w) = w$.

- (d) (Harder question) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Suppose that there exist real numbers x, y, z with $x < y < z$ such that $f(x) = y$, $f(y) = z$ and $f(z) = x$. Show that there exist real numbers u and v with $u \neq v$ such that $f(u) = v$ and $f(v) = u$.

Chapter 9

Proofs

This chapter is about proofs. We define some technical terms: the contrapositive, and the converse; and we look at proofs by contradiction and the role of counterexamples in proofs. We also see how proofs can be made up of a number of steps.

The Study skills are about taking examinations. The supplementary material ties up a loose end in the chapter, and describes truth tables and their connection with sets.

Most of mathematics consists in starting with various assumptions and trying to argue to some conclusion. A basic step on the way is “implication”, which we will study in some detail. If the statement “ P implies Q ” is true, then from P we can deduce Q .

9.1 Implication

The definition appears a little bit odd, at first glance.

Definition 9.1 Let P and Q be statements. Then $P \Rightarrow Q$ is true (read “ P implies Q ”) if either P is false or Q is true.

There are two strange things about this definition. First, it says nothing about the fact that Q is deduced from P by some mathematical or logical argument. Second, the definition itself looks odd: is this what we usually mean by “implies”?

The statement “ $P \Rightarrow Q$ ” can be read in other ways; the most important of these is “if P , then Q ”. Now it is possible to see that the everyday meaning of “if P , then Q ” agrees with the definition we gave. Suppose, for example, that I say to you,

“If it’s fine tomorrow, then I will take you to the Zoo.”

The only circumstance in which I have told you a lie is if it is fine and we don't go to the Zoo. If we go to the Zoo, then my statement is OK; and, if it rains all day, my statement is also OK. So the statement is true if either it rains tomorrow or we go to the Zoo; this is precisely what the definition requires.

This means that, to prove a conditional statement “if P , then Q ”, we may assume that P is true and try to prove that Q is true. Here is an example.

Proposition 9.2 *If n is even, then n^2 is even.*

Proof Remember that a number n is even if it can be written in the form $n = 2k$ for some natural number k . So suppose that $n = 2k$. Then $n^2 = 4k^2 = 2l$, where $l = 2k^2$; so n^2 is even. $\square$

This is the kind of argument that you would probably give without thinking twice.

There is yet another form of words we use to describe $P \Rightarrow Q$: we say that P is a *sufficient condition* for Q . For knowing that P holds is sufficient for us to deduce that Q also holds.

Sometimes we are more interested in the relationship the other way round. if $P \Rightarrow Q$, then we can also write $Q \Leftarrow P$, the reversed arrow showing that the implication is from right to left. We also state this as “ P only if Q ”, or “ Q is a *necessary condition* for P ”. In the last one, we have changed our point of view: P cannot be true unless Q is true, so Q is necessary for P .

To summarise, the following six statements all have exactly the same meaning:

- (a) $P \Rightarrow Q$ (that is, P implies Q);
- (b) $Q \Leftarrow P$ (that is, Q is implied by P);
- (c) if P , then Q ;
- (d) P only if Q ;
- (e) P is a sufficient condition for Q ;
- (f) Q is a necessary condition for P .

9.2 Necessary and sufficient

What about $P \Leftrightarrow Q$? This formula means that both the implications $P \Rightarrow Q$ and $Q \Rightarrow P$ both hold. There are three statements which all mean the same thing here:

- (a) $P \Leftrightarrow Q$;

- (b) P if and only if Q ;
- (c) P is a necessary and sufficient condition for Q .

To prove $P \Leftrightarrow Q$, we have to prove both the implications $P \Rightarrow Q$ and $Q \Rightarrow P$. As we saw earlier, it is not a good idea to write a long chain of statements with $\Leftrightarrow$ between each pair; you write it down forwards, and check that the forward implications hold, but it is all too easy to forget to check the backward implications as well.

Here is an example.

Proposition 9.3 n is even if and only if n^2 is even.

Proof We already saw that, if n is even, then n^2 is even. So we have to prove the other implication: if n^2 is even, then n is even.

This requires a new principle, which we have used many times during this module but haven't yet stated formally: *proof by contradiction*.

Suppose that n is not even; then it is odd, that is, $n = 2k + 1$ for some number k . Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2l + 1$$

where $l = 2k(k + 1)$; so n^2 is also odd, that is, not even. (We saw that every number is either even or odd but not both; so “odd” means the same as “not even”.) But this contradicts our assumption that n^2 is even. So our assumption that n is not even is incorrect; that is, n is even, as we had to show. $\square$

9.3 The contrapositive

Now we have to look more closely at what was going on at the end of the last proof.

Theorem 9.4 The statement $P \Rightarrow Q$ is logically equivalent to the statement $(\text{not } Q) \Rightarrow (\text{not } P)$.

Proof We know that $P \Rightarrow Q$ is true in all cases except when P is true and Q is false.

Hence $(\text{not } Q) \Rightarrow (\text{not } P)$ is true in all cases except where $(\text{not } Q)$ is true and $(\text{not } P)$ is false; that is, where Q is false and P is true. But this is exactly the case when $P \Rightarrow Q$ is false. So the two statements are true in exactly the same situations. $\square$

Definition 9.5 The statement $(\text{not } Q) \Rightarrow (\text{not } P)$ is the *contrapositive* of the statement $P \Rightarrow Q$.

Thus, an implication and its contrapositive are logically equivalent, and proving one is exactly the same thing as proving the other.

Consider again the second half of the proof in the last section. We had to show that n^2 even implies n even; we chose instead to prove the contrapositive form, n odd implies n^2 odd.

So, to summarise, three jobs are all equivalent, and proving any one of them shows the others as well:

- (a) $P \Rightarrow Q$, which is proved by assuming that P holds and deducing that Q holds;
- (b) the contrapositive $(\text{not } Q) \Rightarrow (\text{not } P)$, which is proved by assuming that $(\text{not } Q)$ holds (that is, that Q is false) and deducing that $(\text{not } P)$ holds (that is, that P is also false);
- (c) arguing by contradiction, assume that P is true and Q is false, and reach an impossibility. (This is how we actually phrased the argument in the last section.)

9.4 The converse

The *converse* of an implication is simpler to write down than the contrapositive, but it is *not* logically the same as the original statement.

Definition 9.6 The *converse* of the implication $P \Rightarrow Q$ is the implication $Q \Rightarrow P$.

Thus, the converse of an implication is the implication “the other way round”.

For example, let $P(n)$ be the statement “ n is prime”, and $Q(n)$ the statement “ n is odd”. When restricted to numbers greater than 2, the implication $P(n) \Rightarrow Q(n)$ is true, but the converse implication $Q(n) \Rightarrow P(n)$ is false, as shown by the case $n = 9$.

This is what the 19th century logician Lewis Carroll had to say about a statement and its converse. Alice is at the Mad Hatter’s Tea Party, and the Hatter has just asked her a riddle: “Why is a raven like a writing-desk?”

‘Come, we shall have some fun now!’ thought Alice. ‘I’m glad they’ve begun asking riddles.—I believe I can guess that,’ she added aloud.

‘Do you mean that you think you can find out the answer to it?’ said the March Hare.

‘Exactly so,’ said Alice.

‘Then you should say what you mean,’ the March Hare went on.

‘I do,’ Alice hastily replied; ‘at least—at least I mean what I say—that’s the same thing, you know.’

‘Not the same thing a bit!’ said the Hatter. ‘You might just as well say that “I see what I eat” is the same thing as “I eat what I see”!’ ‘You might just as well say,’ added the March Hare, ‘that “I like what I get” is the same thing as “I get what I like”!’ ‘You might just as well say,’ added the Dormouse, who seemed to be talking in his sleep, ‘that “I breathe when I sleep” is the same thing as “I sleep when I breathe”!’

‘It is the same thing with you,’ said the Hatter, and here the conversation dropped, and the party sat silent for a minute, while Alice thought over all she could remember about ravens and writing-desks, which wasn’t much.

9.5 Longer proofs

Implications, of the sort we have discussed, are the basic ingredients of mathematical proofs. But sometimes we have to string long chains of them together.

If we could prove the implications $P \Rightarrow R_1, R_1 \Rightarrow R_2, R_2 \Rightarrow R_3, \dots, R_{n-1} \Rightarrow R_n$, and $R_n \Rightarrow Q$, then we can deduce that $P \Rightarrow Q$. For assume that P holds. Then from $P \Rightarrow R_1$, we can deduce R_1 ; then we can deduce R_2 ; and so on; and finally we deduce Q .

The reason that finding proofs is hard is that, if you are asked to prove that P implies Q (or, in other words, to assume P and prove Q), it is not at all obvious what all the intermediate statements $R_1, \dots, R_n$ should be.

One very useful technique that can be combined with this is the idea of “working back” from the conclusion. You reason as follows:

I could prove Q if I could only prove R ; now I could prove R if I could only prove S ; now I could prove S if I could only prove T ; ... aha! I see how to prove T !

Here is an example.

Let $a_1, a_2, \dots, a_n$ be positive real numbers. Their *arithmetic mean* is just their average: add them up and divide by n . Their *geometric mean* is found by multiplying the numbers together and taking the n th root. If all the numbers are equal, say $a_1 = \dots = a_n = a$, then the arithmetic and geometric means are both equal to a .

Theorem 9.7 *Let $a_1, a_2, \dots, a_n$ be positive real numbers. Then their arithmetic mean is greater than or equal to their geometric mean. Equality holds if and only if all the numbers are equal.*

Here, we analyse the problem for $n = 2$. The general case is a bit harder; see the supplementary material.

Let the two numbers be a and b . We are trying to prove that

$$\frac{a+b}{2} \geq \sqrt{ab}.$$

This would follow if we could show that

$$\left(\frac{a+b}{2}\right)^2 \geq ab,$$

since we could just take the square root. This would follow if we could show that

$$(a+b)^2 \geq 4ab,$$

since we could simply divide by 4. Now this last inequality can be transformed into

$$\begin{aligned} a^2 + 2ab + b^2 &\geq 4ab, \text{ which follows from} \\ a^2 - 2ab + b^2 &\geq 0, \text{ in other words} \\ (a-b)^2 &\geq 0. \end{aligned}$$

But the last statement is true, since every perfect square is greater than or equal to zero.

So we have the proof, but we still have to write it down clearly; that means, writing the steps in the logical order, not in the order we found them. So write “Proof”, and see what happens:

Proof

$$\begin{aligned} (a-b)^2 &\geq 0 && \text{(since any square is positive or zero)} \\ a^2 - 2ab + b^2 &\geq 0 && \text{(expanding)} \\ a^2 + 2ab + b^2 &\geq 4ab && \text{(adding } 4ab \text{ to each side)} \\ (a+b)^2 &\geq 4ab \\ \left(\frac{a+b}{2}\right)^2 &\geq ab && \text{(dividing by 4)} \\ \frac{a+b}{2} &\geq \sqrt{ab} && \text{(taking the square root)} \end{aligned}$$

which was to be proved.

Moreover, if the argument shows that, if $(a+b)/2 = \sqrt{ab}$, then each $\geq$ in the proof can be replaced by $=$, and we see that $(a-b)^2 = 0$, so that $a = b$, which proves the last part of the theorem. $\square$

You should take special care with this kind of argument. We have worked back from the conclusion to the hypothesis; if you are not careful, you will find that you have proved the converse of what you set out to prove! There is an example of this error in the supplementary material for the next chapter.

9.6 Examples and counterexamples

What about disproving an implication $P \Rightarrow Q$? We have seen that this implication is false if P is true and Q is false.

Suppose we are trying to decide on the truth of a general statement $P \Rightarrow Q$. Here P and Q might be statements about all natural numbers, for example.

A proof of this implication would have to be a general proof, which works for every possible natural number.

On the other hand, if the implication is false, we only have to find one number n for which $P(n)$ is true and $Q(n)$ is false. Such a number is called a *counterexample* to the implication $P \Rightarrow Q$.

For example, the implication “if n is odd, then n is prime” is false; the number $n = 9$ is a counterexample (since it is odd but not prime). Similarly, the implication “if p is prime, then $2^p - 1$ is prime” is false; the number $p = 11$ is a counterexample, as we have seen.

Often a mathematician is in the position of not knowing whether an implication is true (so that she should look for a general proof) or false (so that she should try to find a counterexample).

In Chapter 1, we saw that Euler disproved Fermat’s conjecture that $2^{2^n} + 1$ is prime for all natural numbers n , by finding a counterexample (he showed that 641 is a divisor of $2^{2^5} + 1$).

9.7 Divide and conquer

Sometimes the best way to prove something is to divide the proof into separate cases. If we are sure that any possible situation is covered by (at least) one of the cases, and we can prove our conclusion true in every case, then we are finished.

One example of this strategy might be a question asking you to prove that two real numbers x and y satisfy $x < y$. We know that there are only three possibilities, namely $x < y$; $x = y$; or $x > y$; if we can show that the second and third cannot occur, then we must have the first.

Here is an example. Suppose that you have just seen the proof that, if $x < y$, then $x + z < y + z$, and you are asked to prove the converse:

Theorem 9.8 *If $x + z < y + z$, then $x < y$.*

Proof Assume that $x + z < y + z$. We will show that $x = y$ and $x > y$ are both impossible, so can conclude that $x < y$.

Suppose that $x = y$. Then $x + z = y + z$, which is contrary to our assumption.

Suppose that $x > y$, that is, $y < x$. Then, by what we already proved, it follows that $y + z < x + z$, which again contradicts the assumption. $\square$

Remark Remember that the converse of a statement is not the same as the original statement; there is always some work to do to prove it.

Here is a different example. I will prove a famous theorem known as *Fermat's Little Theorem*. (Same Fermat, different theorem!)

Theorem 9.9 (Fermat's Little Theorem) *Let p be a prime number, and let n be any integer. Then p divides $n^p - n$.*

Proof We divide the proof into three cases:

- (a) $p = 2$;
- (b) $n \geq 0$;
- (c) $n < 0$ and $p > 2$.

Clearly these cases cover any possibility, so if we can prove the theorem in all three cases we are done.

Case (a): $p = 2$. In this case $n^2 - n = n(n - 1)$. Of the two consecutive integers n and $n - 1$, one is even and the other odd; so their product is even, whence $2 \mid n^2 - n$.

Case (b): $n \geq 0$. The proof in this case is by induction on n . Thus, let $P(n)$ be the statement that p divides $n^p - n$. For $n = 0$, we have $0^p - 0 = 0$, and p certainly divides 0, so $P(0)$ is true.

Suppose that $P(n)$ is true, so that p divides $n^p - n$. Now we have to look at $(n + 1)^p - (n + 1)$. We can expand this, using the Binomial Theorem:

$$\begin{aligned} (n + 1)^p - (n + 1) &= n^p + \binom{p}{1}n^{p-1} + \binom{p}{2}n^{p-2} + \cdots + \binom{p}{n-1}n + 1 - (n + 1) \\ &= (n^p - n) + \binom{p}{1}n^{p-1} + \binom{p}{2}n^{p-2} + \cdots + \binom{p}{p-1}n. \end{aligned}$$

By the inductive hypothesis, p divides $n^p - n$. Also, p divides each binomial coefficient $\binom{p}{i}$ for $i = 1, 2, \dots, p - 1$. For

$$\binom{p}{i} = \frac{p!}{i!(p-i)!},$$

and there is a factor p in the numerator but no factor p in the denominator to cancel it.

So p divides all the terms in the expression for $(n + 1)^p - (n + 1)$, and so we have verified $P(n + 1)$.

Thus the proof by induction is complete.

Case (c): $n < 0$ and $p > 2$. In this case, let $n = -m$, so that $m > 0$. Now

$$n^p - n = (-m)^p - (-m) = -m^p - (-m) = -(m^p - m).$$

Here we used the fact that $p > 2$ to deduce that p is odd and hence that $(-m)^p = -m^p$.

Now p divides $m^p - m$ by the second case, so p divides $-(m^p - m) = n^p - n$, and we are finished. $\square$

9.8 Other methods

Induction is a very important technique. Many properties of the natural numbers can be proved by induction, and it is useful in many other places too.

If you have a problem involving a natural number n , it is always worth seeing whether you can make a proof by induction on n . But beware: it doesn't always work!

Cantor's diagonal method Don't be afraid to use other people's good idea. Suppose you were asked to prove the following theorem:

Theorem 9.10 *The set of infinite sequences of 0s and 1s is uncountable.*

This looks very much like a place where Cantor's method can apply. Suppose that you have a countable list of all the sequences, and create a new sequence which differs from all of them.

9.9 A puzzle

On this [Web page](#), Emily Allman poses the following problem:

It's 6am and thirty lockers stand in a long lovely row, closed and neat, ready for the opening of another school day.

- The first student arrives and, what the heck, opens every locker!
- The second student arrives and changes the position of every second locker (i.e. lockers 2, 4, 6, 8, etc all got closed).
- The third student arrives and changes the position of every third locker (from open to closed or vice versa)
- The fourth student changes the position of every fourth locker
- ...

- This continues until the 30th student arrives and changes the position of locker #30.

What is the final configuration of the lockers after all 30 students have passed by?

Try this for yourself; we will talk about how to attack it in the next chapter.

9.10 Study skills: In the exam room

You take a maths module in order to understand the material. In an ideal world, that would be enough; at the end, you would understand it, and everybody would realise that you did. Unfortunately the real world is not like that. We need to have examinations in order that you can demonstrate your understanding; your degree certificate will prove that you have indeed understood the material you studied.

In the section about coursework, I promised to explain another difference between coursework and exams. When you get a coursework sheet, you have a week to tackle it; you have access to your lecture notes, the library, the Internet, your friends, and any other resources that might be useful to you. In the exam, on the other hand, you have none of these resources, and you only have two hours to display what you know. So you will find that exam questions tend to be more straightforward than coursework questions. If you have been able to understand and do the coursework, the exam should hold no terrors for you.

There are many tips about taking exams. Here are just a few.

- (a) Don't stay up late the night before cramming stuff into short-term memory! An exam is partly an endurance test (you have to be able to concentrate for two hours without a break), and you should be fit and rested before you go into the exam room.
- (b) Give the examiners value for money. If you are spending too long on one question, leave it and try another. (But see point (e) also.) If a question is worth five marks and you have only written a couple of words, stop and think: maybe the examiner wanted more detail here.
- (c) An exam question can contain three kinds of material:
 - Bookwork (definitions or statements of theorems).
 - Routine material which you have seen during the module.
 - Questions designed to make you think, probably unseen.

Often all three will occur in the same question. Mathematics is about thinking and reasoning; so it is natural that we should expect you to do this in the exam rather than simply reproduce facts.

- (d) In many questions, the solution to one part depends on the parts before it. So, if you are stuck on part (b) of the question, leave it and go on to part (c), being prepared to assume what you were asked to do in (b).
- (e) If you are running out of time, or if the details of something escape you, write down an outline of how you would answer the question if you had the time. You could get a significant proportion of the marks for doing this.
- (f) If a question asks for a definition, give the definition as precisely and concisely as you can. Don't say "It's like when ...", and don't give an example unless one is asked for.
- (g) If you have some spare time at the end of the exam, don't just sit there; switch your brain from creative mode to critical mode, and read carefully what you have written. If you find an error, point it out, even if you don't have time to put it right.

9.11 Supplement: Arithmetic and geometric mean

Theorem 9.11 *The arithmetic mean of n positive real numbers is at least as great as their geometric mean. Equality holds if and only if all the numbers are equal.*

Proof The proof is by induction. The case $n = 1$ is straightforward (if there is only one number, then it is equal to both its arithmetic mean and its geometric mean); and we proved it in the notes for $n = 2$. So the induction certainly starts.

So let $P(n)$ be the statement that the arithmetic mean of any n positive real numbers is at least as great as their geometric mean. Assume that $P(n)$ is true, and take $n + 1$ positive real numbers $a_1, \dots, a_{n+1}$; let α be their arithmetic mean. We are going to replace two of the numbers by a single number, so that the arithmetic mean is not changed; then we show that the geometric mean is decreased, which will prove the inductive step.

If the numbers are all equal, then the theorem is true; so we may suppose not. Then, at least one of the numbers, say a_n , is greater than α ; and at least one, say a_{n+1} , is smaller than α . So $(a_n - \alpha)(\alpha - a_{n+1}) > 0$. I will call this equation (*), since I will need to refer back to it.

Let $b = a_n + a_{n+1} - \alpha$. Note that $b > 0$ since $a_n > \alpha$. Consider the n numbers $a_1, \dots, a_{n-1}, b$. Since $(n + 1)\alpha = a_1 + \dots + a_{n+1}$, we have

$$n\alpha = a_1 + \dots + a_{n-1} + (a_n + a_{n+1} - \alpha) = a_1 + \dots + a_{n-1} + b,$$

so α is the arithmetic mean of these n numbers. By the induction hypothesis, it is at least as big as their geometric mean $(a_1 a_2 \cdots a_{n-1} b)^{1/n}$, and so

$$\alpha^{n+1} = \alpha^n \cdot \alpha \geq a_1 a_2 \cdots a_{n-1} b \alpha.$$

But from equation (*), we know that

$$b\alpha - a_n a_{n+1} = (a_n + a_{n+1} - \alpha)\alpha - a_n a_{n+1} = (a_n - \alpha)(\alpha - a_{n+1}) > 0,$$

so $b\alpha > a_n a_{n+1}$. Then we have

$$\alpha^{n+1} > a_1 a_2 \cdots a_{n-1} a_n a_{n+1},$$

and (at last!) the proof is finished. $\square$

9.12 Supplement: Truth tables

Truth tables are a convenient way of dealing with simple logical relations between statements. The truth of each of the combinations P and Q , P or Q , $P \Rightarrow Q$ and $P \Leftrightarrow Q$ only depend on the truth or falsity of P and Q , and we can express this dependence in tables as shown. Here T and F stand for “true” and “false” respectively.

P	Q	P and Q	P	Q	P or Q
T	T	T	T	T	T
T	F	F	T	F	T
F	T	F	F	T	T
F	F	F	F	F	F

P	Q	$P \Rightarrow Q$	P	Q	$P \Leftrightarrow Q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	T	F
F	F	T	F	F	T

These tables capture the informal definitions

- (a) P and Q is true if and only if both P and Q are true;
- (b) P or Q is true if and only if either P or Q is true (or possibly both);
- (c) $P \Rightarrow Q$ is true in all cases except when P is true and Q false;
- (d) $P \Leftrightarrow Q$ is true if both $P \Rightarrow Q$ and $Q \Rightarrow P$ are true.

There is also a truth table for “not”, which is simpler since only one statement is involved:

P	not P
T	F
F	T

Using these truth tables, one can check that certain expressions are logically equivalent. For example, P and Q is equivalent to $(\text{not } P) \Rightarrow Q$.

There is also a connection with set theory. Let A and B be two sets, and let P be the statement $x \in A$ and Q the statement $x \in B$. Then P and Q is the statement $x \in (A \cap B)$, and P or Q is the statement $x \in (A \cup B)$. Thus, we can use truth tables to prove identities about sets. Work out for yourself what sets correspond to $P \Rightarrow Q$ and $P \Leftrightarrow Q$.

The connection between “and” and intersection, “or” and union, has led us to use a similar notation for these logical connectives:

- (a) P and Q is written as $P \wedge Q$;
- (b) P or Q is written as $P \vee Q$.

Exercises

9.1 All the statements below were given as answers to test questions by students. For each statement, explain what (if any) mistakes it contains, and write down what you would say to the person who wrote it to clear up their misunderstandings.

- (a) We have to prove the statement is true for $n = n + 1$.
- (b) one-to-one if the function produces only one unique value from $A \rightarrow B$; onto if the function produces more than one unique value.
- (c) The contrapositive of the statement

“If n is an even integer greater than 2, then n is the sum of two prime numbers.”

is

“If n is an odd integer less than 2, then n is the sum of two non-prime numbers.”

9.2 One of the following is not a function; which one, and why? In the other cases, is the function one-to-one? is it onto?

- (a) The function $F : \mathbb{C} \rightarrow \mathbb{R}$ given by $F(z) = \arg(z)$.
- (b) The function $F : \mathbb{R} \rightarrow \mathbb{R}$ given by $F(x) = x^3 - x$.
- (c) The function $F : \mathbb{R} \rightarrow \mathbb{R}$ given by $F(x) = e^x$.

9.3 Is this proof valid?

Theorem $\cos^2 x + \sin^2 x = 1$.

Proof Let $f(x) = \cos^2 x + \sin^2 x$. Differentiating, $f'(x) = 2\cos x(-\sin x) + 2\sin x \cos x = 0$. Hence $f(x)$ is constant. But $f(0) = 1$; so $f(x) = 1$ for all x . $\square$

9.4 (Puzzle) What are the last two digits of the number obtained by multiplying together all the odd numbers between 1 and 10000?

9.5 (Project) Let r be an irrational real number greater than 1. The *Beatty sequence* B_r is the sequence of natural numbers whose n th term is $\lfloor nr \rfloor$.

For example, if $r = \sqrt{2} = 1.41421\dots$, then the terms of B_r are

$$\begin{aligned} \lfloor 1.41421\dots \rfloor &= 1, \lfloor 2.82843\dots \rfloor = 2, \lfloor 4.24264\dots \rfloor = 4, \lfloor 5.65685\dots \rfloor = 5, \\ \lfloor 7.07107\dots \rfloor &= 7, \lfloor 8.48528\dots \rfloor = 8, \lfloor 9.89949\dots \rfloor = 9, \dots \end{aligned}$$

The gaps in the sequence, namely 3, 6, 10, $\dots$, form another interesting sequence.

- (a) *Rayleigh's Theorem* asserts that the gaps in the Beatty sequence B_r form the Beatty sequence B_s , where $1/r + 1/s = 1$. Find a proof of this theorem and express it in your own words.
- (b) In the case where $r = \sqrt{2}$, let p_n be the n th term in the sequence B_r , and q_n the n th term in the sequence of gaps. Prove that $q_n = p_n + 2n$.

Chapter 10

More on proofs

In this final chapter, we turn to more practical matters: How do you go about proving something yourself? And how can proofs be wrong, and how can you spot that? The Study skills mark the end of the course and simply wish you well with your mathematics in the future. The supplementary material has more examples of proofs for you to consider.

10.1 Constructing proofs

It is difficult for me to teach you how to construct proofs. The best I can do is to give you examples, and derive general principles from them. The best you can do is practice!

[This web page](#) contains stories that famous mathematicians have told about how they went about finding proofs.

Here are a few examples and tips.

10.1.1 How to start

If the question asks you to prove something, it is very important to write down clearly what you are assuming and what you have to prove. One of my colleagues suggests writing the assumptions at the top of the page and the conclusion at the bottom, and then thinking about how to fill in the blank space between.

10.1.2 Contrapositive and contradiction

We already discussed this, but it is one of the most powerful techniques available, and always worth a try. If you are trying to prove that P implies Q , or if you are

given that P holds and trying to prove that Q holds, try to prove that not Q implies not P , or that P together with not Q yield a contradiction.

10.1.3 Counterexample

If you are trying to prove a general statement, then you have to give a general proof. However, to disprove a general statement, a specific counterexample is enough.

For example, consider *Goldbach's Conjecture* from Chapter 1: every even integer greater than 2 is the sum of two primes. To prove this, you would have to give an argument which works for every single even integer. But to show that it is false, it would suffice to give one even number which could not be expressed as the sum of two primes. (Nobody has succeeded in either of these tasks yet.)

10.1.4 Proof both ways round

Remember that to prove that $P \Leftrightarrow Q$, or that P is a necessary and sufficient condition for Q , you have to prove both the implications $P \Rightarrow Q$ and $Q \Rightarrow P$.

Sometimes, if you are lucky, you can prove $P \Rightarrow Q$, and then just reverse the steps to get the other implication. But *don't count on this*: usually it is not so simple and sometimes it goes badly wrong. It is best to think of the two proofs as entirely separate.

You may remember from Chapter 2 that there is an important situation where this occurs: *proving that two sets are equal*. Remember that

- (a) $A \subseteq B$ means that $(x \in A) \Rightarrow (x \in B)$; and
- (b) $A = B$ means that $A \subseteq B$ and $B \subseteq A$.

So to prove that $A = B$, you have to show both that $(x \in A) \Rightarrow (x \in B)$ and also that $(x \in B) \Rightarrow (x \in A)$.

10.1.5 Proving a disjunction

Suppose you are trying to prove that one of two possible alternatives, say Q and R , holds (maybe assuming some hypothesis P). How do you do it?

The statement “ Q or R ” is true if one or other of Q and R (possibly both) is true. It is logically equivalent to “not Q implies R ”. So you can prove it by adding not Q to the hypotheses and trying to prove R .

Some people worry about the argument, since you never seem to show that Q might actually occur. So let us have an example.

Theorem 10.1 *Let a and b be real numbers. If $ab = 0$, then $a = 0$ or $b = 0$.*

Proof According to the above, we assume that $ab = 0$ and that $a \neq 0$. Then the inverse a^{-1} exists; this is a number c with the property that $ca = 1$. Then

$$b = 1 \cdot b = (ca)b = c(ab) = c \cdot 0 = 0,$$

and we are done. □

Here is an explanation of what is going on. You are told that $ab = 0$, and required to fill in a form where you tick at least one of the two boxes:

$a = 0$	☐
$b = 0$	☐

When you have ticked a box, you have finished your job and can hand in the form. So let us consider your options.

If $a = 0$, then you can tick the first box and hand in the form.

If $a \neq 0$, then the argument in the proof of the theorem applies, and shows that $b = 0$; so you can tick the second box and hand in the form.

There are other strategies too. If you are trying to prove that either Q or R holds, you might assume that they are both false and try to reach a contradiction. (The negation of “either Q or R holds” is “ Q and R both fail to hold”.) Yet another approach is illustrated by the solution of the party problem in the supplementary material.

10.1.6 Induction

Simple examples of induction start at 0 or 1; usually it is easy to prove the base case of the induction and harder to prove the inductive step. But sometimes both of these assumptions have to be questioned.

Recall that $n!$ (“ n factorial”) is the product of all the integers from 1 to n inclusive.

Theorem 10.2 *Let k be a positive integer. Then for all but finitely many positive integers n , we have $n! > k^n$.*

This is a problem about natural numbers, so it looks like a proof by induction. So let us try. The problem with the base case is that we don’t know where to start, since the statement is allowed to fail for the first few values: for example,

$$\begin{aligned} 1! = 1 &\leq 2^1 = 2, \\ 2! = 2 &\leq 2^2 = 4, \\ 3! = 6 &\leq 2^3 = 8. \end{aligned}$$

But the inductive step is easy. Let $P(n)$ be the statement $n! > k^n$. Remember that n can be as large as we like, so let us assume that $n > k$ and that $P(n)$ holds; we will prove that $P(n+1)$ holds. We have

$$(n+1)! = (n+1) \cdot n! > (n+1) \cdot k^n > k \cdot k^n = k^{n+1},$$

where the second step uses the induction hypothesis and the third step uses the fact that $n \geq k$. So $P(n+1)$ holds.

However, starting the induction is much more difficult. For any specific value of k , we can do this by calculation. For example, with $k = 2$, we have $4! = 24 > 16 = 2^4$, so the statement holds with $n = 4$, and hence for all $n \geq 4$.

I will leave it to you to try to start the induction. A couple of alternative proofs of the theorem appear in the supplementary material.

10.1.7 Proof by exhaustion

This doesn't refer to the fact that doing mathematics is a very tiring occupation!

Sometimes you reach a stage in the argument where only a finite number of choices are possible. In this case, you can simply try them all; either one of them works and the result comes out, or none works and what you are trying to show is impossible.

Sudoku players will certainly recognise this situation. I recommend to you an [article on Sudoku, mathematics and logic](#) by Catherine Greenhill, which discusses how several proof techniques arise in solving Sudoku puzzles.

10.1.8 Working back from the conclusion

This is another very important technique in constructing proofs. You are trying to reach some conclusion Q , and you guess that a number of steps will be required to take you from where you are (your assumptions) to Q . There is an example in the preceding section, where we applied this technique to prove the Arithmetic-Geometric Mean inequality for two numbers.

10.2 Solution to the lockers puzzle

How do you attack a puzzle like the one about the lockers that I set in the last chapter? The first two tips are, work an example, and take a simple case. Let us replace 30 by 10. We make a table with the locker numbers along the top and students down the side; each time a student changes a locker (opens a closed

locker or closes an open one) we put a star. This is what we get.

	1	2	3	4	5	6	7	8	9	10
1	*	*	*	*	*	*	*	*	*	*
2		*		*		*		*		*
3			*			*			*	
4				*				*		
5					*					*
6						*				
7							*			
8								*		
9									*	
10										*

Since all the lockers are closed at the start of the day, a column with an even number of stars will mean that locker is closed at the end, whereas an odd number of stars mean that it will be open. From the table we see that lockers 1, 4 and 9 are open, the remainder are closed.

Next, is there a pattern? Our first guess is that the lockers whose numbers are perfect squares are open, the rest are closed.

Finally, is there a reason for this; can we explain it? This is the part where we have to do some mathematics. Again, take an example.

Look at locker number 10. This is changed by students numbers 1, 2, 5 and 10. These are all the numbers that divide 10. Now it is clear why this must be so; student number n changes all the lockers which are multiples of n , so locker m is changed by the students whose numbers divide m .

Now to say that 2 divides 10 means that $10 = 2 \times 5$, and so 5 also divides 10. So the divisors of 10 pair up: $10 = 1 \times 10$ and $10 = 2 \times 5$, and there are an even number of them.

Is this true for any number? Let's look at 9. It is changed by students 1, 3 and 9. now $9 = 1 \times 9$, so 1 and 9 pair up; but $9 = 3 \times 3$, and there is nobody else to pair up with 3. So the numbers that have unpaired divisors are the squares.

Now we understand why it is true, the final job is to write it down as a mathematical argument (this means, a convincing argument).

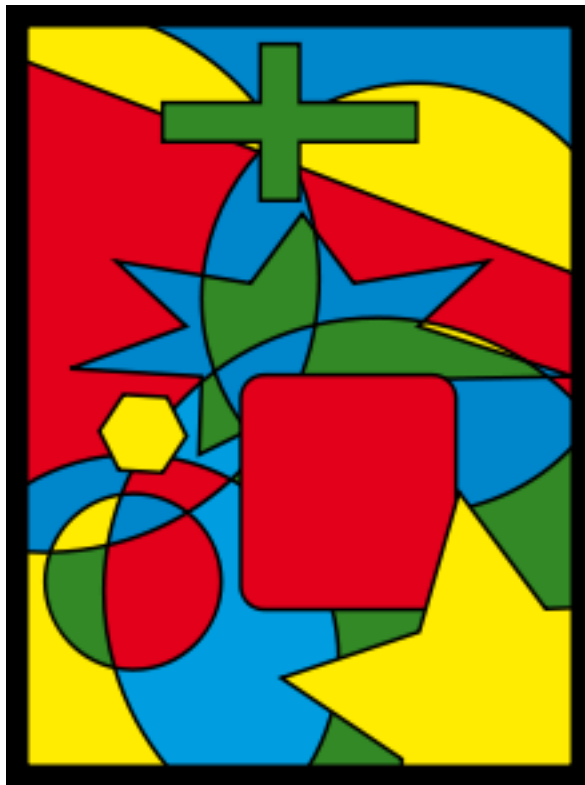
10.3 Debugging false proofs

In the final section of the notes, we look at the other aspect of proofs. I am going to present you with a few proofs which don't stand up to examination. Read them carefully and try to find the flaw before you read my commentary on them.

Anything called a theorem in this section of the notes should be taken with a very large grain of salt. You will do yourself no favours by quoting these theorems in the exam!

But in general it is important to remember that, just because the proof is wrong, it does not follow that the theorem is false. Over the centuries, many false proofs of important theorems were published. Sometimes it took many years for the mistake to be spotted. In many cases, someone came along later and did a better job.

An example was the *Four-Colour Theorem*. Suppose you have a map which you want to colour, so that each country gets a single colour, and two countries which share a common boundary. Here is an example from Wikipedia (this is based on an image by Chas zzz brown, used under the Gnu Public Licence):



Theorem 10.3 (Four-Colour Theorem) *Every planar map can be coloured with four colours.*

This theorem was conjectured by Francis Guthrie in 1862. P. G. Tait and Alfred Kempe gave separate proofs in 1879–1880; both were wrong, but stood unchallenged for more than ten years. When Percy Heawood exposed the mistake in Kempe’s proof, he was able to save enough of the argument to prove that you

could manage with five colours. There matters stood until, in 1976, Kenneth Appel and Wolfgang Haken gave a correct proof (but one which could not easily be checked, since it made extensive use of the computer).

This was the first time that a major theorem had been proved with the help of a computer.

10.3.1 A theorem on primes

Theorem 10.4 *Let p be a prime number. Then $2^p - 1$ is prime.*

Proof Suppose, for a contradiction, that p is not prime, say $p = ab$, where $a > 1$ and $b > 1$. Now we use the algebraic fact that

$$x^b - 1 = (x - 1)(x^{b-1} + x^{b-2} + \cdots + x + 1).$$

Substituting $x = 2^a$, we see that

$$2^p - 1 = 2^{ab} - 1 = (2^a - 1)(2^{a(b-1)} + 2^{a(b-2)} + \cdots + 2^a + 1).$$

Thus we have a factorisation of $2^p - 1$, which is therefore not prime. This contradiction proves the theorem.

Comment The theorem is in fact false. If you take $p = 11$, then you can compute that

$$2^{11} - 1 = 2047 = 23 \cdot 89.$$

Where could the mistake in the proof be? If you are faced with this, and can't see the problem, the first thing you should do is to check all the details. Is the algebraic fact true? Yes it is, since if you multiply out you get

$$\begin{aligned} (x - 1)(x^{b-1} + x^{b-2} + \cdots + x + 1) &= (x^b - x^{b-1}) + (x^{b-1} - x^{b-2}) + \cdots + (x^2 - x) + (x - 1) \\ &= x^b - 1, \end{aligned}$$

since all the intermediate terms cancel.

Probably you already spotted that the theorem we actually proved is the *converse* of the one stated, namely

Theorem 10.5 (a true theorem!) *If $2^p - 1$ is prime, then p is prime.*

or, in contrapositive form (which the argument actually proves), “if p is not prime, then $2^p - 1$ is not prime.”

Remark A prime number of the form $2^p - 1$ is called a *Mersenne prime*, after the French mathematician Marin Mersenne (a contemporary of Fermat) who first investigated them. He knew that numbers of the form $2^p - 1$ (for prime p) are not all prime, but was slightly mistaken about exactly which ones are prime.

There are very efficient tests for primality of numbers of Mersenne's form. As a result, the largest known primes are all of this form.

Incidentally, it once happened to me (when a large number of this form had just been proved prime) that a journalist contacted me about a story that “the largest prime number” had been found. What would you have said to the journalist?

10.3.2 The colour of horses

Theorem 10.6 *All horses have the same colour.*



Proof This is a proof by induction. Let $P(n)$ be the statement:

In any set of n horses, all the horses have the same colour.

We prove $P(n)$ by induction.

First, to start the induction, $P(1)$ is obviously true; in a set containing only one horse, clearly all the horses in the set have the same colour!

Next, the inductive step: We assume that $P(n-1)$ is true and prove $P(n)$. Accordingly, let $\{H_1, H_2, \dots, H_n\}$ be a set of n horses. Then

- the subset $\{H_1, H_2, \dots, H_{n-1}\}$ is a set of $n-1$ horses, so horses $H_1, H_2, \dots, H_{n-1}$ all have the same colour, by the induction hypothesis;
- the subset $\{H_2, \dots, H_{n-1}, H_n\}$ is also a set of $n-1$ horses, so that the horses $H_2, \dots, H_{n-1}, H_n$ all have the same colour, again by the induction hypothesis.

It follows that all of $H_1, \dots, H_n$ have the same colour, and so the inductive step is complete, and with it the proof.

Comment In this case, the conclusion is obviously false; nobody seriously thinks that all horses have the same colour. So there must be a mistake in the proof.

There are two parts of the proof: beginning the induction, and the inductive step. Beginning the induction is OK. (You might be a bit worried about whether a statement about all elements of a set with a single element, but there is no problem with this.) So the problem must be in the inductive step.

So at this stage, you don't believe the result, but you believe that all horses in a set of size 1 have the same colour. So, to keep things simple, let's ask whether all horses in a set of size 2 have the same colour. Obviously not, you say. So let us look at the inductive step in the particular case $n = 2$. We are given the set $\{H_1, H_2\}$ of two horses. We are assuming that $P(1)$ is true – indeed, we agreed that it is true!

- In the set $\{H_1\}$, all the horses have the same colour.
- In the set $\{H_2\}$, all the horses have the same colour.

Does it follow that H_1 and H_2 have the same colour? Of course not. The inductive step was carefully written to make you think that there is an overlap between the sets $\{H_1, H_2, \dots, H_{n-1}\}$ and $\{H_2, \dots, H_{n-1}, H_n\}$, from which the conclusion would follow. But if $n = 2$, these sets don't overlap, and the argument falls down.

Of course the argument would be valid for larger n . So if $P(2)$ were true, then so would be $P(3)$, and $P(4)$, and so on. This is another way to see that $P(2)$ must fail.

10.3.3 Troubles with infinity

Here is a proof that I learnt at school.

Theorem 10.7 $0 > 0$.

Proof The proof uses the infinite series

$$S = \sum_{n=1}^{\infty} \frac{1}{n} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \dots$$

known to mathematicians as the *harmonic series*.

We have

$$\begin{aligned} 0 &= S - S \\ &= \frac{1}{1} + \left(\frac{1}{2} - \frac{1}{1}\right) + \frac{1}{3} + \left(\frac{1}{4} - \frac{1}{2}\right) + \frac{1}{5} + \left(\frac{1}{6} - \frac{1}{3}\right) + \dots \\ &= \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots \\ &> 0. \end{aligned}$$

(In the second line, we subtract the n th term $1/n$ of S from the term $1/(2n)$ twice as far along. In the fourth line we use the fact that in the series in the third line, the positive terms are all bigger than the following negative terms, so the sum is strictly positive. (In fact the last series is another important one; its value happens to be $\log_e 2 = 0.693147181 \dots$)

Comment The problem arises because the harmonic series is not convergent: its sum is infinite, and it is not valid to say that $\infty - \infty = 0$.

10.3.4 Troubles with complex numbers

Complex numbers also need to be handled with care. It is problems like this which made mathematicians very suspicious of using them at first.

Theorem 10.8 $-1 = +1$.

Proof We have $i^2 = -1$, so

$$-i = \frac{1}{i} = \frac{1}{\sqrt{-1}} = \sqrt{\frac{1}{-1}} = \sqrt{-1} = i.$$

Dividing both sides by i gives the result.

Comment Every non-zero complex number has two square roots, as we saw. If x is a positive real number, then we can define $\sqrt{x}$ to be the positive square root of x ; but there is no consistent way to extend the definition of $\sqrt{x}$ to complex numbers x .

10.3.5 More

There are more examples of false proofs (for you to find the flaw) in the supplementary material.

10.4 Study skills: And finally ...

We have come to the end of the module. It is a brand new module; you are the first students to have the opportunity to study it.

I know that, if you have taken to heart what you have learned in this module, you will be able to cope with whatever the rest of your degree course brings. I'm also confident that you will think that maths is enjoyable, one of the greatest pleasures that life can bring.

Good luck with the rest of your degree course. If you want to, please do keep in touch with me and let me know how you have used what you learned here.

10.5 Supplement: The party problem

Theorem 10.9 *At any party with at least six guests, either there are three guests who all know each other, or there are three guests who are all strangers to each other.*

We assume that the relation of knowing each other is symmetric (if A knows B then B knows A).

Proof We have to prove a disjunction, so we could assume that one of the two possible conclusions is false and try to prove the other, or we could assume that both are false and try to reach a contradiction. Instead, we will just plunge in and see what happens. We repeatedly divide into cases; in each case we have to come up with a set of three guests who either all know each other or are all strangers to each other.

Let A be one of the guests at the party. There are at least five more guests; so choose five of them, say B, C, D, E, F. There are various possibilities for how many of these A knows: it could be 0, or 1, or 2, or ..., or 5. But let us just divide into two cases: A knows 3 or more of B, ..., F; or A knows at most 2 of B, ..., F.

Case 1: A knows at least three of B, ..., F. Let us suppose that A knows B, C and D. (There are really lots more cases here, since the three guests known to A might be B, C, E, or ..., or D, E, F; but the argument is the same in each case.) If any two of B, C, D are acquaintances, say B and C, then A, B, C are three guests who are mutual acquaintances. (Again there are three cases, but the proof is the same in each case.) But if none of them are, then B, C, D, are three guests who are mutual strangers.

Case 2: A knows at most two of B, ..., F; so there are at least three of these five who are strangers to A. The argument is now the mirror-image of Case 1. Suppose that B, C, D are all strangers to A. If any two of B, C, D are strangers to each other, say B and C, then we have found three mutual strangers A, B, C; otherwise B, C, D are mutual acquaintances. $\square$

Remark The assertion would not be true for a party with five or fewer guests. Suppose that the guests were A, B, C, D, E; suppose that A knows B and E, B knows A and C, C knows B and D, D knows C and E, and E knows D and A,

while all other pairs are strangers. (Draw a diagram, with acquaintances joined by red lines and other pairs by blue lines.) Then there is no group of three mutual acquaintances, and no group of three mutual strangers.

10.6 Supplement: Proof that $n! > k^n$ for almost all n

We have some unfinished business from the notes. I promised you a proof that for any fixed number k , the inequality $n! > k^n$ holds for all but finitely many n . We saw that the inductive step works fine, but starting the induction is a problem.

Here are two approaches. Both are a bit unexpected. You never know what will come in handy in proving a theorem!

First proof Suppose that $n > k^2$. Then

$$\begin{aligned} n! &= 1 \cdot 2 \cdots (k^2 - 1) \cdot k^2 \cdots n \\ &> k^2! (k^2)^{n-k^2} \\ &= \frac{k^2!}{k^{2k^2}} k^{2n}. \end{aligned}$$

Call the first factor on the last line $c(k)$; note that it only depends on k . Now we have $n! > c(k)k^{2n}$, so if we could show that $c(k)k^{2n} > k^n$ for large enough n , we would be done.

Cancelling a factor k^n , what we have to prove is that $k^n > 1/c(k)$. Now k^n is an increasing function of n tending to infinity as n does, so for some n it will indeed be bigger than $1/c(k)$.

Now, as in the proof in Chapter 9, we simply have to rearrange the argument so that it goes forward to the conclusion.

Second proof Stir a little calculus into the mix. We know that

$$e^k = \sum_{n=0}^{\infty} \frac{k^n}{n!}.$$

Now if $k^n \geq n!$, then the k th term in the sum is bigger than 1. But there can't be more than e^k terms bigger than 1 in the sum, or they would add up to more than the quantity on the left-hand side. So all but at most e^k of the terms in the sum are less than 1, which means that $k^n < n!$ for all but finitely many values of n .

10.7 Supplement: More false proofs

For these proofs, it is up to you to find the mistake. Remember that a mistake in the proof doesn't necessarily mean that the theorem is wrong.

The first few examples are taken from the module MTH5117, Mathematical Writing. If you enjoyed this module, then take that one next year!

Theorem 10.10 *For any prime number p , p divides $2^p - 2$.*

Proof

$$2^2 - 2 = 2 \cdot 1, \quad 2^3 - 2 = 3 \cdot 2, \quad 2^5 - 2 = 5 \cdot 6, \quad 2^7 - 2 = 7 \cdot 18, \dots$$

Theorem 10.11 *For any real numbers x, y, z , if $x + z < y + z$, then $x < y$.*

Proof Take $z = 0$. Then $x + 0 < y + 0$, that is, $x < y$.

Theorem 10.12 $\sqrt{2} + \sqrt{3} < \sqrt{10}$.

Proof

$$\begin{aligned} \sqrt{2} + \sqrt{3} < \sqrt{10} &\Rightarrow (\sqrt{2} + \sqrt{3})^2 < 10 \\ &\Rightarrow 5 + 2\sqrt{6} < 10 \\ &\Rightarrow 2\sqrt{6} < 5 \\ &\Rightarrow 24 < 25, \end{aligned}$$

which is true, so the proof is complete.

Comment Can you fix this proof?

Theorem 10.13 $1 + 2 + 4 + 8 + 16 + \dots = -1$.

Proof Let $S = 1 + 2 + 4 + 8 + 16 + \dots$. Then

$$S = 1 + 2(1 + 2 + 4 + 8 + \dots) = 1 + 2S,$$

so $S = -1$.

To make it more convincing, here is a second proof:

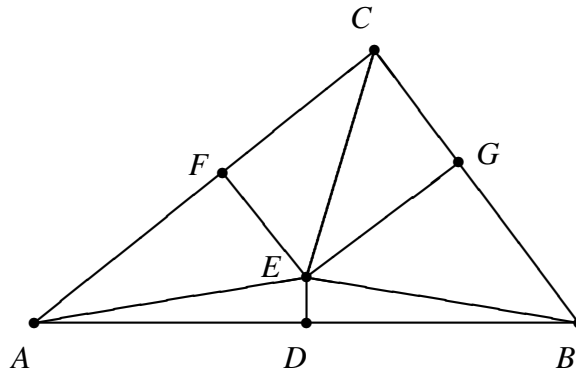
Proof We are summing a geometric progression with first term $a = 1$ and common ratio $r = 2$. Using the formula, the sum is $a/(1 - r) = 1/(1 - 2) = -1$.

The next theorem uses the kind of methods that Euclid used in his geometry textbook. The argument uses logical reasoning, but there is always a diagram to help the reader understand how the argument goes. The problem is that sometimes, instead, the diagram misleads ...

Theorem 10.14 *All triangles are isosceles.*

This proof is due to W. W. Rouse Ball in 1892. I have based this account on [Jim Loy's web page](#).

Proof



Let ABC be the triangle. Bisect the angle C , and draw the perpendicular bisector of the side AB (bisecting AB at D). If these two lines were the same or be parallel, the triangle ABC would be isosceles. So suppose not. Then they must meet at some point E . This point E must be either inside the triangle, or outside the triangle, or on the line segment AB . I will do the case where they meet inside the triangle; the other cases are similar.

Draw the lines AE and BE . Draw the lines EF and EG , perpendicular to sides AC and BC respectively, as shown in the diagram.

For each of the three pairs of congruent triangles below, you should explain why they are congruent.

The right triangles CEF and CEG are congruent; so $EF = EG$ and $CF = CG$. The right triangles ADE and BDE are congruent; so $AE = BE$. The right triangles AEF and BEG are congruent; so $AF = BG$. Now $AC = BC$ by adding of equals to equals. In other words, triangle ABC is isosceles.

10.8 Supplement: The ontological argument

Almost everyone accepts that the question of whether God exists or not cannot be decided by rational argument, but requires an act of faith. However, this has not

stopped theologians from trying to give “proofs” that God exists. Here is a version of the most famous, the *ontological argument*; it is associated with the name of Anselm, former archbishop of Canterbury. Do you agree with the argument or not?

Theorem 10.15 *God exists.*

Proof

- Our understanding of God is a being than which no greater can be conceived.
- The idea of God exists in the mind.
- A being which exists both in the mind and in reality is greater than a being that exists only in the mind.
- If God only exists in the mind, then we can conceive of a greater being — that which exists in reality.
- We cannot be imagining something that is greater than God.
- Therefore, God exists.

In thinking about this argument, you may be helped by first finding the error in the next theorem.

Theorem 10.16 *1 is the largest natural number.*

Proof Let n be the largest natural number. We have to prove that $n = 1$, which we do by arguing by contradiction. So suppose that $n \neq 1$. Then, obviously, $n > 1$. But then $n^2 > n$, and so n is not the largest natural number. The contradiction proves that $n = 1$.

Exercises

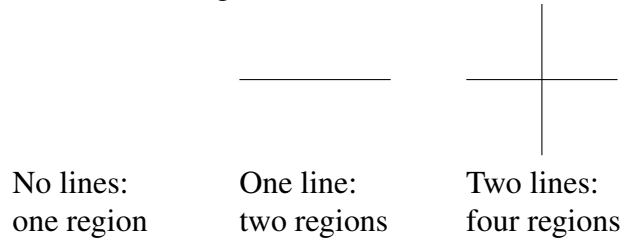
10.1 What is wrong with the following arguments?

(a) **Proposition** *If $a \neq 0$ and $ab = ac$, then $b = c$.*

Proof Take $a = 1$: then certainly $a \neq 0$. Now $ab = b$ and $ac = c$, so if $ab = ac$, then $b = c$.

- (b) **Proposition** n lines, no two parallel and no three concurrent, divide the plane into 2^n regions.

Proof Each new line divides each existing region into two parts and so doubles the number of regions:



- (c) **Proposition** If $2 + 2 = 5$, then I am the Pope.

Proof We are given that $2 + 2 = 5$. Subtract 2 from both sides: $2 = 3$. Now subtract 1 from both sides: $1 = 2$. Now the Pope and I are two; therefore we are one.

- (d) **Proposition** $+1 = -1$.

Proof Let $x = +1$. Then

$$\begin{aligned}
 x^2 &= 1 \\
 \Rightarrow x^2 - 1 &= 0 \\
 \Rightarrow \frac{x^2 - 1}{x - 1} &= 0 \\
 \Rightarrow \frac{x - 1}{x + 1} &= 0 \\
 \Rightarrow x &= -1,
 \end{aligned}$$

and substituting $x = +1$ we get $+1 = -1$ as required.

10.2 In each of (a), (b) and (c) below, decide which (if any) of the given statements is the contrapositive, and which (if any) is the converse, of the given statement. (You are not asked whether the statements are right or wrong, but can you decide?)

- (a) If $a < b$, then $a^2 < b^2$.

- (i) If $a^2 < b^2$, then $a < b$.
- (ii) If $a > b$, then $a^2 > b^2$.
- (iii) If $a^2 > b^2$, then $a > b$.

- (b) If p is prime and p divides ab , then $p \mid a$ or $p \mid b$.

- (i) If $p \mid a$ or $p \mid b$, then either $p \mid ab$ or p is prime.

- (ii) If $p \mid a$ or $p \mid b$, then p is prime and $p \mid ab$.
 - (iii) If $p \nmid a$ and $p \nmid b$, then either $p \nmid ab$ or p is not prime. (Here $a \nmid b$ means “ a does not divide b ”.)
- (c) If it is raining, the bus is late.
- (i) If the bus is late, then it is raining.
 - (ii) If the bus is not late, then it is not raining.
 - (iii) If the bus is on time, then it is not raining.

10.3 (Puzzle) How many squares are there on a standard chessboard? (I mean the total number of squares, not just the 64 single black and white squares.)

10.4 (Project) At the end of the course, look back on your definition or description of mathematics in the questions for the first chapter. Would you answer it differently now?